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Business models and strategies for industrial symbiosis based on water-smart solutions

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Business models and strategies for industrial symbiosis based on water-smart solutions

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ABSTRACT
This report assesses the industrial networks and new business models under development in WIDER UPTAKE. Perspectives from previous research on industrial symbiosis, social networks and circular business model experimentation are applied. We find that a mix of stakeholders and the presence of a network orchestrator or bridging actor are characteristic of the networks where industrial symbiosis is emerging. For water reuse, there are pronounced regulatory and organisational barriers, and the networks have a higher share of public stakeholders. While users are represented in some networks, more direct contact with end-users is generally recommended. The identified business models are all associated with positive environmental and social impacts. The economic model is in place and well aligned in some of the cases. However, for water reuse and biocomposite for riverbank protection this layer is less defined, and the balance between costs and revenues is challenging. All the models are to some extent associated with resource dependency, which must be managed carefully, to avoid lock-in scenarios. Furthermore, they interact closely with other value chains and business models. For a large part, this is how they generate economic, environmental, and social value, but it can also be challenging, in terms of value chain management and strategic planning.

KEYWORDS
Industrial symbiosis, networks, business models, circular economy, sustainability

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1 Executive summary

Water scarcity and quality are pressing sustainability challenges in our time, linked with climate change, biodiversity, and the need to feed a growing population. Against this background, WIDER UPTAKE works to facilitate industrial symbiosis based on reuse of treated wastewater, nutrient recovery, and production and use of biochar and biocomposite materials in five countries. This report presents results from two activities: An analysis of the industrial networks in the demonstration cases in the project, and a process to explore and assess new business models, for upscaling the innovations.

The assessment of industrial networks combined qualitative analysis linked to the concept of Industrial Symbiosis (IS), and Social Network Analysis (SNA). The analysis shows that the number of stakeholders considered relevant has increased during the project. The most central actors are water utilities and business companies utilising resources from water and wastewater treatment. However, public authorities, target users, research organizations and technology providers are also connected, in varying degrees. While several of the networks emerge as anchor-tenant systems, there are also cases where pre-existing relationships are the entry point, and of green-twinning, where new exchanges are developed based on previous exchange of resources. Most of the networks have a clearly defined core. In some, a key "orchestrator" stands out, while one has a less central, but well-connected actor in a bridging role, and one network appears to be more distributed.

The exploration of new business models (BMs) used the Triple Layered Business Model Canvas (TLBMC) in combination with elements of Ecologies of Business Model Experimentation (EBME). As expected, the TLBMC assessment suggests that the identified BMs are associated with environmental gains, both through direct impacts of the products and services provided, and through system interactions, e.g., saving potable water, substituting fossil phosphorous, and limiting soil erosion, deforestation, and use of tropical hardwood. In terms of social value, there are also strongly positive effects, e.g., employment, awareness, job satisfaction, urban liveability, and reduced health risk, as well as wider impacts related to food security, economic stability, social cohesion, and community revival.

However, the alignment between the environmental, social, and economic layers of the BMs is variable. For the soil improvement products from Sirkula, struvite from Hias and organic fertiliser from IVAR and Terramarine AS in Norway, most elements are in place and the models are operating or to be realised within few months. For biochar produced by SSGL, Ghana, the economic model is also sorted out and promising in a short-term perspective. When it comes to biocomposite material for riverbank protection in the Dutch case, not all relationships are clear and high costs remain a challenge. Thus, this model may more likely be implemented in a medium-term perspective. While process solutions for reuse of treated wastewater are mature, the BMs for this are less developed. In Sicily, an ultrafiltration unit is in place and the local stakeholders are committed, but the economic model for full scale operations is not clear. In Prague and Accra, no full-scale infrastructure has been built, and the distribution of responsibilities and costs has not been much considered yet.

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These observations are linked to dependencies between the BMs and their wider "ecosystems": Dependency on infrastructure development is a key issue for water reuse, and to a lower degree in the other cases. Resource dependency is a factor in all cases. However, the soil improvement and organic fertiliser production can draw on multiple sources. This is also the case for biocomposite materials in the Netherlands, but here competition for biomass is an issue already. In terms of dependency on existing value chains, the costs of chemicals are of strong influence, both in the water reuse cases and for the recovery of phosphorus as struvite. The BMs for struvite and soil improvement involve marketing and distribution through others. Terramarine, NPSP, and SSGL will partly handle this themselves, but the considered BM for biocomposite from NPSP requires collaboration with an external producer. Furthermore, we find interdependence and future synergies with other circular bioeconomy activities in both of the Norwegian cases. The biocomposite material in the Dutch case competes with tropical hardwood and recycled plastics, and the market for the biochar from SSGL is linked to use of charcoal stoves, competing with regular charcoal and LPG (liquid petroleum gas). In the water reuse cases, the alternative cost of pumping groundwater or using other sources will be of influence, and in Prague there may be synergies with blue-green infrastructure providers.

Looking towards the future, the viability of the new BMs and their impacts will be influenced by wider trends, including climate change, population development, and future prices on energy and fertilisers. Public acceptance is not seen as a challenge for the soil and fertiliser products, nor for the biochar and biocomposite material. When it comes to water reuse, there are positive attitudes to industrial reuse in Prague. However, farmers in Sicily request certification of the water quality, and the stakeholders in Ghana expect a certain scepticism from consumers. Market acceptance will be strongly influenced by costs, as well as usability and compatibility with existing practices. In terms of socio-political acceptance, circular economy is a stated aim in all the case countries, but there are still uncertainties regarding future framework conditions. In the Czech Republic and Ghana, treated wastewater for reuse is not legally recognised, and in Sicily there are strong debates over quality testing requirements, which will influence the costs significantly. Moreover, several of the cases find that limited coordination across scales and levels is a challenge.

These findings underscore the need to build awareness and knowledge about industrial network development, to enable wider uptake of water-smart solutions. The cases with more mature BMs are also the cases with a higher share of business actors, while those facing severe regulatory barriers have a focus on engaging public decision-makers. In the cases with more mature BMs, we also find a network orchestrator or someone in a bridging role, which may be crucial for industrial symbiosis. A closely-knit core with reciprocal relations may be positive in terms of coordination and exchange, but a more distributed structure where peripheral actors can provide links to other networks, as in the Dutch case, may also be beneficial from an innovation point of view. While user representatives are engaged in some of the networks, all the cases note a need for more involvement of end-users, to better meet their needs and preferences.

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2 Introduction

WIDER UPTAKE aims to facilitate industrial symbiosis as a means to increase resource efficiency, limit climate gas emissions and develop sustainable business based on water-smart solutions. Recognizing that the barriers for wider uptake of water-smart solutions are not only technological but also of organizational, regulatory, social, and economic character, the project aims to identify and demonstrate means to overcome barriers related to:

1. Monitoring and control of health and quality risks
2. Circular-economy and efficiency potential
3. Governance and business models for industrial symbiosis
4. Measuring water smartness and progress towards achievement of the SDGs.

The project runs demonstrations of wastewater reuse for agriculture and urban greening, phosphorus recycling, biogas, digestate/organic fertilizer and biochar utilisation, and production of biocomposites for manufacturing materials with resources recovered from the whole water cycle. The partnership includes eleven water utilities and industry partners, as well as seven research institutions, distributed across five countries (Norway, Netherlands, Czech Republic, Italy, and Ghana), as illustrated in Figure 1.

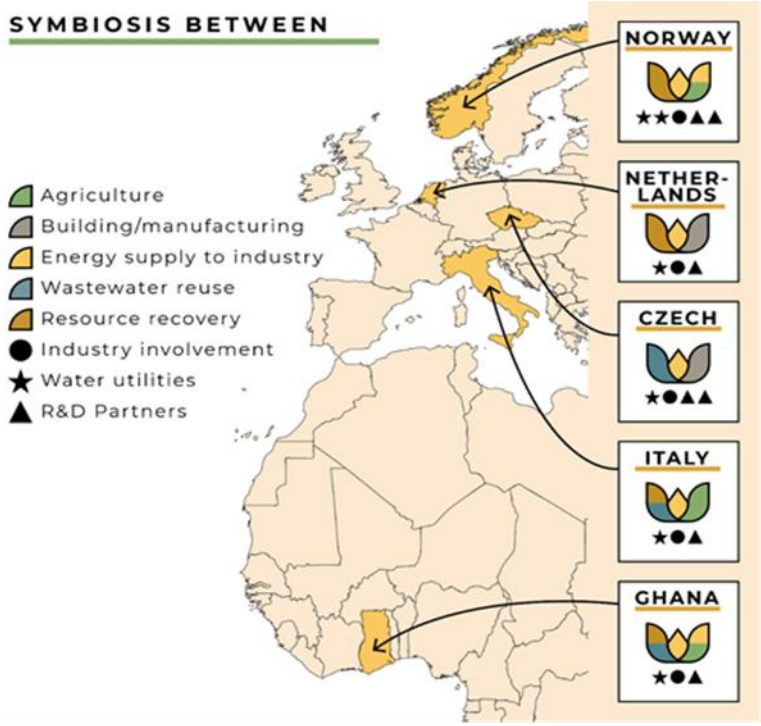


Figure 1: WIDER UPTAKE demonstration cases.

This report presents results from activity in Work Package 4: Governance and business models, carried out between Month 12 and Month 36. This activity consisted of three tasks:

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a) A mapping of stakeholders, linked with joint efforts to engage the industry partners and other key stakeholders in the demonstration cases in industrial network development for industrial symbiosis based on water-smart solutions.
b) A process to explore innovative BMs in order to enable wider uptake of water-smart solutions, designed and implemented based on a review of existing methods and perspectives on sustainable and circular BM development.
c) Testing of the most promising models identified, by assessing their potential impacts in future scenarios.

To some extent, these tasks have built on information about the contextual interactions, drivers, barriers, and opportunities prevailing in each demonstration case, collected during the test application of the Governance assessment for Circular Economy based on Water resources (GOCIWA), presented in D4.1.

It is important to distinguish between this work and the activities related to exploitation and dissemination, which are part of WP6. In line with Bocken et al. (2018), the purpose of the business model experimentation conducted here has been to identify and build legitimacy for WIDER UPTAKE solutions across stakeholder groups (internal and external), by joint collection and sharing of information, identify best practices and test assumptions about the future feasibility and impact of the considered BMs in a simple, low-cost way. Thus, this work does not concern business plans or business development as such.

In final phase of the project, the findings from the initial governance assessments and the results communicated here will be used to identify pathways to industrial symbiosis based on water-smart solutions in different contexts (T4.4).

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3 Objective and structure of the report

The objective of this report is to provide new knowledge on the development of industrial networks to enable uptake of water-smart solutions and shed light on the novel BMs that currently are being developed and tested by water utilities and other actors. Based on the results of the tasks listed above, as well as on the experience[s] gained from running the respective processes in the different case studies, we address the following research questions:

1. What characterises the industrial networks in the WIDER UPTAKE demonstration cases?
2. How can these networks be strengthened or expanded, to enable wider uptake of the water-smart solutions in focus?
3. What kind of business models are the key actors aiming for?
4. To what extent are they dependent on external factors and future trends?

Our assessment of industrial networks uses the literature on Industrial Symbiosis (IS) as point of departure, while also drawing on insights from Social Network Analysis (SNA). For the work on BMs a review of recent literature was carried out, to ensure that the most recent perspectives and methods were considered.

The following chapter provides an overview of the background for this work, in terms of global water challenges and resource scarcity, as well as policies, regulations and previous research emphasizing the opportunities and barriers to circular economy based on water resources. Chapter 5 zooms in on Industrial Symbiosis (IS), a concept that is central to both the WIDER UPTAKE and in the Horizon 2020 funding call. We focus especially on perspectives and tools to assess and facilitate industrial network development, to enhance the uptake of symbiotic solutions based on water resources. In chapter 6, we apply some of the approaches deemed most relevant for WIDER UPTAKE, to characterise the networks surrounding the demonstration cases, and discuss steps to further develop and strengthen the existing networks.

Chapter 7 moves on to discuss BMs. Here, we are not concerned with detailing individual business plans, but with sustainable business model innovation, as a process where a set of actors jointly identify or envision the key elements in a new, collaborative model, whereby they can create value from addressing societal and environmental needs (Boons & Lüdeke-Freund, 2013). Such innovation is surrounded with increasing interest in academic research, as well as from business practitioners, investors, policy makers and development partners dealing with green solutions and transition strategies. The first part presents the findings from a literature review that was carried out to assess the state of the art and identify possible new perspectives on how to stimulate and frame BM exploration. The second part describes the approach selected for business model exploration in WIDER UPTAKE.

The following chapters (8 to 13) present and discuss the results of our business model exploration with the key partners and selected stakeholders in the respective demonstration cases. Each chapter provides an overview of the identified BM, together with its actual and potential economic, environmental, and social impacts. The discussion addresses the maturity of the BM, as well as its interaction with contextual factors, including dependencies on

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infrastructure, resources, and other BMs, as well as social acceptance and political feasibility in a longer-term perspective.

Chapter 14 provides an overarching discussion, where we investigate the commonalities and differences across the cases, in terms of business model maturity, alignment between the economic, environmental, and social layers of the BMs, as well as their dependencies in relation to infrastructure development, resource availability, and other value chains and BMs. We also provide a brief discussion of wider societal trends and current policies, and regulations may influence their development and impacts in a long-term perspective. Chapter 15, finally, is a summary conclusion.

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4 Background

4.1 Water challenges and resource scarcity

Water availability and quality are two of the main challenges for society in the 21st century. 2.3 billion people live in water-stressed countries, of which 733 million people are in high and critically water-stressed countries (UN-Water 2021). Water availability and quality are strongly related with human health, food production, adequate ecosystem functions, and global economic growth (UNESCO, 2021).

Water availability can be measured by the UN Sustainable Development Goal (SDG) indicator 6.4.2, *level of water stress*, which is the freshwater withdrawal as a proportion of available freshwater resources. While this indicator globally is at a safe value, water availability is said to be stressed if this index is higher than 25 %, and water scarcity is a challenge in many countries. Lack of water may have several causes: demand for water may be exceeding available water resources with sufficient quantity and/or quality, water supply infrastructure may be inadequate, or institutions may be failing to balance everyone's needs (UN-Water 2023).

Climate change affects freshwater resources negatively both in terms of quantity and quality (Mannina et al., 2022). Reduced rainfall and higher temperatures lead to diminishing groundwater resources because of less infiltration and increased evaporation, even without increased demand for domestic, industrial, or agricultural water supply. Scarcity due to water quality can also be made worse, e.g., because there is less water for dilution and exchange in water bodies receiving runoff and/or discharges of pollutants resulting in reduced availability of water resources with required quality for various uses. Climate change is also associated with other water challenges, e.g., increased precipitation with high intensity rain storms resulting in flooding and problems with water quality and stormwater management.

Global water use is growing at more than twice the rate of population increase, and a 60 percent surge in food demand may be expected by 2050 (FAO, 2020). Behind this figure there are many uncertainties. According to a recent meta-analysis of 57 global scenarios, the total global food demand is expected to increase by 35% to 56% between 2010 and 2050 (van Dijk et al., 2021). At any rate, there is an urgent need to address water scarcity (FAO, 2020).

In the context of water scarcity, the agriculture sector is both a cause and a victim. Agriculture is the main consumer of water with about 70% of the total world water consumption (Bakari et al., 2022). While the demand for water for agriculture and other uses is increasing rapidly, more frequent, and severe droughts are having an impact on agricultural production, while rising temperatures translate into increased crop water demand. Moreover, climate change impacts are making water availability and quality more unpredictable in many parts of the world.

Another important issue stressing the agriculture field is the availability of phosphorus (P). Fertilization in agriculture has fully dominated the use of P resources since more than 90% of current use of phosphorus resources (and >80% of fossil P resources) is as fertilizer in agriculture (Koppelaar and Weikard, 2013; Chowdhury et al., 2014). However, ongoing mining may deplete and reduce the quality of the remaining, fossil P resources. While there are

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different views on how imminent this risk is, phosphate rock and phosphorus have been placed on EU's list of critical raw materials (European Commission, 2018). Moreover, the supply and demand of fossil P are unevenly distributed in geographical terms. This has geopolitical consequences and may well affect security of supply.

The current use of P has also got environmental impacts. Phosphorus and nitrogen compounds are the main cause of eutrophication, which involves various risks for humans, animals, and plants (Almanassra et al., 2021). Further, mining of phosphate rock gives rise to negative environmental impacts due to P contaminants derived from fossil P resources. Reducing the demand for virgin fossil phosphate may reduce environmental burdens and may improve the future security of supply. Technically speaking, the demand for fossil P resources may be reduced in multiple ways. Consumption of P may be restricted to essential uses, and agricultural use may be made more efficient, by technical improvements and/or substituting meat protein with plant protein. Moreover, substantial amounts of P may be recycled in mining, industry and from agricultural and (peri-)urban residues (Reijnders, 2014). Currently, the scope related to P recovery from wastewater, seems to be around 10 % of agricultural requirements (Matayeva et al., 2022).

Furthermore, the agriculture field is distressed by the soil erosion issue. Soil erosion is due to a number of physical and human processes (such as, wind, rainfall and surface runoff, mass movement, climate, deforestation, agricultural practices etc.), some of which may increase with increasing climate change. Thus, smarter land management is a necessity and sustainable soil improvement products have a crucial role to play (WRI, 2020).

Against this background, Water Europe has launched a vision of a water-smart society that highlights the need for radical system change. The vision was first presented in 2017 but has since been revised. The most recent definition reads as follows: *"A Water-Smart Society is one in which the value of water is recognised and realised to ensure water security, sustainability, and resilience; all available water sources are managed so that water scarcity and pollution are avoided; water and resource loops are largely closed to foster a circular economy and optimal resource efficiency; the water system is resilient against the impact of climate and demographic change; and all relevant stakeholders are engaged in guaranteeing sustainable water governance"* (Water Europe 2023:4).

Within this context, wastewater valorisation has a key role in providing water and resources coming from secondary sources. As summarised by Kundu et al (2022) the valorisation of wastewater is strictly linked with 11 of the 17 UN SDGs (Figure 2). Specifically, the wastewater valorisation is directly linked with SDG6 “clean water and sanitation” providing for example clean water for reusing, thus promoting life on land (SDG15) and favouring sustainable cities and communities (SDG 11). This aggregated interlinking information is vital for policymakers, researchers, practitioners, public institutions on pollution control, nutrient and resource recovery, energy recovery and the development of national action plans (Qadir et al., 2020).

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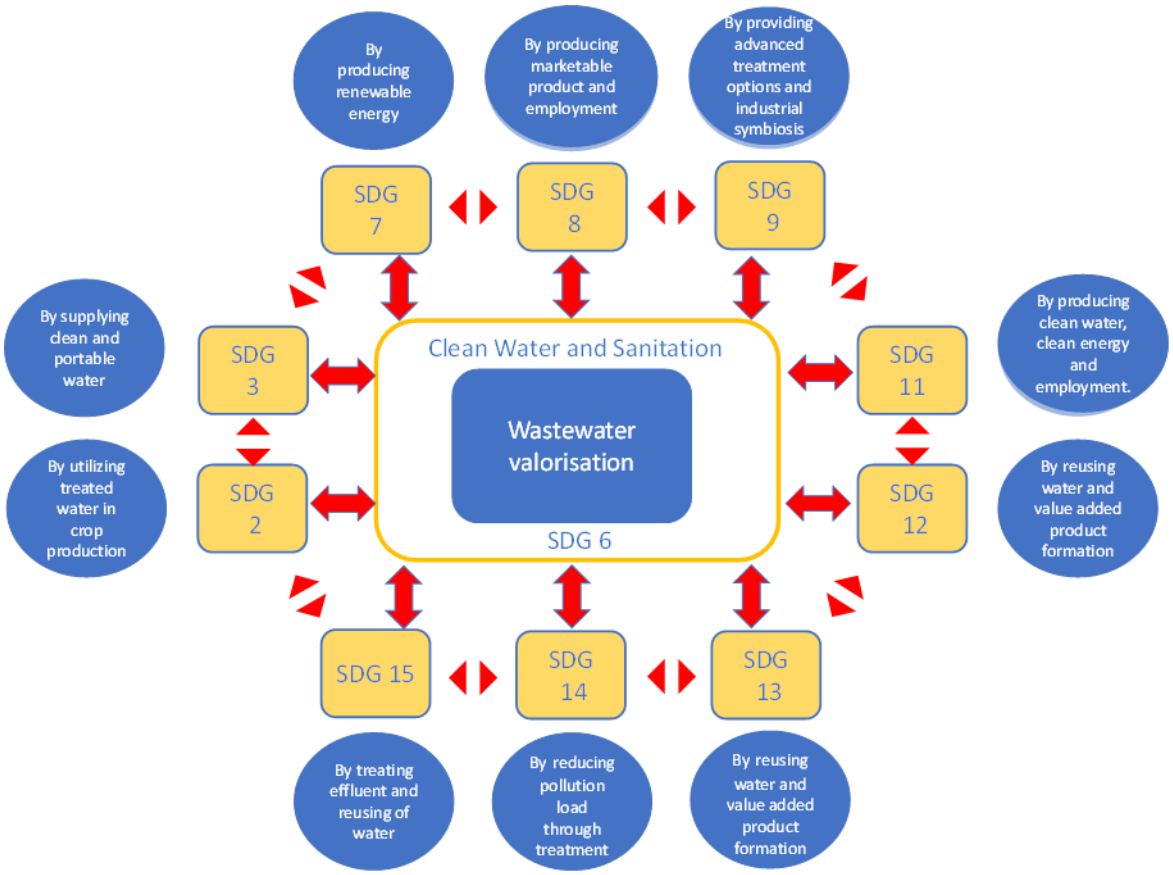


Figure 2: SDGs as basis for wastewater valorisation.

4.2 Circular economy based on water resources

The transition to a circular economy (CE) implies a fundamental change from the traditional logic of business. Overall, the concept emphasizes less resource use and eliminating negative environmental externalities while delivering more value for stakeholders and society. Thus, CE can be defined as *"an economic system in which resource input and waste, emission, and energy leakages are minimized by cycling, extending, intensifying, and dematerializing material and energy loops. This can be achieved through digitalization, servitization, sharing solutions, long-lasting product design, maintenance, repair, reuse, remanufacturing, refurbishing, and recycling"* (Geissdoerfer et al., 2020b: 3).

While there are multiple definitions, most of these focus on materials and energy, and pay little attention to water and water-related ecosystems (Salminen et al., 2020). Against this background, Salminen et al. (2022) provide a definition of water-smart CE, as an economic concept whereby water is abstracted within the ecological boundaries of surface and groundwater bodies and used efficiently, with a focus on reducing losses and recovering energy and other resources, and without significant risks to ecosystems and human health (Salminen et al., 2022:4).

A growing body of research shows that water reuse can be safe and sustainable. Reclamation and reuse of wastewater is increasingly prioritised, especially in water-scarce countries (Wilcox

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et al., 2016). Resource recovery is also recommended, to limit the use of critical raw materials and make valuable nutrients available for human use (IWA, 2016). Technologies for struvite formation for recovery of phosphorus and nitrogen are presently at technology readiness level 7 to 9 (Gherghel et al., 2019). However, costs are high, and a market is yet to develop. Other products that can be recovered are nitrate, ammonium sulphate and calcium phosphate. From sewage sludge, adsorbents, and heavy metals, as well as proteins and enzymes may be extracted and provide added value (Guerra-Rodriguez et al., 2020). Furthermore, production of raw material for bioplastics can be combined with wastewater treatment (Morgan-Sagastume et al., 2014). While biogas production is implemented widely, other energy recovery options are explored (Gherghel et al., 2019). As demonstrated in WIDER UPTAKE, new biocomposite materials are also being developed.

With a view to these opportunities, Smol et al. (2020) propose a CE framework for the water and wastewater sector, based on pre-established waste prevention rules or so-called "R-strategies": reduction, reuse, recycling, and recovery, as well as two additional aspects: reclamation (removal)—focused on highly effective removal of pollutants, and rethink—as a basis for introducing systematic changes in the whole value chain. However, not all forms of CE necessarily lead to positive environmental impact.

Key considerations are thus to ensure that a circular product substitutes a primary production alternative and does not lead to increased overall consumption, and that customers actually select it (Zink and Geyer, 2017). Water-smart solutions based on wastewater reuse and resource recovery are well positioned to achieve such goals, as the products are direct alternatives to comparable primary products. However, to make a positive environmental impact, water-smart circular products must be competitive in terms of price, quality, and availability to customers (Zink and Geyer, 2017).

4.3 System change and need for new mindset

Transitioning towards circularity implies changes in some of the key (taken-for-granted) principles of wastewater management, from linear treat-and-dispose towards resource recovery and creating synergies between water management and other sectors, e.g., agriculture, energy, building and construction. This, in turn, requires various changes in structure and actor behaviours, including new value chains and BMs, changes in infrastructure and other technologies (wastewater treatment plants, etc.), institutions (including regulatory change, development of standards, seeing wastewater as a resource, etc.), technological and commercial knowledge regarding CE, etc. (Geissdorfer et al., 2018). In other words, a transition towards CE implies a change in the prevailing 'socio-technical regime' or the *"semi-coherent set of rules that orient and coordinate the activities"* of key actors (Geels, 2011). A key part of this regime are the prevailing engineering practices, skills, procedures, and ways of defining problems.

Hence, a notable dimension in CE transition is change in the institutional logics guiding its key actors. Arguably, the water sector has been dominated by a so-called 'hydraulic logic' which prioritizes security of supply and safe quality of water and wastewater. While security and safety remain a priority, competing logics have emerged in recent years. A 'water market logic'

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highlighting economic efficiency and rationalization is gaining traction. Another and sometimes competing institutional logic has been the 'water sensitive logic' influenced by community and ecologist perspectives, which is oriented towards decentralisation, small-scale recycling, and local wellbeing (Fuenfshilling and Truffer, 2014).

Interlinked with these developments is an increasing call for integrated water resources management (IWRM), to promote the coordinated development and management of water, land, and related resources in order to maximize economic and social welfare without compromising the sustainability of vital ecosystems. The Water-Energy-Food (WEF) nexus highlights the need for an integrated approach to water, food and energy distribution, and consumption, which implies a stronger focus on users and the locations of their activities, to align with CE principles, environmental health, and human well-being (MacGrane et al., 2019). Moreover, wastewater treatment plants are increasingly encouraged to see themselves as wastewater resource recovery facilities (WRRFs). This, indeed, requires a paradigm shift, where systems thinking, industrial networks, and new business models play a central role.

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5 Industrial symbiosis networks

5.1 The concept of Industrial Symbiosis

Broadly speaking, industrial symbiosis (IS) refers to the process by which wastes, or by-products of an industry or industrial process become the raw materials for another (EC, 2018). IS can be defined as engaging *"traditionally separate industries in a collective approach to competitive advantage involving physical exchange of material, energy, water, and by-products"* (Chertow 2000:313). Furthermore, Chertow (2007) has introduced the '3-2 heuristic' (see Figure 3) as a criterion for what can be deemed as an industrial symbiosis rather than a bi-lateral sharing/transaction of waste/materials. The 3-2 heuristic asserts that there must be at least three different actors involved in the exchange of at least two different resources for the exchange to be described as a basic type of IS (ibid.:12).

Whereas the early research on IS mostly focused on the exchange of tangible objects such as materials, infrastructure, and by-products, more recent studies have also highlighted the importance of sharing intangible resources, such as knowledge (Fraccascia and Giannoccaro, 2020). In the work on IS networks in WIDER UPTAKE, we have mainly emphasised the exchange of tangible objects between different industrial actors involved in the industrial networks where the demonstration cases and main actors in WIDER UPTAKE are involved.

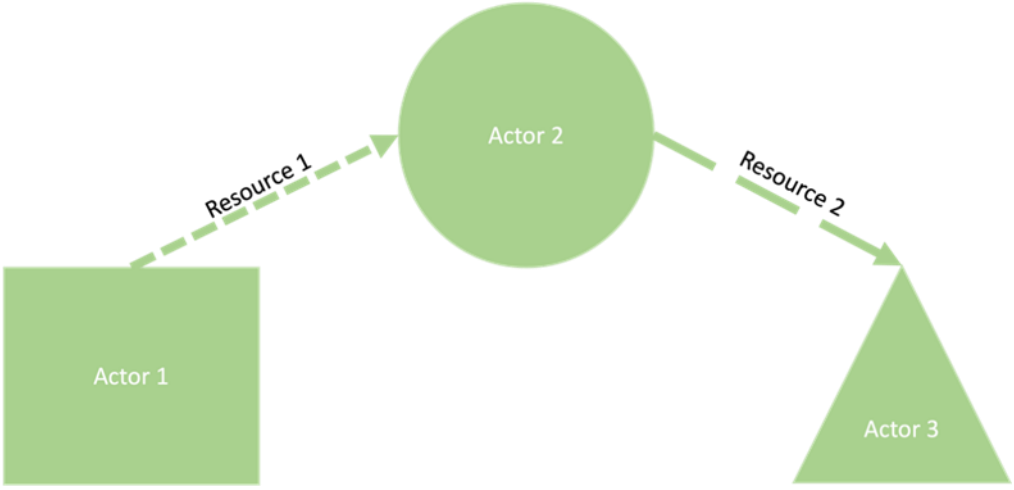


Figure 3: The 3-2 heuristic of industrial symbiosis (adapted from Chertow, 2007).

A key factor in IS is close coordination and collaboration between the actors that share or exchange the tangible (waste, by-products, energy etc.) or intangible (knowledge, experience, information etc.) resources required (Chertow 2000). Thus, the industrial networks of the industry partners in WIDER UPTAKE are essential, as it is through interaction and collaboration with their respective stakeholders that the partners work towards achieving IS. To develop more knowledge of these processes and how they may be strengthened, it is necessary to get more detailed insight into the characteristics of the industrial networks around the demonstration cases and their key actors.

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Several studies have provided insights on factors that facilitate IS in different industries. Walls and Paquin (2015) argue that there are four levels of analysis at which IS operates; 1) the institutional level, 2) the network level, 3) the organizational level, and 4) the individual level. The institutional level refers to how IS can be coordinated by governments through policies or governmental agencies that promote IS. The network level deals with how relationships and exchanges between actors develop, what roles and responsibilities actors within these networks can take on, and how these are influenced by power relations, social ties, and proximity (e.g., geographical collocation). The organizational level refers the intra-firm factors that influence whether firms engage in IS networks and collaborations. The individual level relates to how certain individuals within firms push for IS and engage the broader community, e.g., through utilizing their social relationships and networks (Walls and Paquin 2015). Although all four levels are relevant in studies of IS, the work in WIDER UPTAKE is aimed at developing the IS involving specific actors in relation to the project demonstration cases. Therefore, we pay particular attention to factors that influence the current workings of these networks and their development.

According to Chertow (2000) two key elements for IS sustainability are cooperation and perseverance: The development of IS networks and relationships require work overtime and that evolutionary approaches therefore are key. She outlines three different approaches to developing IS networks:

- Green-twinning/by-product synergy: the identification of new possible exchanges (e.g., of by-products) is easier when one already exists. The existing exchange then works as a springboard for further IS network development.

- Pre-existing organizational relationships and networks can be the starting point for the development of industrial symbiosis, as existing relationships between two parties make the development of new joint avenues more likely. Thus, trust is a key factor.

- The anchor-tenant model: the existence of large companies (anchors) that can provide stable flows of by-products that can be exchanged and utilized by other firms (tenants). These firms' core business is dependent on the anchor and its provision of by-products which these "scavengers" can reuse, repair, remanufacture or recycle (Walls and Paquin 2015). Walls and Paquin (2015) highlight the issue of resource dependence and asymmetric power relationships in such IS cases, where the scavenger firm is dependent on the anchor firm in terms of resource supplies and argues that discussions of these aspects still are limited in IS literature.

Resource dependency within an IS network could potentially lead to lock-in scenarios where the entire network depends on one core anchor firm, and the result of that core actor leaving the network could cripple the remaining network. To avoid resource dependency, a diversity of actors who contribute with different resource inputs and outputs, technologies, and processes is key (Walls and Paquin 2015). Diverse IS networks are also more flexible, meaning that if one actor leaves the network another can take its place, and they are therefore more resilient to

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shocks/disruption to the network. At the same time, IS networks profit from being robust, meaning that the configuration of the network is stable.

Interaction with policymakers, public administrations, health and safety agencies, and other public actors with a stake in water management has been identified as key issues for succeeding with industrial symbiosis (Oughton et al., 2021). As such, governance aspects are important in the realisation of IS and should be a core concern for utilities and companies that strive to develop industrial symbiosis networks.

Lock-in situations in IS networks can also relate to institutional lock-in, also known as path dependence. In such a situation, the network has locked into the established technologies and practices and fails to make progress through innovation. These lock-ins can inhibit the IS network from utilizing the resources in the most environmentally friendly and efficient way.

5.2 IS network mapping approach

The mapping of stakeholders and assessment of industrial networks in WIDER UPTAKE is intended to provide insights on the existing industrial networks and how they may be developed further in order to enable IS and sustainable business models for the solutions being tested in the demonstration cases.

The data collection for the mapping exercise was a two-step process, where the first part was done through the distribution of Excel sheets for all key industrial partners to fill out (see Table 1). This mapping was done in December 2021. The objective was to identify stakeholders and stakeholder interactions in their industrial networks (i.e., public actors, interest organizations, private actors) relevant for the uptake of the solution(s) demonstrated in the different cases (e.g., biocomposites, water reuse, fertilizer production, etc.). Examples of interactions can be business transactions, collaboration(s), funding (research or other), lobbying, awareness creation, etc.

Table 1: Information to be filled out by partners for IS network mapping.

Categories	Examples/questions for consideration
Stakeholder name	Full name of organisation, website if applicable
Type of stakeholder	Government bodies, policy makers, regional/local authorities, owners, private companies, end users, research/knowledge partners, interest organizations, NGOs, etc.
Degree and form of interaction	Are you in contact with this stakeholder? How often? How does the interaction take place (e.g., in project meetings, shareholder meetings, through letters, reports, media, industry events, etc.)?
Direction of interaction	Between you and the mentioned stakeholder, who mostly initiates interactions? How?
Purpose of interaction	What is the reason(s) for your interaction? What stake or role does the respective stakeholder have I your activity and/or

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	the demonstrated solution? (co-development, financing, supporting, regulating, monitoring, etc.)? Are there specific objectives or gains you want to achieve, concerning your CE solution?
Quality of interaction	How do you rate your present relationship? Is it satisfactory? Does it hinder or contribute towards realising your CE solution? Would you like to see it improve in any way? (Have more/less interaction, improved information flow or response time, initiate more formal collaboration, get more attention, etc.?
Potential actions for improvement	What could be done to improve the relationship and/or use it to promote your CE solution? By whom? Can WIDER UPTAKE contribute? If so, how?

After the partners had completed the Excel sheets, the information was translated into stakeholder maps (using PowerPoint) where the interactions in the respective industrial networks were mapped out. The second step of the process entailed showing the partners the developed stakeholder maps and discuss whether these were in accordance with reality, and add any missing actors, or remove those who were identified as unimportant. This verification process was done in spring 2022.

5.3 Social network analysis (SNA)

Social Network Analysis (SNA) was further used to unpack the connections in the stakeholder networks and learn more about their power relations and positions in the various country cases. SNA analyses relationships among a wide range of actors through quantitative measurement and visualization (Borgatti et al., 2009). The tool has mostly been applied in the industrial sector but is widely applicable and adaptable to other sectors (Spielman et al., 2011).

Responses regarding actor connections were used to generate the network data for this analysis, especially the questions pertaining to stakeholders in the network and the direction of interactions. The data was then compiled in a square (nxn) matrix. A relational score of 1 or 0 was assigned to actors depending on whether there is an interaction or not between them. Assuming there is a relation between stakeholders k and j, and either of them initiate an interaction, a value of 1 is given; nkj=1 or njk=1. However, if there is no relationship, a value of 0 is given; nkj=0 or njk=0. There are also instances where the relation is unidirectional and that renders the matrix asymmetric. nkj will be assigned a value of 1 (nkj=1) if the interactions are initiated by stakeholder k. Therefore, njk will be assigned a value of 0 (njk=0). This implies that the relational matrix in not necessarily symmetric and these details were carefully considered by each country case before the matrices were drawn.

Actors in SNA are referred to as nodes, and the connection between them are called ties. The degree of connectedness of an actor is indicated by the node size in the network map. To further measure the strength of relations among actors, the Freeman degree of centrality was adopted. Centrality indices including betweenness, effect size, coreness, and in/out degrees were adopted for this study. High betweenness score is an indication of the degree to which actors provide a bridge for connecting others in the industrial symbiosis networks. Effect size

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measures how strong impact an actor has on the network structure; how much would change that would be caused if this actor was to leave/be removed. Thus, it gives an indication of the source of a network's structural holes. How close an actor is to the core of the network is given by the degree of coreness. The in-degree and out-degree measure the level of linkages an actor receives from and gives to others, in a network, respectively. Prominent actors are those with high in-degree scores whilst influential actors are those with high out-degree scores (Borgatti et al., 2009; Freeman, 2004). A network's overall centralization measures the degree of unequal distribution of interaction among actors. The UCINET software was used to generate the industrial symbiosis network maps and network attributes (centralization indices).

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6 Industrial networks in the studied cases

6.1 The Hamar case

6.1.1 Initial, qualitative network description

In the demonstration case at Hamar, the industrial network consists of both private and public actors. Hias, an intermunicipal company for water and wastewater management owned by four municipalities, is the key partner. Hias is working very closely with their subsidiary Hias How2O, which has been established as a limited company specialised in technology for biological wastewater treatment. A third core actor is Sirkula, previously part of Hias but in 2016, established as a separate intermunicipal company for municipal waste management. In WIDER UPTAKE, Hias and Hias How2O have worked together to establish a full-scale unit for struvite production, and Hias and Sirkula collaborate to build business around soil improvement products based on co-composting of the digestate from biogas production at Hias and organic waste fractions from Sirkula.

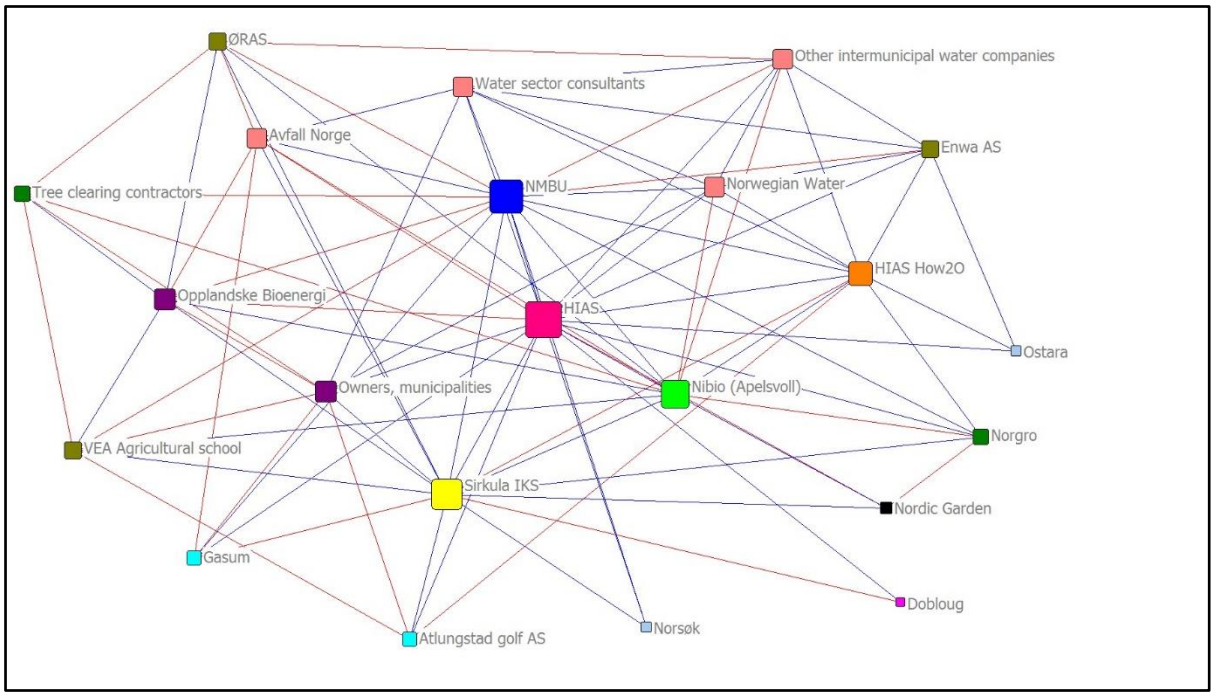
Important partners in the testing and promotion of struvite as a fertiliser have been the Norwegian Centre for Ecological Farming (NORSØK) and Atlungstad Golf, a neighbouring golf course. Hias is also in dialogue with the local industries; meat and dairy production; to secure a stable and clean "input" to the wastewater treatment facility, ensuring to secure the quality and safety of the products/resources that they recover from the wastewater. This also includes the County Governor, who is in charge of discharge permits.

6.1.2 SNA for the Hamar case

When the initial stakeholder mapping was carried out, the above-mentioned actors were mentioned as the most important ones. However, further interaction revealed that a number of other actors and stakeholders also are involved. Some were there all along, but not considered so central, others were to be confidential before, but can now be named, and others have been brought in during the project.

SNA based on the information we have by 2023, shows a quite large network. Unlike the Stavanger network where quite a number of stakeholders jointly occupied various positions in the network, few stakeholders are characterised with a particular network attribute and the degree of centralization vary across stakeholders, as shown in the various node colours and sizes. The visualization in Figure 4 shows that Hias has more network connections with other stakeholders in the network compared to the others. Closely following is the Norwegian University of Life Sciences (NMBU), with others such as Sirkula and Nibio (research institute, life sciences) having various degrees of centralisation.

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The colours of the nodes are used to visualize their respective sizes, where same colour represents same node size.

Figure 4: Network map for the Hamar case.

Stakeholders with fewer connections in the network are Ostara, the international provider of the struvite reactor, Norgro, a producer and distributor of soil products, and a local entrepreneur, Nordic Garden, Nowegian Centre for Ecological Farming (NORSØK), Gasum, which is a distributor of biogas and other energy products, Atlungstand Golf, and Enwa AS. While there are many reciprocal connections (blue ties), there are also many, and indeed more non-reciprocal connections (red ties). Hence, there could be potentially more uni-directional interactions than bi-directional ones in the network. This may be natural, since the network is linked to a specific economic activity, but since the strength of every network depends on the level of interactions within it, more interaction could be beneficial. Currently, few stakeholders drive the interactions.

In terms of SNA network indices (Table 2), Hias dominates in all the measures, similar to the case/to what is observed for SSGL in Ghana. Indegree centrality measuring the prominent role of actors saw Hias (16), NMBU (16), Sirkula (14) and Nibio (Apelsvoll) (13) as the top four actors who receive a lot of interactions in the network. Their prominent role in the network also implies they are very key actors that are likely to be consulted on any issue regarding the networks. The stakeholders with the least connections in the network evidenced by their smaller node sizes in the network map were also confirmed by the network attributes in Table 2 as receiving and giving out less interactions. With the exception of Nibio (Apelsvoll), all the aforementioned stakeholders together with Hias How2O were found to be the more influential stakeholders in the network, as they possess high indegree. Hias How2O uniquely has a similar attribute score on indegree and outdegree, e.g., they contribute and receive equally in their interactions, suggesting a degree of power sharing. The foregoing discussion suggests that these stakeholders have strong power positions in the network, though to various degrees.

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However as indicated earlier, Hias ranks highest in all the network attributes and is a core actor in the network.

Table 2: Social Network Indices of the network in the Hamar case.

	Indegree	Outdegree	Coreness	Betweenness	Effect size
Hias	16.00	17.00	0.45	113.52	12.92
Sirkula	14.00	13.00	0.32	70.85	10.57
Hias How2O	10.00	10.00	0.29	22.68	5.85
NMBU	16.00	10.00	0.29	44.35	10.37
Norwegian Water	7.00	8.00	0.25	4.72	3.03
Other intermunicipal water companies	5.00	8.00	0.25	6.97	2.92
Water sector consultants	8.00	8.00	0.24	9.35	3.63
Nibio (Apelsvoll)	13.00	8.00	0.23	36.38	7.88
Enwa Group	6.00	7.00	0.22	2.68	2.27
Opplandske Bioenergi	6.00	8.00	0.22	12.93	4.61
Norgro	5.00	5.00	0.18	3.44	1.95
VEA Agricultural school	3.00	7.00	0.16	5.18	3.55
Gasum	2.00	5.00	0.15	0.20	1.93
ØRAS	4.00	6.00	0.15	9.13	3.30
Avfall Norge (NWMRA)	6.00	5.00	0.15	5.96	3.68
Owners, municipalities	9.00	4.00	0.14	8.07	5.92
NORSØK	3.00	3.00	0.12	-	1.00
Ostara	3.00	3.00	0.11	-	1.00
Atlungstad Golf AS	4.00	3.00	0.10	1.05	2.36
Tree clearing contractors	3.00	4.00	0.10	0.33	2.79
Dobloug (local entrepreneur)	1.00	2.00	0.09	-	1.00
Nordic Garden	3.00	3.00	0.08	0.20	1.25

6.1.3 Summary characterisation of the network

Hias is a strong core network stakeholder, with Sirkula following. In line with the initial description of the network, other core stakeholders include Hias How2O and NMBU. The degree of coreness comes to support their respective node sizes observed in Figure 4.

For the betweenness attribute, the very high score of Hias (113.5) indicates that Hias has very high bridging power, e.g., it appears Hias performs all the connecting and linking roles with new and existing stakeholders in the network. Following is Sirkula IKS, with a score of 70.85, NMBU (44.35) and Nibio (Apelsvoll), with 36.38. Efforts to attract new stakeholders to the network should target the actions of Hias and Sirkula and to some large extent, NMBU and Nibio

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(Apelsvoll). Finally, in assessing the potential existence of structural holes in the HIAS network should a stakeholder exit, Hias, Sirkula, NMBU and to some degree Nibio (Apelsvoll) were found to be the holders of the network. This is not surprising given the consistent evidence of power roles played by these stakeholders.

While this network includes a wider group of stakeholders, the core industrial network is built on pre-existing organisational relationships and networks. It may also be characterised as a resource-dependent network, where Hias so far is a key contributor of resources. In IS terminology, Hias may thus be seen as the anchor firm of the network. Furthermore, the network has a majority of private actors, but also includes the two most relevant sector associations, and collaboration with some relevant knowledge institutions, and public authorities (e.g., the County Governor). The knowledge institutions are anchored in the same part of Norway as the core actors, and many of the other network members are also local/regional, e.g., proximity seems to be a factor. Looking towards the future, more dialogue with the target end users is recommended, especially for struvite as a new fertiliser product. The partners, especially Hias, is also interested in expanding their network in other directions, seeing themselves as a wastewater resource recovery facility (WRRF). A stated ambition linked to the demonstration case and local Community of Practice (CoP) in WIDER UPTAKE is to become a national competence centre for circular wastewater solutions.

6.2 The Stavanger case

6.2.1 Initial, qualitative case network description

IVAR, as the key partner, is an intermunicipal company, which handles both water cycle services and waste management for its twelve owning municipalities. The company is also a producer of biogas. The form of interaction with users varies between municipalities depending on their agreement with IVAR. As early as in 2011, a joint venture was established with Grønn Vekst, a company specialised in recycling of organic and mineral biproducts. The objective was to provide an organic fertiliser named Minorga, based on digestate from the biogas production. After setting up production, a Norwegian market was targeted, but for several reasons this was challenging at the time. Eventually, the product found its way into the market in Vietnam. In 2020, a change of organizational structure of Grønn Vekst, resulted in the establishment of Terramarine AS (Terramarine), as a private entity that is continuing the collaboration in a seller-customer relationship IVAR. Grønn Vekst continues to exist and provide other soil improvement products in Norway.

The regional energy company Lyse Neo is a long-term collaborator when it comes to biogas, which currently is upgraded and injected into their local gas grid. There is also dialogue with the farming community in the region, and plans for a new, large biogas plant together with other partners under a recently established entity called Bio Jæren AS. The potential for using CO_2 for greenhouses has likewise been mentioned, but these activities are outside WIDER UPTAKE. The partners collaborate with several knowledge institutions, and IVAR plays a central role in the national associations Norwegian Water and Norwegian Waste Management and Recycling Association (NWMRA).

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6.2.2 SNA for the Stavanger case

Figure 5 provides an overview of the connections between key stakeholders in the Stavanger case by 2023. The group of stakeholders represented as one in "Various research facilities" includes a number of R&D partners, e.g., the University of Stavanger, Norce Research, and Nibio (represented by a local research station). The role of the "Various research facilities" must therefore be read of as the sum of multiple stakeholders' interactions within the IS network.

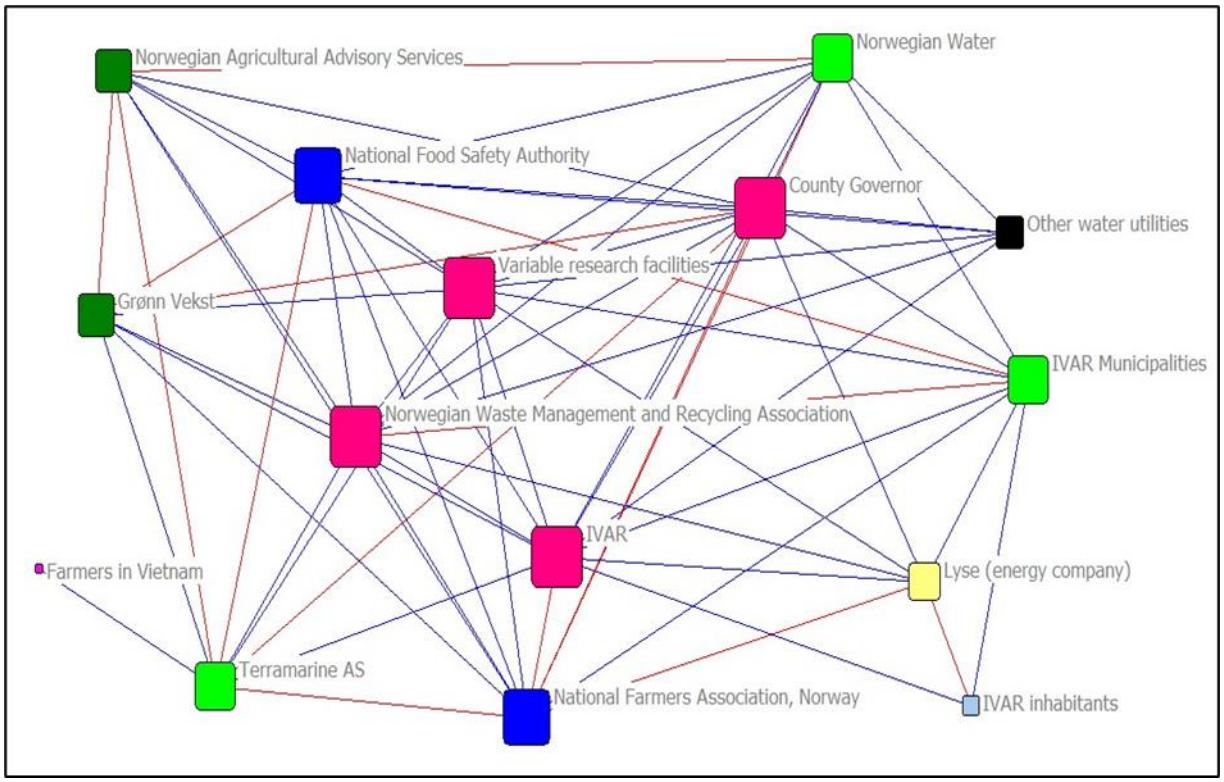


Figure 5: Network map for the Stavanger case.

Typically, one is likely to find one or two stakeholders with mostly connected ties in these types of networks, constituting a central core, but the network in the Stavanger case shows more than two stakeholders at the core. The SNA suggests that there are fewer connections between IVAR and inhabitants in their coverage area, other water utilities, Lyse Neo (labelled Lyse in Figure 5), and the least of all, farmers in Vietnam. The positions and node sizes of these stakeholders do in no way imply they are less important in the functioning of the network. It only implies that given their functions and roles, they interact less with other stakeholders in the network. The NWMRA, IVAR, Variable Research Facilities and County Governor have the most connections of same magnitude and node size. Closely following are the National Food Safety Authority and National Farmers' Association in Norway. This implies that these actors have more connections and are closely connected to other stakeholders within and potentially beyond the network.

A positive characteristic in terms of network functioning, is that this network structure has more reciprocal ties (interactions) than non-reciprocal ones. This suggests that not one single

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stakeholder mostly initiate interactions and that there are more bidirectional than unidirectional interactions. This can also be understood as a result of the type of stakeholders that are part of the network, as several of them are non-economic actors. The power relations within the IS network can be seen in Table 3.

Table 3: Social Network Indices of the network in the Stavanger case.

Stakeholder	Indegree	Outdegree	Coreness	Betweenness	Effect size
Variable research facilities	12.00	12.00	0.39	15.64	5.00
NWMRA	12.00	11.00	0.35	14.36	4.89
IVAR	12.00	11.00	0.33	24.46	5.61
National Food Safety Authority	10.00	9.00	0.29	4.57	3.68
Norwegian Water	7.00	9.00	0.29	1.94	2.43
County Governor	12.00	9.00	0.28	7.62	4.64
Grønn Vekst	5.00	8.00	0.26	0.85	2.00
Terramarine AS	5.00	9.00	0.26	26.35	3.71
National Farmers' Association, Norway	9.00	8.00	0.26	3.59	3.77
IVAR Municipalities	8.00	8.00	0.24	7.65	3.44
Other water utilities	6.00	6.00	0.21	-	1.00
Lyse Neo (energy company)	5.00	7.00	0.21	1.83	2.33
Norwegian Agricultural Advisory Services	8.00	5.00	0.17	0.14	2.15
IVAR inhabitants	3.00	2.00	0.06	-	1.00
Farmers in Vietnam	1.00	1.00	0.03	-	-

Just as shown in the network map in Figure 4, the social network indices presented in Table 3 indicate that Various research facilities, the NWMRA, County Governor and IVAR are the prominent stakeholders in the network. Their high indegree scores imply that these stakeholders have more inbound links and in this sense are more central in the network compared to other stakeholders. With the exception of the County Governor, the remaining three also scored high in outdegree (number of outbound links), indicating that these stakeholders are prominent in terms of interaction. This suggests that various research facilities, the NWMRA and IVAR also hold influential positions in the network. It is positive, in many ways, that influence does not rest in the hands of just one, but of multiple stakeholders who are playing different roles. Potential tension or challenges linked to resource dependency

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may be balanced or better handled with such a network structure, than if the main stakeholder is dominating in terms of all or most network scores.

Regarding the degree of coreness (relative distance to all other actors), variable research facilities is top on the score, closely followed by the NWMRA and IVAR. In this sense, variable research facilities can be said to be closer to the network core. Peripheral stakeholders on the other hand include the IVAR area inhabitants and Farmers in Vietnam. The reason for this can be that the connection between IVAR and the inhabitants is service based. In the case of Vietnamese farmers, there is geographical distance, and the connection is through fertilizer vendors in Vietnam, meaning there is no direct contact. These stakeholders can be categorized as end-customer of services and products, which can limit their potential and need for interacting with other stakeholders in the network.

The Betweenness score (based on the number of shortest paths an actor is on), is a measure of stakeholders that serve as bridging links in a network, shows that Terramarine AS, is seen as having the most bridging power (26.35) in the network. This was closely followed by IVAR and variable research facilities. This suggests that Terramarine and IVAR (24.46) have more influence in connecting and attracting new stakeholders to the IVAR network. This is also understandable, as the IS network is focused on the economic activity of fertilizer production, where IVAR provides the organic matter (sludge) and produces final product, whereas Terramarine provides the recipes and sells the product.

6.2.3 Summary characterisation of the network

In SNA terms, the key stakeholders found to hold the network structure together were IVAR (5.61), Variable research facilities (5.00), Norwegian Waste Management and Recycling Association (4.89) and County Governor (4.64), in decreasing order of influence. However, we would argue that IVAR, as one of the two core stakeholders in the networks, plays a comparatively more important role than the other three. Furthermore, the SNA demonstrates that Terramarine is an important actor with a key bridging function. As Terramarine relies on interactions with a range of actors to source materials and produce the organic fertilizer, it is understandable that they also can play a role in attracting even more actors to the industrial network, e.g., through finding new sources of organic matter that can be utilized for fertilizer production. The SNA characterisation of the NWMRA as a key actor in the network can be understood as a result of the position and role such an industry organization has in connecting actors within the sector.

The network can be characterised as a resource-dependent network, where IVAR holds an anchor role, being in charge of the biogas production, as well as providing a key resource and being the one producing the organic fertilizer for Terramarine. However, Terramarine is also experimenting with multiple sources for their fertiliser products, such as sludge from fish-farming and poultry. As such, they are aware of this dependency and working towards multiple sources. Furthermore, the network has a majority of public actors, but also includes the two most relevant sector associations, and collaboration with some relevant knowledge institutions, and public authorities. The key knowledge institutions are located in the same part of Norway as the core actors, and several of the other members are also local/regional, e.g., proximity seems to be a factor. Additionally, the representatives from IVAR identify a potential

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lock-in regarding their own operations, in that there can be different views on how circular economy and the sustainable development goals best can be achieved.

6.3 The Sicily case

6.3.1 Initial, qualitative description of the network

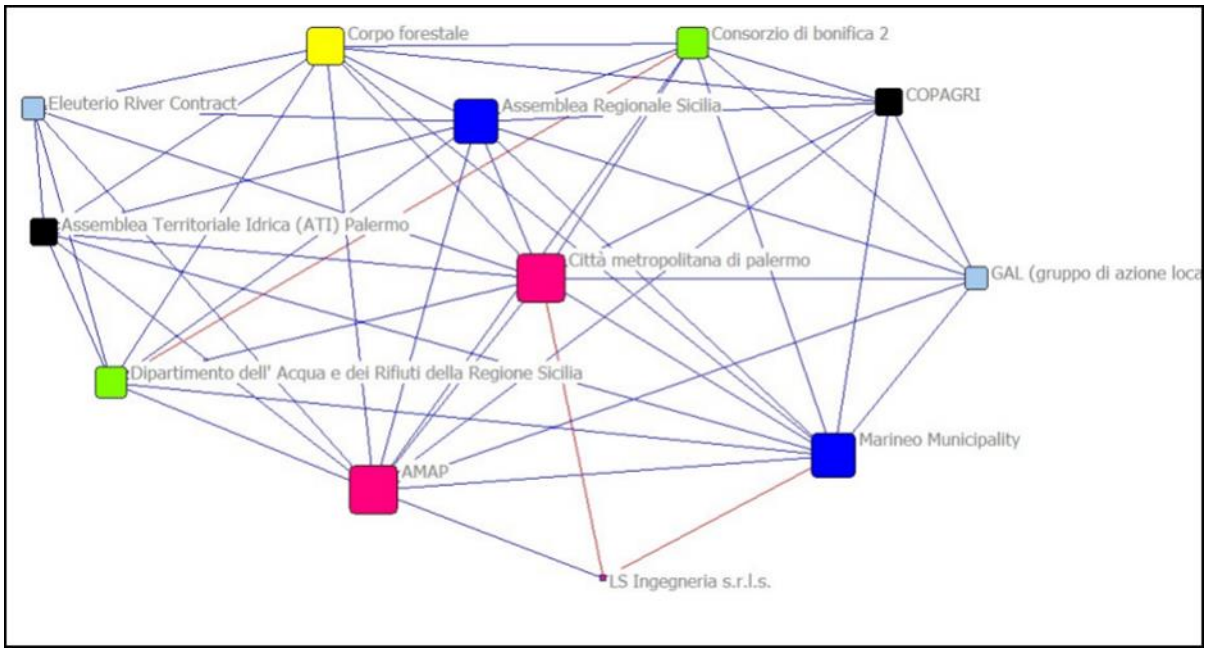
In the case in Sicily, AMAP Spa (AMAP), as the manager of the integrated water service in thirty-five municipalities in the Metropolitan City of Palermo, interacts with the municipalities which are the owners of the company, as well as with the Metropolitan City administration. Furthermore, AMAP collaborates closely with UNIPA, amongst other on the demonstration in Marineo, where the aim is to produce biopolymers and recover nutrients by ab-/desorption columns, and in Corleone, where the aim is full-scale reuse of treated wastewater for agriculture and reduction/reuse of wastewater sludge. The local branch of the farmers' association COPAGRI is therefore an important dialogue partner. It should be noted, however, that a relatively low share of the local farmers are active in this organisation. Other key stakeholders are the Consorzio di Bonifica di Palermo (water reclamation consortium of Palermo), ATI Palermo (regional water authority). The department of Water and Waste of the Sicilian regional Assembly has been involved in view of pushing the proposal of a new regulatory framework supporting water reuse.

Initially, a lack of interaction with associations and potential users of fertilisers (e.g., COPAGRI, Consortium of Reclamation) was detected, related to the absence of a market dealing with products obtained from recovery through the application of water-smart solutions. Currently, the fertilizer production depends on whether the fertilizer producer MUGAVERO finds the quality of the recovered resources from AMAP to be of sufficient quality. No dialogue on the creation of a brand exists between the actors, and the development process is currently in a laboratory stage. Moreover, the lack of direct interaction with citizens was revealed, who are among the main users and end-users of the water obtained from the wastewater treatment plants involved in the case study.

6.3.2 SNA for the Sicily case

The SNA carried out in 2023 provide a more detailed picture of the industrial network. Since the case involves demonstrations of different solutions in two different towns, two network maps are provided, even though many of the same stakeholders and indeed the same network is engaged in both cases. Even though UNIPA plays an active role in the network, as noted above, it is not included in the SNA, simply because the institution was not among those listed in the matrix used for the SNA. In a similar vein, the local research institutions that participate in WP4 are also not included in the network maps related to the other demonstration cases.

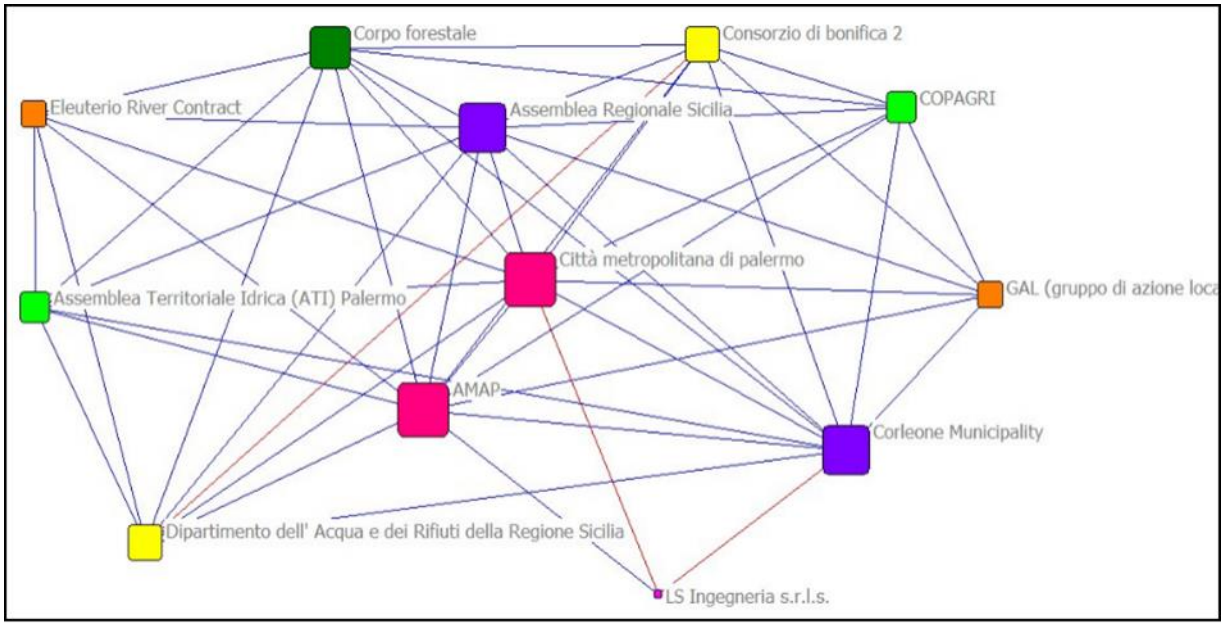
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Reciprocal ties Non-reciprocal ties

The colours of the nodes are used to visualize their respective sizes, where same colour represents same node size.

Figure 6: Social network map for Marineo, Sicily case.



Reciprocal ties Non-reciprocal ties

The colours of the nodes are used to visualize their respective sizes, where same colour represents same node size.

Figure 7: Social network map for Corleone, Sicily case.

As illustrated in Figure 6 and Figure 7, stakeholders with relatively more connections in the Sicily case are AMAP and Citta metropolitana di Palermo, closely followed by Assemblea Regionale Sicilia, Marineo and Corleono Municipalities. These simply highlight the active participation of the Italian local/regional government in the network, which is commendable. Stakeholders with comparatively fewer connections in the network were found to be LS Ingegneria s.r.l.s., Elurterio River Contract and GAL (Gruppo Azione Locale - local partnership to

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promote the development of the rural area). Then again, this is not to say they are less important in the network. Their respective roles may not require them to have many interactions in the network as compared to others. Nonetheless, if their roles require that they interact more in the network, then action needs to be taken by the network leads to put strategies that will encourage them to build more bridges with other stakeholders.

Another thing to note is the domination of reciprocal ties in the network as compared to non-reciprocal ones. This is to say that most stakeholders give and receive interactions within the network and hence fostering forward and backward linkages in interactions. The Italy case is quite a unique one compared to the rest of the networks that have been studied so far and it will be important to learn lessons and strategies from that case in order to promote proactive interactions among all other network stakeholders.

In terms of network attributes presented in Table 4, AMAP, ATI Palermo, Assemblea Regionale Sicilia and Città metropolitana di Palermo recorded higher level of coreness. These stakeholders are hence closer to the core and likely to receive more interactions compared to others, as seen in Figure 6 and Figure 7. AMAP dominates in all the network attribute scores However, the gap between them and the others is not very wide except for the betweenness score.

Table 4: Social Network Indices for the Sicily case.

Stakeholder	Indegree	Outdegree	Coreness	Betweenness	Effect size
AMAP	11.00	11.00	0.37	17.00	4.55
Assemblea Territoriale Idrica (ATI) Palermo	7.00	7.00	0.36	0.33	1.29
Assemblea Regionale Sicilia	10.00	10.00	0.36	4.57	3.10
Città metropolitana di Palermo	11.00	10.00	0.36	7.40	4.21
Marineo Municipality	10.00	9.00	0.33	4.95	3.55
Corpo forestale	9.00	9.00	0.33	2.62	2.44
Corleone Municipality	10.00	9.00	0.33	4.95	3.55
Consorzio di bonifica 2	7.00	8.00	0.29	0.70	1.67
COPAGRI	7.00	7.00	0.26	0.33	1.29
Dipartimento dell' Acqua e dei Rifiuti della Regione Sicilia	8.00	7.00	0.26	0.70	1.67
Eleuterio River Contract	6.00	6.00	0.22	-	1.00
GAL (gruppo di azione locale)	6.00	6.00	0.22	-	1.00
LS Ingegneria s.r.l.s.	1.00	3.00	0.12	-	1.00

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Stakeholders that receive more interactions in the Sicily network were seen to be AMAP and Città metropolitana di Palermo, closely followed by Assemblea Regionale Sicilia, Marineo and Corleone Municipalities. For stakeholders that mostly give initiate interactions with other network members, these same stakeholders were found. Moreover, the indegree and outdegree scores are closely related which only confirms the many reciprocal ties we observed in the network maps in Figure 6 and Figure 7. For bridging stakeholders, AMAP, Assemblea Regionale Sicilia, Assemblea Regionale Sicilia, Marineo and Corleone Municipalities recorded high scores for the betweenness attribute, highlighting their strong bridging role in the network. Like other networks, these stakeholders have the power to connect members of the network and attract potentially new and important stakeholders to broaden the network. In terms of effect size, AMAP and Città metropolitana di Palermo stand out. As mentioned above, this is a measure of how large a stakeholder effect is on a network such that their absence could likely collapse the network gives an indication of the possible sources of network structural holes.

6.3.3 Summary characterisation of the network

The observations noted above suggest that AMAP and Città metropolitana di Palermo, closely followed by Assemblea Regionale Sicilia, Marineo and Corleone Municipalities hold more power in the network structure. The network is particularly characterised by bi-directional linkages which is a good indicator of network strength. The unlikely absence of AMAP, Città metropolitana di Palermo, Assemblea, Regionale Sicilia, Marineo and Corleone Municipalities would likely to cause a structural breakdown, as several holes will be left in the network. Thus, for the Sicily case, the network is mainly composed of public entities. These interactions are mostly aimed to address legislative and organisational issues. An ambition for the future is to develop closer contact with more of the local farmers, to increase their knowledge and acceptance of the solution.

6.4 The Prague case

6.4.1 Initial, qualitative network description

PVS a.s. (PVS) is the main industrial partner in the demonstration case in Prague. PVS is owned by the City of Prague, which underlines the links between the City and the WIDER UPTAKE project. PVS holds 49 % of the shares in the operator company PVK a.s. (PVK), the rest is owned by Veolia Group in the Czech Republic. PVK currently operates both parts of the Central WWTP Prague including the New water Line (NWL), which provides the effluent for the demonstration units of water reclamation in Prague. The industrial symbiosis involves the reuse of treated effluent from Prague Central WWTP for irrigation of urban greenery in the City of Prague. PVK is also responsible, based on a contract from PVS, for the everyday control and maintenance of the facilities for tertiary treatment of NWL effluent (water reclamation unit) and for four irrigation demonstration units with three different types of urban greenery, i.e., grass, flowers, and bushes/small trees.

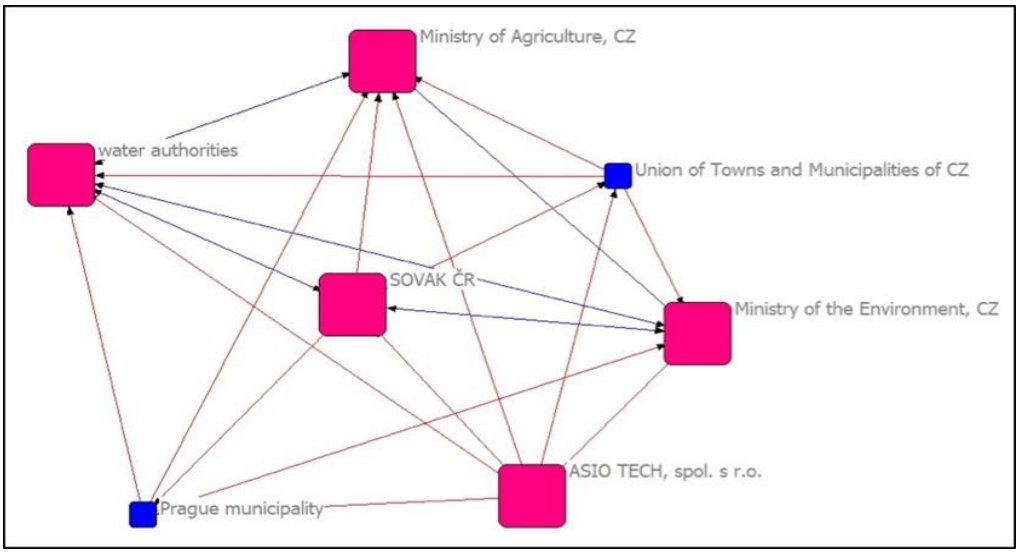
Regular sampling, analyses (both chemical and microbiological) and data processing are performed by two other research partners of the Wider Uptake project, i.e., the University of Chemistry and Technology (UCT) and the Czech Technical University (CVUT). The two universities cooperate closely with experts of PVS and PVK. However, a core challenge in the

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Prague case is that utilisation of treated effluent currently is not allowed by law. Although there is already interaction with policy makers and the City of Prague, the network could potentially be developed further by including further organisations that can influence decision-makers in a positive direction. Although the industrial symbiosis network in Prague currently is under development it, it is resource-dependent, as PVS and PVK jointly provide the treated effluent for reuse in urban greenery. The potential users of the treated effluent will be the City of Prague. If PVS was to stop producing treated effluent, there would be no alternative effluent providers who could cover potential needs. During the project period, the technology provider ASIO TECH has also become involved in the network. Besides being a relevant technology provider, this private company has been central in another water reuse project, located in Brno. Other companies may also get involved, either directly, indirectly through the contact with corporate members of SOVAK ČR (Water Supply and Sewerage Association of the Czech Republic, in the following denoted as SOVAK) and The Czech Water Association (CzWA).

6.4.2 SNA of the Prague case

A unique characteristic of the network in the Czech case is that unlike other networks that have stakeholders with varying network ties and node sizes, five out of the seven key stakeholders assessed had the same level of interaction among the network actors (Figure 8). This is indicated by the same node size and colour for five stakeholders and similar for the remaining two stakeholders, as seen in Figure 8. The Union of Towns and Municipalities of Czech and the Prague municipality are found to receive less interactions within the network (blue nodes), suggesting that they are not strongly connected in the network. Notwithstanding the equal level of interactions among the top five stakeholders, Water authorities and the Ministry of Environment had more forward and backward linkages with other actors in their interactions (see from the direction of the blue arrows). For the entire network, there were more non-reciprocal linkages (red ties) than there were reciprocal (blue ties), signalling that more unidirectional interactions are prevalent in the network.



The colours of the nodes are used to visualize their respective sizes, where same colour represents same node size.

Figure 8: Network map of the Prague case.

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As shown in Table 5, network coreness was a bit missing in the Czech structure with only ASIO TECH showing some degree of coreness. Water authorities, ministries of environment and agriculture on the other hand play more prominent roles in the network as they are likely to receive more interactions compared to others. Surprisingly, ASIO TECH, was not observed to be recipient of interactions but dominates in the initiation and giving out of interactions. This is also evident in the network map (Figure 8) as more unidirectional interactions (arrow heads) are observed from ASIO TECH to the other stakeholders. They can be said to the more influential stakeholder in the network compared to the others. Another close influential stakeholder is SOVAK, making ASIO TECH together with them the more influential stakeholders. This is not to say the others do not give out interactions as all other stakeholders recorded some level of outdegree scores.

SOVAK is seen to be the stakeholder with the most bridging power to connect and attract other stakeholders to the network. This is followed by water authorities and the Ministry of the Environment. The remaining stakeholders, including ASIO TECH, recorded zero scores for the betweenness attribute. ASIO TECH only gives out interactions which are unidirectional and so are unable to serve as a connection between stakeholders.

Table 5: Social Network Indices for the Prague case.

Stakeholder	Indegree	Outdegree	Coreness	Betweenness	Effect size
ASIO TECH	0.00	6.00	1.00	0.00	2.18
Water authorities	6.00	3.00	0.00	3.50	1.94
Ministry of Environment, CZ	6.00	3.00	0.00	3.50	1.94
Ministry of Agriculture, CZ	6.00	2.00	0.00	0.00	1.83
SOVAK	3.00	5.00	0.00	8.00	1.83
Prague Municipality	2.00	3.00	0.00	0.00	1.60
Union of Towns and Municipalities, CZ	2.00	3.00	0.00	0.00	1.60

However, for the effect size, ASIO TECH, is the stakeholder with the highest score, implying that they hold most of the connections in the network. Following ASIO TECH, are water authorities and the Ministry of Environment. These suggest that they have an important position in the industrial symbiosis network of Czech and their absence is likely to cause structural breakdown of the network. It must be noted that the closeness of the effect size values of the stakeholders also suggest that all the stakeholders have some potential of causing holes in the network structure should they exit.

6.4.3 Summary characterisation of the network

As it seems in the SNA, this network holds few stakeholders and hence the exit of either one of them will be a loss to the entire network structure and function. SOVAK ČR then has a huge responsibility of attracting and connecting new stakeholders to the network since few active stakeholders are observed for the network. Apart from ASIO Tech, there are also mainly public stakeholders. As we have seen, there are more uni-directional than bi-directional ties, suggesting there are relatively few reciprocal relations. This is in line with our qualitative

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observations, which suggest that this network, when it comes to water reuse, is an emerging one, where more active dialogue is needed. One the other hand, the Czech network structure is such that no single stakeholder seems to have a very dominant position. This may be seen as positive, in the way that power, or influence, is distributed among the various stakeholders.

It must be noted, however, that the SNA results rely on the data that was entered, how wide the partners cast the net when filling the Excel sheet and SNA matrix. It could well be that the Czech network has more stakeholders involved, e.g., CzWA was recognised as a stakeholder during the initial mapping but does not figure in the SNA. The local research partners also play an active role in this case. At any rate, the identified structural conditions are interesting, and the findings underscore that the efforts of the Czech project partners to increase the level of interaction and engage more stakeholders is crucially important.

6.5 The Dutch case

6.5.1 Initial, qualitative network description

NPSP has a quite diverse network, where they collect and use resources from several different suppliers and collaborate with a range of public and private actors. In relation to the WU project, NPSP sources grass from Waternet (the Water Board responsible for the greater Amsterdam region). There is also a possibility to source reed from Waternet, but there are currently challenges related to getting the reed to meet the specifications required for NPSP to utilize them in their production.

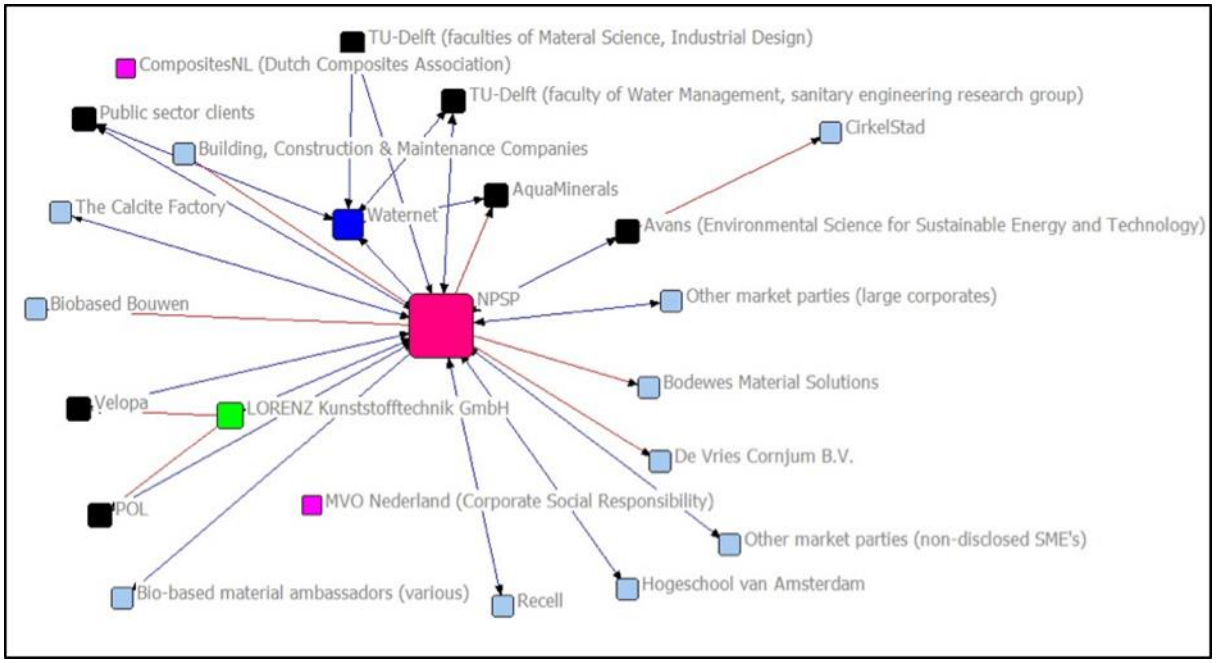
The IS network is quite diverse in terms of resource input, which makes it quite flexible and enables NPSP to source resources from actors that can provide them. Key actors in terms of access to resources is AquaMinerals (a resource intermediary owned by drinking water companies and water boards) who provides NPSP with calcite from drinking water treatment, and Recell who provides cellulose from recovered toilet paper. The diversity and flexibility of suppliers of resources contributes to avoiding potential resource lock-ins. Figure 9 shows how the IS network is made up of a range of public and private actors. Regarding public actors, NPSP relies on knowledge institutions such as Delft University of Technology (TU Delft), the Amsterdam University of Applied Sciences, and Avans University of Applied Sciences for research and development.

6.5.2 SNA for the network in the Netherlands case

Further assessment of the NPSP network using SNA also reveals that most of the interactions in the network is centred around NPSP. From Figure 9, it can be observed that the stakeholder with the most connections in the network is NPSP. The size of the NPSP node relative to the other nodes clearly shows that not much of interactions are led or received by other stakeholders, leaving NPSP as a dominant stakeholder in the network. Though interactions are centred around NPSP, these interactions are mostly reciprocal with few non–reciprocal ones. Whilst there are more interactions happening between NPSP and all other stakeholders, there is not much or nearly non-existent interactions happening between other stakeholders. This implies that stakeholders do not interact among themselves except with NPSP. Nonetheless, some few stakeholders such as Waternet, Aquaminerals, TU-Delft, Avans, LORENZ, and

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Building, construction and maintenance companies had some level of interactions among themselves aside the ones they shared with NPSP.



The colours of the nodes are used to visualize their respective sizes, where same colour represents same node size.

Figure 9: Map of the network in the Netherlands case.

Given that most of the interactions as seen in Figure 9 were centred around NPSP, it is not surprising that the SNA indices in Table 6 shows similar trends regarding power relations, in terms of upholding and developing the structure of the network. NPSP gives and receives most interactions. In terms of prominence and influential attributes, NSPS appears to both be serving as a bridging and core stakeholder in the network. NPSP hence have the power to connect other stakeholders to the network and their absence from the network will create a huge structural hole (effect size) in the network. All these point to the important role that NPSP play in the Netherlands case.

Other stakeholders worth mentioning include Waternet and Avans (Environmental Science for Sustainable Energy and Technology), as they are positioned to connect other stakeholders in the network as well as attract new stakeholders. They are also recorded as the top two stakeholders after NPSP, with the power to receive and give out interactions in the network. They also join NPSP with big effect size where their absence might cause the network to malfunction. The large number of stakeholders on the periphery is associated with potential for further development and innovation in the network, since they provide openings towards other actor networks that could be of relevance for NPSPs activity related to biocomposite for riverbank protection.

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Table 6: Social Network Indices for the Netherlands case.

Stakeholder	Outdegree	Indegree	Betweenness	Coreness	Effect size
NPSP	18.00	15.00	270.50	0.94	18.42
Waternet	5.00	5.00	22.50	0.11	3.60
TU-Delft (Water Management, sanitary engineering)	2.00	2.00	-	0.10	1.00
TU-Delft (faculties of Material Science, Industrial Design)	2.00	2.00	-	0.10	1.00
LORENZ Kunststofftechnik GmbH	3.00	1.00	-	0.10	1.50
Public sector clients	2.00	2.00	-	0.10	1.00
Avans (Environmental Science for Sustainable Energy and Technology)	2.00	1.00	16.00	0.09	2.00
Hogeschool van Amsterdam	1.00	1.00	-	0.09	-
Recell	1.00	1.00	-	0.09	-
POL	1.00	2.00	-	0.09	1.00
Velopa	1.00	2.00	-	0.09	1.00
Other market parties (non-disclosed SME's)	1.00	1.00	-	0.09	-
Other market parties (large corporates)	1.00	1.00	-	0.09	-
Bio-based material ambassadors (various)	1.00	1.00	-	0.09	-
Building, Construction & Maintenance Companies	1.00	-	-	0.09	-
Biobased Bouwen	-	1.00	-	0.09	-
The Calcite Factory	1.00	1.00	-	0.09	-
AquaMinerals	1.00	2.00	-	0.01	1.00
De Vries Cornjum B.V.	-	1.00	-	-	-
Bodewes Material Solutions	-	1.00	-	-	-
CompositesNL (Dutch Composites Association)	-	-	-	-	-
CirkelStad	-	1.00	-	-	-
MVO Nederland (Corporate Social Responsibility)	-	-	-	-	-

6.5.3 Summary characterisation of the network

NPSP's IS network has developed through a green-twinning approach, where the initial development of the network started during a period when NPSP and Waternet used the same

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coworking space, whereafter it was identified that Waternet could provide calcite for NPSP's biocomposite production, delivered through Aquaminerals.

In terms of developing their industrial network further, NPSP has identified several actors that can be useful to collaborate with in terms of gaining access to knowledge networks, where they can both market their solution and learn/share knowledge concerning their work on CE. The work within WP4 and on CoPs in WIDER UPTAKE can contribute to the enhancement of the network through triggering reflections concerning the existing interactions and collaborations that NPSP engages in and contribute to the discussion on how new potential collaborations can be developed and to what ends.

6.6 The Ghana case

6.6.1 Initial, qualitative description of the network

The industrial symbiosis (IS) network in Ghana has the presence of all key stakeholders that support the functioning of the network. There are stakeholders who mainly became part of the industrial symbiosis network because of the WU project. For example, research stakeholders from the CSIR-Water Research Institute (CSIR-WRI) connected with SSGL, as the main stakeholder of the network, and also with the Ghana Standards Authority (GSA) for this reason. Interactions between SSGL and the CSIR-Institute of Industrial Research (CSIR-IIR) and the CSIR-WRI have been touted as part of the strongest interactions created out of the WU project, though the interactions initially did not exist. Their main role in the network is to conduct studies towards improving the quality and safety of the innovations generated by SSGL (biochar and treated wastewater).

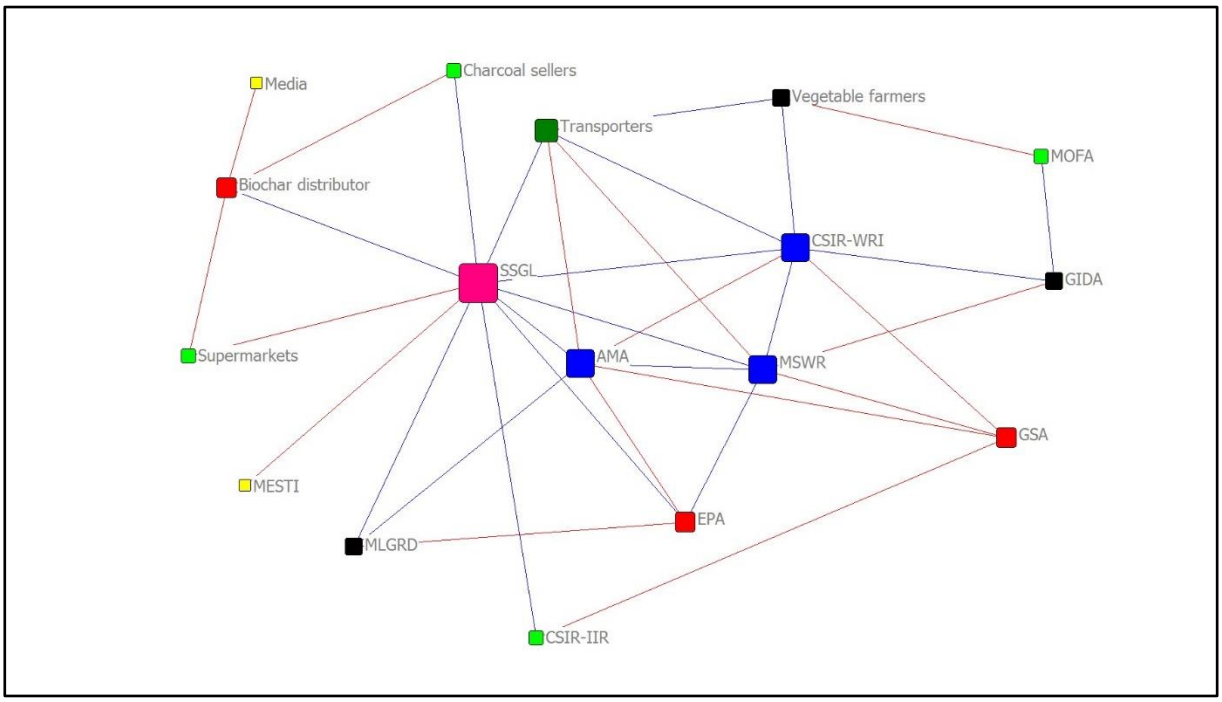
The structure of the network provides a clear case of resource dependency challenge in the network. Though there are a couple of waste management organisations in the country, SSGL is the establishment undertaking the production of biochar from the faecal sludge and the treatment of wastewater. SSGL can be described as having a "pseudo monopolistic" power in the wastewater management market. Hence, once SSGL leaves the IS network, there will be little left for the network to function in producing its innovative products. The resource dependency in this instance does not lie in the raw material as the supply sources are enormous, but in the technological capacity to generate innovative products from the waste being produced. While some of the stakeholders play overlapping roles, for example, the policy and regulatory agencies, others are unique. The Accra Metropolitan Assembly (AMA) is one such unique stakeholder in the network whose absence is likely to create a huge structural hole as they own all the waste generated in the metropolis, thus no wastewater treatment company can operate without their approval.

6.6.2 SNA of the Ghana case

In Ghana, a typical case of resource dependency as earlier indicated is also observed in the social network analysis framework. Unlike the Stavanger and Czech cases, where one could find a couple of actors with similar node sizes or closely sized nodes, that of Ghana was different. SSGL has more connections in the network compared to any other stakeholder as indicated by their node size in the network map in Figure 10. The other set of actors after SSGL with more connections in the network are the Accra Metropolitan Authority (AMA), a regulatory body, the

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Water Research Institute (CSIR-WRI), a research institution and the Ministry of Sanitation and Water Resources (MSWR), a policy support body. Having these different bodies in the network is promising as their individual mandate and function in the network complement each other.



The colours of the nodes are used to visualize their respective sizes, where same colour represents same node size.

Figure 10: Map of the network in the Ghana case.

Stakeholders with fewer connections included the vegetable farmers, charcoal sellers, supermarkets, the Industrial Research Institute (CSIR-IIR) the Ministries of Food and Agriculture (MOFA), Environment, Science, Technology, and Innovation (MESTI), and Local Government and Rural Development (MLGRD), among other regulatory stakeholders. Given the importance of advocacy and public awareness on the resource reuse, the role of media is very important, and it is encouraging to see the media in the IS network of Ghana, though with very little interaction with other network stakeholders.

There were more reciprocal ties than non-reciprocal ones and SSGL appears to be the stakeholder with more reciprocal interactions with other stakeholders. Further analysis from the social network indices (Table 7) show SSGL dominating in all the attributes as seen in the HIAS case of Norway. Stakeholders with the power to give out more interactions with others in the network include SSGL, CSIR-WRI and the MSWR. This implies that in terms of influential position within the IS network in Ghana, these stakeholders in the industry, research and policy making spaces hold such power. For influential actors, the MSWR, AMA, CSIR-WRI and Water Transporters were found to have such power in addition to SSGL. The position of Water Transporters needs to be highlighted given the crucial role they play in the Ghana water reuse network. They solve a major logistical challenge for the Ghana case, and it is good to know they are among the top actors that receive more interactions in the network, placing them side by side with other influential actors including research, regulatory and policy spaces.

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Table 7: Social Network Indices for the Ghana case.

Stakeholder	Outdegree	Indegree	Betweenness	Coreness	Effect size
SSGL	10.00	10.00	118.14	0.59	9.28
EPA	3.00	3.00	0.33	0.42	1.50
CSIR-WRI	7.00	5.00	55.06	0.41	4.71
MSWR	5.00	6.00	20.52	0.37	4.14
AMA	4.00	6.00	8.79	0.28	3.80
Transporters	3.00	5.00	12.24	0.22	2.44
MLGRD	3.00	2.00	0.33	0.21	1.30
GIDA	3.00	2.00	20.74	0.15	2.40
Charcoal sellers	2.00	1.00	-	0.13	1.00
CSIR-IIR	2.00	1.00	3.58	0.13	2.00
GSA	2.00	2.00	1.00	0.12	2.75
Vegetable farmers	2.00	3.00	5.26	0.12	2.20
Biochar distributor	2.00	3.00	14.50	0.11	2.90
MESTI	1.00	-	-	0.11	-
MOFA	2.00	1.00	0.50	0.05	2.00
Media	1.00	-	-	0.02	-
Supermarkets	-	2.00	-	-	1.00

Stakeholders with much power and ability to connect other stakeholders in and to the Ghana IS network include SSGL, CSIR-WRI, GIDA and MSWR. All of these are very active stakeholders in the water and irrigation space. Ghana Irrigation Development Authority (GIDA) for example is a stakeholder you cannot overlook when it comes to the agricultural irrigation space. In the same vein, just as SSGL, the Ministry of Sanitation and Water Resources (MSWR) and CSIR-WRI are prominent in the water quality and treatment space. These have the potential and capacity to connect other stakeholders and also bring in new and vital ones to the network. This suggests, e.g., that if much interaction and engagement has not been done with MSWR, it will be imperative for the Ghana case to engage them further in order to tap in their ability to connect more stakeholders. As the Betweenness score of 14.5 indicates, there is also a potential for the biochar distributor to serve as a bridge in connecting other stakeholders. She should therefore be kept closely to the network through conscious engagements.

SSGL, the Environmental Protection Agency (EPA), CSIR-WRI and the MSWR were also found to form the core of the network. Juxtaposing this finding with the node sizes observed in the network map, one would wonder why EPA is the second core actor in the network. It must be noted however that it could be possible that EPA may not have interactions with many of the stakeholders in the network but their close relation with SSGL puts them as a core stakeholder in the network. It also highlights that EPA should begin to interact more with the stakeholders since they form the core of the IS network for Ghana. Network structural holes are inherent with the exit of SSGL mostly. CSIR-WRI, MSWR and AMA are also potential sources of structural holes in the IS network of Ghana should they exit the network.

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6.6.3 Summary characterisation of the network

The network in this case has more reciprocal than non-reciprocal ties, and SSGL is in a position where they can take a role as "network orchestrator". However, given that the level and frequency of interactions were not as expected of a well-functioning network, researchers and other partners should create opportunities and avenues that will encourage more interactions among stakeholders. It will also be important for network members to be sensitized and motivated to own whatever platform is established to increase interactions. The Community of Practice established in WIDER UPTAKE is one such platform, which should be kept active also after the project period.

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7 Business model exploration

7.1 Circular business models – a literature review

To ensure that the work on business models in WIDER UPTAKE takes into account the most recent academic research on circular business model innovation, a literature review was carried out in October-December 2021. We used multiple search words in Scopus search engine to identify relevant papers.[1] We particularly sought to identify papers that discussed the water sector and/or the processing industry. We reviewed in total thirty-seven peer-reviewed articles and created a database by answering the following questions in each of the articles:

- What is the empirical context?
- What is/are the circular business model(s) being discussed?
- What are the main findings/arguments of the paper?
- How does the paper help us to better understand the implementation of new circular business models in WIDER UPTAKE?

In the rest of this sub-section, we present the key findings of this review.

A business model (BM) can be defined as *"the rationale of how an organization creates, delivers and captures value"* (Osterwalder and Pigneur, 2010, p.14). Thus, it deals with strategy and interconnections between a set of internal and external resources that often are tacitly understood and making them explicit, can shed new light on opportunities and challenges the actors face. The value in question may often be profit, for the shareholders of a commercial firm, but it may also be of other kinds, e.g., safe, rational, and well-functioning water services for the municipalities owning a water utility. Circular business models (CBMs) are *"business models that are cycling, extending, intensifying, and/or dematerializing material and energy loops to reduce the resource inputs into and the waste and emission leakage out of an organizational system"* (Geissdoerfer et al., 2020b: 7). In other words, CBMs operationalize CE into a plan of action that makes circularity a reality. A CBM may still be focused on business and have negative as well as positive side-effects. However, CBMs may also be coupled with a major shift in the logic of an organization (Geissdoerfer et al., 2020b) towards a broader understanding of value, including the benefits and costs for the overall society and environment (Bocken et al., 2019).

When it comes to CBMs, four archetypes can be identified: cycling, extending, intensifying, and dematerializing (Geissdoerfer et al., 2020a). Cycling means that materials and energy are reused, remanufactured, refurbished, and recycled. Extending involves prolonging the lifetime of materials and products through design and repairment. Next, intensifying resource loops aims to reduce the idle periods of products through e.g., sharing economy principles. The fourth type, dematerializing, seeks to avoid hardware use altogether, by substituting the consumer need with a service or software. So far, recycling practices and user-oriented product-service systems have received most attention in the research literature (Rosa et al., 2019). CBMs can

[1] "circular business model innovation" AND "process", "circular business model innovation", "circular business model experi*" and "circular business model"

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be implemented by transforming a current BM of an organization firms into a circular one. Moreover, new start-ups may be formed with a CBM from the start. In addition, an established organization may diversify and employ a CBM in parallel with its existing BM or acquire or merge with a company with a CBM (Geissdoerfer et al., 2020b). CBMs can be limited to operations within a single organization or be based on a collaboration with other organizations (Bocken and Ritala, 2021), as was seen in sections 4 and 5.

CBM design has been linked to sustainable business model design, adapted from the original business canvas approach by Osterwalder and Pigneur (2010). Bocken et al. (2018) identify several key foci areas. First, value creation highlights which resources (physical, human, and financial) are utilized through which kind of processes and technologies, and which actors are involved in this activity. Second, value proposition considers the positive sustainability impact of the CBM in terms of environmental, social, and economic benefits. Third, value capture assesses the costs of the CBM in terms of investment and maintenance expenses as well as the revenues in terms of profits. Fourth, value delivery acknowledges the relationship between customers and the services providers, the interfaces between them, as well as the customer groups (Bocken et al., 2018; Viva et al., 2020). All the above foci areas should have features supporting circular economy (Geissdoerfer et al., 2018).

Moreover, the systemic nature of CBMs warrants attention to networks (Kanda et al., 2021; Zucchella and Previtali, 2019). Hence, fifth, value network consists of the actors that create the environmental, social, and economic value of the solution. Sixth, coordination is necessary as circular solutions are dependent on the alignment between the different interdependent actors CBM (Kanda et al., 2021). Indeed, the "network orchestrator" or lead organization can be particularly important for this task in facilitating the building of joint vision, trust-building, ensuring the commitment of resources and providing transformational leadership (Zucchella and Previtali, 2019). Seventh, centralization of control is relevant, for considering the extent to which a central actor may manage the system of relationship in the circular solution (Kanda et al., 2021).

A common form of CBM with sustainability potential is waste valorisation, e.g., turning waste, such as by-products and residue, into valuable resources such as energy, materials, or products within the same or a different organization or industry (Leder et al., 2020). However, it has been argued that CE practices within waste valorisation that only include organization(s) from a single industry often lead to mere optimization of resource flows and only limited sustainability. Coupling resource flows between different industries, as in industrial symbiosis, is more likely to lead to closing of resources loops, and thus higher sustainability (Oliveira Pavan et al., 2021; Schultz et al., 2021).

Drivers for CBM adoption can be external or internal. For instance, technological advancement may be an important external driver. In Asian and European contexts, the governmental regulation and incentives have been important. In the case of producing biogas from agro-waste, public policy support, such as feed-in tariffs, have been important as investment costs are high and the demand and price may be uncertain (Donner et al., 2020). In countries where CE is not a political priority or policy, or where regulations and other formal institutions around

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CE are weak, the will and priorities of internal shareholders can be important drivers (Chiappetta Jabbour et al., 2020).

Typical external challenges to upcycling have been e.g., securing continuous and reliable raw materials supply from other organizations (Donner et al., 2020; Vermunt et al., 2019). Moreover, virgin resources may be cheaper and easier to manage due to more consistent quality and more reliable supply (Guldmann and Huulgaard, 2020; Vermunt et al., 2019). Demand for circular products may be unclear, there may also be limited understanding of users' needs and demands, and there can be issues with low acceptability among customers in the case of circular products based on waste (Donner et al., 2020; Guldmann and Huulgaard, 2020; Vermunt et al., 2019). Moreover, CBMs may require a change in prevailing consumer habits and institutions, creating inertia for change (Singh and Giacosa, 2019).

Dependence and lack of trust in other organizations may also be challenging (Guldmann and Huulgaard, 2020), as well as poor collaboration between key actors (Reim et al., 2019). Organizations in the circular value chain may be interdependent with one another, and thus problems in one actor's operations may lead to problems at other actors' (Donner et al., 2020). Moreover, legislation may also hinder e.g., the recovery of resources from waste streams, and there may be lack of policy incentives to utilize them (Vermunt et al., 2019).

Internal barriers can be for instance uncertainty regarding the overall economic and environmental benefits of CBMs and lower performance in traditional business indicators such as in return of investment. Moreover, organizations may lack relevant knowledge and experience regarding topics central to a given CBM, including regulation (Guldmann and Huulgaard, 2020; Vermunt et al., 2019). One other barrier also has to do with the high investment costs that often lead to firms' dependence on subsidies (Reim et al., 2019). CBMs may be circular only to a limited degree, reducing the potential of CBMs to address societal issues (Reim et al., 2019).

As already implied, CBMs also lead to changes in supply chain management. Circular supply chain management can be understood as *"the design and control of a network of organizations and end-users that strives for economic, environmental and social benefits by reducing, maintaining and recovering resources"* (Vegter et al., 2020: 12). The management of a circular supply chain is more encompassing and has more systemic features than the linear one (Viva et al., 2020). Linear supply chains have the functions of planning, sourcing, making, delivering, and enabling. In the management of a circular supply chain, there is also the need for a function of return, meaning moving products and resources back from end-user to the supply chain. This means that the use of the products also becomes a part of circular supply cains. Moreover, recovering is needed in order to remanufacture, refurbish, upcycle, recycle etc. the resources and products to complete the circular supply chain (Vegter et al., 2020). For circularities linked to water, the resources will in many cases re-enter the natural cycle, hence there is no function of return as such, but the use phase and end-of-life will still be crucially important. In other cases, for example for biocomposite materials based on resources from water management, the return function may come into play. Management of such full supply chains also requires more attention to the specific elements of the supply chain system (Viva et al., 2020).

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7.1.1 Business model experimentation

BMs can be developed through outside-in or inside-out approaches (Joyce and Paquin, 2016). Outside-in entails exploring innovation opportunities by considering one's own enterprise through different idealised BMs, or BM archetypes (Bocken et al., 2014). Inside-out, on the other hand, starts by detailing the organization's existing BM and explores potential changes to this model, for example using the well-known business model canvas (BMC) approach (Osterwalder and Pigneur, 2010). CBM experiments can be helpful to get real-life experiences and prepare an organization and its partners for CBM's implementation (Bocken et al., 2018).

CBM experimentation is used to get actual data of possible circular BMs and try them out in collaboration with the key stakeholders (Antikainen et al., 2017). Possible outcomes are for instance better understanding of the acceptability of the CBM, better knowledge regarding potential collaborations, and initiation of a mind-set change (Antikainen and Bocken, 2019; Bocken et al., 2018). Organizationally, it has been suggested that CBM experiments should be granted an autonomous space with the organization which is segregated but nevertheless coupled with the broader context of the organization. This space should allow for testing, negotiating, reflecting, and evaluating the new CBM (Hofmann and Jaeger-Erben, 2020).

CBM experimentation may have five iterative phases: ideating, designing, practical planning, implementation and analysis and decision-making (Antikainen and Bocken, 2019; Bocken et al., 2019). The initial ideation and design phases may include the formation of the initial "circular proposition", outlining the ambitions, desired outcomes, design requirements and available capabilities. It is also important to include the "right" actors with suitable capabilities and resource flows, cultural fits, congruent vision, and motivation, etc. Moreover, circular value capture and shared circular purpose are needed, as well as suitable governance structures for the experiment (Brown et al., 2021). In practice, many experiments focus solely on these ideation and design phases (Bocken et al., 2019). Indeed, the implementation phase implies transformations in the organizations' operations and capabilities, making this a critical and challenging phase (Pieroni et al., 2019), which so far has received limited attention in the research literature (Rosa et al., 2019).

BM experimentation may create internal and external engagements for sustainable business transitions. In order for such processes to have a wider sustainable impact (e.g., in terms of the SDGs), the replicability and scalability of CBMs are of central importance. This may not always be straightforward, since CBMs also are likely to have context dependent features, because of the importance of establishing close collaboration with organizations within the circular value chain and pursuing context-dependent customer acceptance, etc. CBM design choice should therefore seek to strike a balance between fitting the specific conditions in each experimental location and the opportunities for wider uptake in other contexts (Zucchella and Previtali, 2019).

Couplings between different actors are at the heart of CBMs, as functioning supply and demand cycles are essential (Antikainen et al., 2017). CBM experimentation may also contribute to testing necessary collaborations (Antikainen and Bocken, 2019). Brown et al. (2021) developed a tool to help identify and design collaborations and collaborative value chains within CBMs, called Circular Collaboration Canvas. In practice, this is used to facilitate workshops to explore

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such collaborations, and includes analytical questions regarding the challenge, resources, customers, and partners of a CBM idea. A similar tool was developed by Boldrini and Antheaume (2021) which seeks to ease collaboration through explicit connection and alignment of individual BMs of several organizations under one joint analytical tool. These tools draw extensively on pre-existing tools for BM mapping and development, in particular the Business Model Canvas of Osterwalder and Pigneur (2010). They also build on insights from the partly preceding and partly overlapping literature on sustainable business model development, where key works such as the Triple Layer Business Model Canvas (Joyce & Paquin, 2016), the Flourishing business canvas (Upward & Jones, 2016) and the Value Mapping Tool (Bocken et al., 2013) have been developed to support a systemic view on business modelling. One approach that may be of particular relevance when it comes to water-smart solutions is the Ecology of Business Models Experimentation (EBME) map (Bocken et al., 2019), which pays special attention to the interaction and dependencies between the focal business model and a wider 'business model ecology', including other BMs and value chains, as well as infrastructure and possible rebound effects.

7.1.2 Five lessons to learn for the water sector

Water-smart solutions, such as recovery of materials from and reuse of wastewater, can be seen as cycling, which is a key CBM strategy. Hence, the rapidly growing literature on CBMs can help to inform the implementation of water-smart solutions. We identify at least five important insights:

First, the CBM literature provides a framework for systematic evaluation and experimentation with novel BMs linked to water-smart solutions. Several process tools have been developed, that may help organisations identify current and future opportunities. The application of such tools may also reveal potential challenges, and thus help organizations foresee and avoid costly mistakes in the design and implementation of novel BMs based on water resources.

Second, CBM thinking provides a holistic perspective on novel BMs not only in terms of economic value, but also in terms of environmental and social features and impacts. As sustainability aspects continue to permeate to the doctrine of (among others) wastewater treatment actors, it is becoming increasingly important to be able to evaluate novel BMs from the perspectives of environmental, social, and economic sustainability.

Third, CBM literature highlights the importance of coordination and well-functioning collaborations between organizations. Water-smart solutions imply novel couplings between organizations. These actors often come from different sectors and thus may have differing operational practices, regulations, norms, and rationales. This may cause mismatches in expectations, values, and operational preferences, which may develop into challenges for the implementation of new BMs. Moreover, there may be a need for trust-building. The CBM literature, as we have seen, encourages inclusion of key partners in BM experimentations, and provides various techniques for this.

Fourth, the call for systemic supply chain management is relevant. Extension of a wastewater supply chain may also have implications for the other stages of wastewater treatment (e.g., new wastewater treatment requirements, or requirements for the traditional drinking water

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and/or wastewater services), meaning that the implications and feasibility of novel BMs must be re-evaluated throughout the supply chain. In terms of coupling together different supply chains, water-smart solutions may tie wastewater treatment together with e.g., energy production, manufacturing, or agriculture. Such novel interdependencies between supply chains require coordination.

Fifth, CBM literature highlights the importance of adequately recognizing the specific local context elements when implementing water-smart solutions. While water treatment principles are quite generic, the novel BMs based on water-smart solutions need to take into account the different conditions for their implementation in different geographical locations. Both the key drivers for implementation, market demand and acceptance of water-smart solutions may differ in different locations.

7.2 Approach selected for WU

7.2.1 Triple Layered Business Model Canvas (TLBMC)

As noted above, the Triple Layered Business Model Canvas (TLBMC) is a well-established tool for sustainability-oriented BM innovation (Joyce and Paquin, 2016). It extends Osterwalder and Pigneur's (2010) original BM canvas by adding two layers: an environmental layer and a social layer. While many of the more recent studies and conceptualizations provide important insights as to what circular BMs entail and require, the canvas approach is a practical tool that has proved to work well in processes to explore and discuss alternative BMs with different kinds of stakeholders.

In WIDER UPTAKE, many of the stakeholders are concerned with implementation of new and more sustainable solutions, but less concerned with business development as such. The TLBMC was selected as a main tool, since it places equal weight on environmental, social, and economic aspects. By drawing attention to the environmental and social benefits and challenges from the beginning, we wanted to attract the interests and make use of the competencies of partners and stakeholders who are not business actors but still have crucial roles to play for the uptake of the focal solutions, such as local authorities, environmental and food safety agencies, and potential end users.

As illustrated in Figure 11, the TLBMC offers a clear way to visualize and discuss a business model's multiple and diverse impacts. Instead of attempting to reduce multiple types of value into a single canvas, it allows economic, environmental, and social value to be explored horizontally within their own layer and in relationship to each other through vertical integration of these (Joyce and Paquin 2016: 1475).

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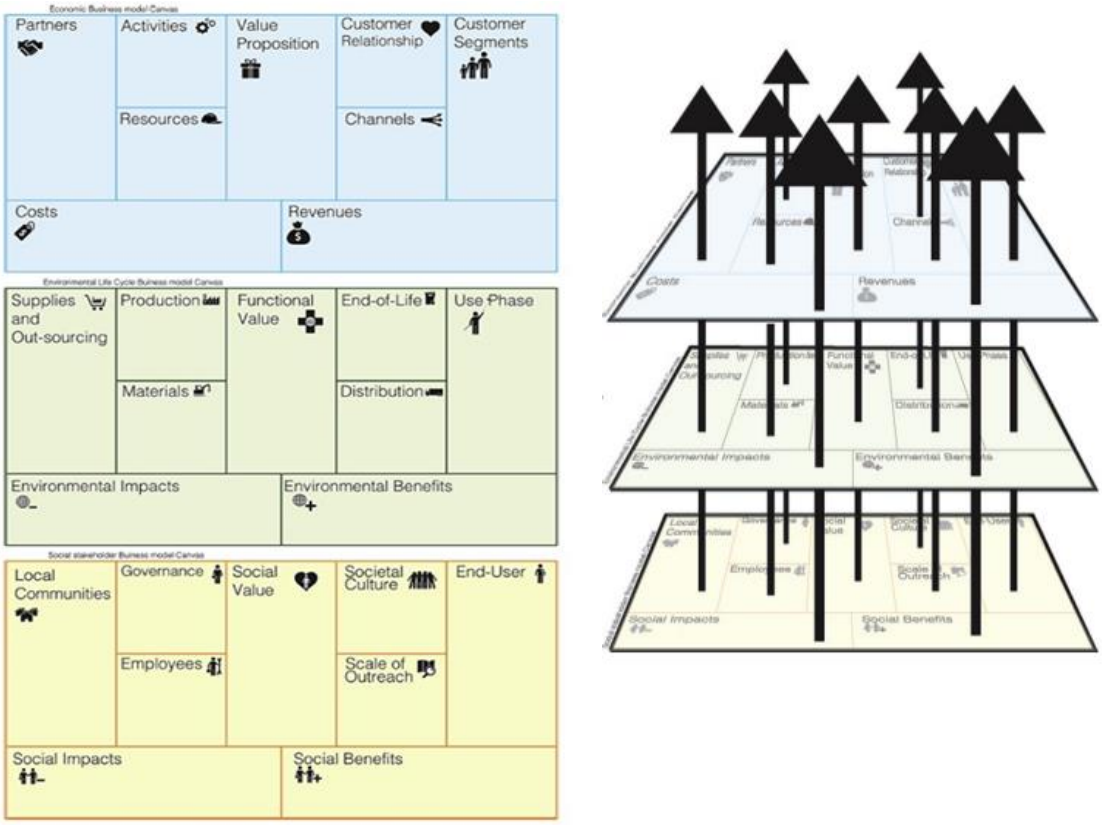


Figure 11: Triple Layered Business Model Canvas (Joyce and Paquin, 2016:1482-1483).

The environmental layer is laid out in parallel with the economic layer but builds on a life cycle perspective on environmental impact. It does not integrate a formal LCA into the canvas but ensures that the full product life cycle is taken into account, starting with identification of the functional value, which is a quantitative description of the functional unit (e.g., 1 kg butter) and the total of these units consumed by customers in a given timeframe such as a year. Next, the required materials, production activities, supplies and outsourcing, distribution, use phase and end-of-life and their potential environmental impacts are mapped. While different indicators may be used, Joyce and Paquin (2016) focus on carbon footprint.

The social layer builds on a stakeholder management approach (Freeman, 1984). It starts from the social value that the actor/s want to create, which in some cases may be explicitly stated in their mission or may be defined more specifically for the product or service in question. Paralleling the other layers, social impacts are traced through different impact categories; impacts on employees, governance (decision-making, stakeholder involvement, etc.), local communities, and wider, societal culture. The category scale of outreach describes the depth and breadth of the relationships built with stakeholders through the action/s (e.g., geographical outreach, social programs, etc.), and finally the actual or potential impacts on end-users are included.

In the process to explore new BMs in WIDER UPTAKE, we focused mostly on populating and discussing the three canvases. However, the alignment between components across the three layers, illustrated on the right side of Figure 2, is also important. To be sustainable, e.g., the value proposition and functional value should be aligned, and there should be alignment with

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the identified customers and social benefits for end-users. Therefore, the degree of alignment is also considered in our analysis of the canvases.

7.2.2 Addressing wider contexts and dependencies

While drawing attention to "the triple bottom line" and offering a relatively easy way to approach new partnerships and transactions, the TLBMC may help communicate change to diverse audiences. However, the TLBMC does not pay much attention to the wider business environment, or the so-called business ecosystem, of the BMs. Osterwalder and Pigneur (2010), noted that BMs are influenced by an external environment, which also should be assessed systematically. In their original framework, this environment consists of wider societal trends (regulatory, technological, cultural, etc.), industry forces (including competitors, substitute products and services, other actors in the value chain, etc.), macroeconomic forces, and market forces, such as changing needs and demands.

Bocken et al. (2018) emphasize that a BM may have unexpected direct and indirect impacts, depending on how the system boundaries are drawn, and there is a danger of undesired rebound effects. As noted above, the Ecology of Business Model Experimentation (EBME) approach suggests that BMs must be understood in a wider context, including their institutional setting, as well as other business models, value chains and infrastructures. With this, they partly reiterate Osterwalder and Pigneur (2010). However, they emphasize more strongly that in a dynamic and complex environment, consideration of wider context factors is needed at the outset.

The EBME procedure includes many of the same elements as TLBMC but is less elaborate on impacts and has more focus on dependencies. These can be dependencies in relation to other products (e.g., mobile phones for apps), existing value chains (e.g., delivery networks), infrastructures (e.g., public spaces), or resources (e.g., water, energy). As discussed in chapter 5, three high-level forms of dependencies are possible: 1) Neutrality, when both business models do not affect each other, 2) competitive dependency, when business models compete for the same resources, and 3) symbiotic or mutual dependency, when the output of one business model constitutes an input for another or the presence of one business model leads to growth in another (Bocken et al., 2018).

However, the influence of relevant policies and regulations, demography, climatic changes, technology trends and cultural developments at a wider societal level are not explicitly considered. When it comes to circular water-smart solutions, such factors are highly influential, both as regards the scope for implementation currently, and for their long-term feasibility and impacts (Damman et al., 2022). In WIDER UPTAKE, therefore, the exploration of new BMs has included consideration of wider developments and structural conditions identified as central in T4.1, as well as dependencies, as initially addressed in T4.2.1 (see chapter 5 above).

7.2.3 BM exploration process

The experimentation procedure proposed by Bocken et al. (2018) has four steps: 1) Defining the sustainability aims of the business, 2) analysing if the BM is associated with any kind of dependencies, 3) identifying or characterising the types of dependencies, in terms of existing infrastructure, products/ services and resources, and 4) exploring how positive value can be

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increased and negative value reduced. The experimentation activity can take different forms, such as interviews, co-creation sessions or other forms of brainstorming, and involve various tools, such as physical or digital prototyping, testing through different websites or mock advertisements, or ethnographic observation (Bocken et al., 2018).

As noted above, WIDER UPTAKE decided to apply the TLBMC as basis. TLBMC forces the actors to think in quite concrete terms, about the various elements in a BM that must be put in place, identify strengths and areas where more work is needed to reach the implementation stage, and also share expectations about costs, benefits, and wider impacts. This was seen as necessary before relating the BMs to the dependencies and contextual interactions that already had been broached in preceding tasks (T4.1).

The TLBMC was applied in two workshops with the key partners and stakeholders in each demonstration case. In most cases, these were 3-hour sessions as part of 1 or 2-day workshops on site where other issues also were discussed: drivers, barriers, and industrial networks (in relation to WP4), and the goals, objectives, and criteria for assessment of sustainability and "water smartness" relevant for the specific case (related to WP5). For each case, we decided to focus on one circularity and "product" initially. In the Hamar case, we decided to add a BM for another circularity during the second workshop, since the exploration process was seen as quite relevant both for the soil improvement selected initially and for the struvite harvested from the biological phosphorus removal process. In Ghana, we also ended up with two, one for biochar and one for water reuse, since the former is more mature in terms of business model development, while many of the stakeholders have a stronger focus on the latter.

To limit CO_2 emissions and travel load, some of the second workshops were carried out online. In all cases, TLBMC templates in Miro were used to guide the discussion. During the physical workshops, participants were split in smaller groups, mixing persons from the respective industry partners and research organisations. For the cases linked to Hias/Hamar and IVAR/Stavanger, the second workshops were organised at the same time, in Hamar, with some joint discussion and parallel group work on the respective BMs. This was according to the partners' own wishes, with a view to learning and experience-sharing. There were preparatory meetings for the partners to familiarize with the methodology, and information material in the form of PowerPoints and Joyce and Paquin's (2016) article was provided. Table 8 provides an overview of the focal circularities and workshops conducted for each case.

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Table 8: Overview of business model workshops.

Case	Circularity in focus	Workshop 1	Workshop 2
Netherlands	Biocomposite for riverbank protection	10.03.-11.03. 2022	24.03.2023 (online)
Norway – Hamar	Soil improvement products Struvite	31.03.-01.04. 2022	01.03.-02.03. 2023
Norway – Stavanger	Organic fertilisers	04.10.-05.10. 2021	01.03-02-03. 2023
Sicily	Water reuse for agriculture	17.10.-19.10. 2022	27.03. 2023 (online)
Ghana	Biochar for energy Water reuse for urban agriculture	22.03.2022	09.02.2023
Czech Republic	Water reuse for urban blue-green infrastructure	03.05.-04.05. 2022	29.03.-30.03. 2023

In addition to the workshops, there were follow-up meetings with the key partners (case owners), to sort out unclarities remaining from the group discussions and get more specific data where such was available. After the first round of workshops, we also had meetings with research partners from WP3 (Assessment and optimisation of circularity and efficiency of symbiotic solutions), to identify possible indicators and preliminary results from their work, which could be used to test and discuss the identified BMs during and after the second workshop, in the final documentation provided.

In the following chapters, we provide an overview of the BMs developed for each case. Each chapter first presents the BM and discusses its sustainability, in terms of impacts and alignment across layers in TLBMC. Subsequently, we analyse its feasibility in terms of four dimensions, namely:

- Maturity
  - Technological maturity
  - To what extent are the elements in place and clearly defined
  - To what extent do the partners agree about the model
  - What is the time perspective – is the model already being tested, will it be realised in near future, or will it take time to make agreements and/or sort out other elements
- Dependencies
  - Dependency on existing infrastructures (e.g., facilities, pipelines, public spaces)
  - Building 'sustainable infrastructures' (e.g., distribution system for treated wastewater)
  - Dependency on other resources (e.g., waste streams)
  - Dependency on existing value chains and networks (e.g., delivery networks)
  - Interactions with existing business models
- Social acceptance
  - Customer/user acceptance
  - Public acceptance
- Political feasibility

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8 BM exploration in the Hamar case

8.1 Solutions in focus

When the work on BMs with the Hamar case started, we decided to focus on the soil products, since they involve resources from both Hias and Sirkula and collaboration between all the three case partners. The harvesting of struvite was barely starting, and Hias had an initial agreement with technology provider Enwa and their international partner Ostara concerning distribution and sale of the struvite. In the following months, however, Hias found a strong interest in the struvite also from Norwegian actors. This must be seen in the context of uncertainty linked to the Ukraine war and European energy crisis and increasing prices of conventional fertilisers. At the time of the second workshop, Hias had left the initial agreement with Enwa and was exploring different alternatives with Norwegian partners. Hence it seemed highly relevant to discuss alternative BMs also for the struvite as such. In the following, we first discuss the BM for soil improvement products.

8.2 BM for soil improvement from Sirkula

8.2.1 Economic layer

As illustrated in Figure 12, several **value propositions** are identified. A main one highlighted in Sirkula's own advertising is that the soil product is locally produced, peat-free, compost-based, and of high nutritional value. Secondly, they add that it is sold at their recycling stations and thus easily available. So far, the research and development carried out locally has resulted in registration of four standardised products:

1. Sirkula soil for lawns, green space covering (with very good drainage properties)
2. Sirkula soil for flower beds and garden use (airy and extra nutrition-rich)
3. Sirkula pure compost (fertiliser not based on sludge, for ecological production)
4. Sirkula soil for herbs and vegetable production (not based on sludge)

All are approved by the Norwegian Food Safety Authority and have their own product declarations. There are also experiments and plans for other specialised products, hence "standardised and designed for purpose" is considered as a value proposition. Another point highlighted, is that unlike most other soil products, the soil from Sirkula and Hias includes sludge that is sterilised. Thus, it prevents proliferation of unwanted species and has an advantage in the face of expected regulatory changes, which will imply stricter purity requirements.

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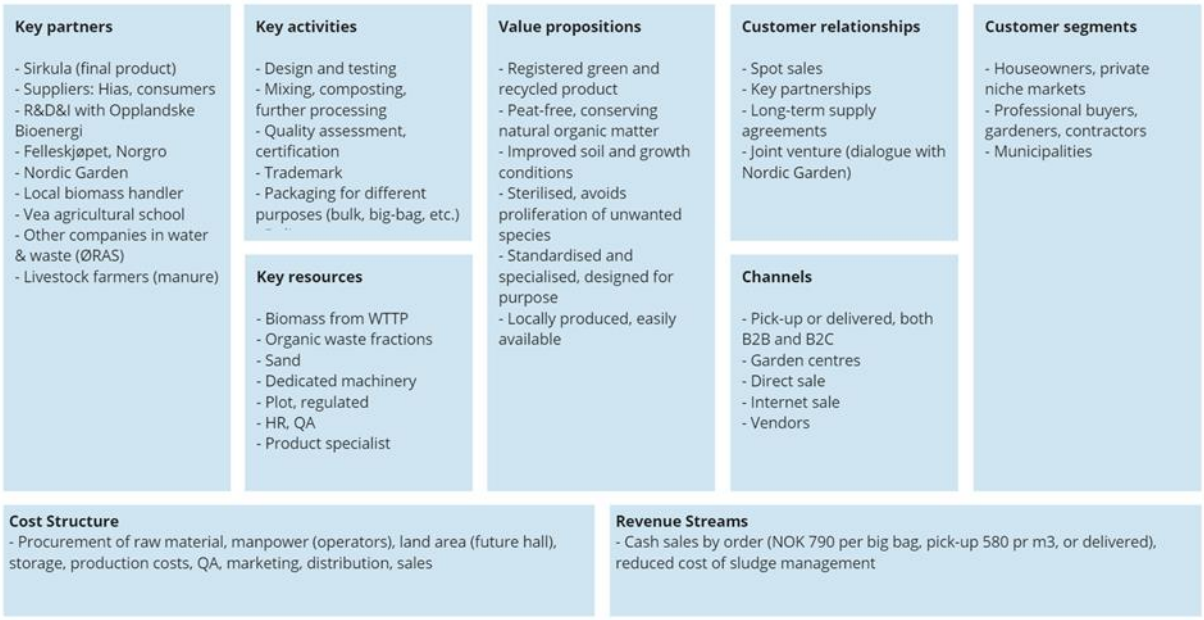


Figure 12: Economic layer, TLBMC for Sirkula soil improvement products.

The customers are individual house and garden owners, professional contractors, as well as municipalities (for geographical reasons mainly the four municipalities owning both Hias and Sirkula). While the business started with spot sales at the recycling stations, more long-term customer relationships have been developed with Hamar municipality and some of the professional buyers. Sirkula is also in dialogue with Felleskjøpet, as a dominant brand and distributor of fertilisers, seeds, machinery, and equipment for agriculture in Norway. Their subsidiary Norgro, is specialised in products for horticulture. Felleskjøpet also owns 50% of Nordic Garden, which is a large supplier of garden products serving the nation-wide chains of garden centres and construction supply stores in Norway.

The channels so far are mainly owned direct, e.g., spot sale at the recycling stations, as well as online sale through Sirkula's website. The product is sold in bulk as well as big bags, uploaded to the trailer or truck of the customer, or delivered. Bucket sale was also done before, but from 2023 replaced with 18-liter bags, which together with increased marketing are expected to increase sales considerably. The key activities, listed in Figure 12, include design and testing; mixing, composting and further processing, as well as quality assurance and registration, packaging, marketing, and sales. The key resources are biomass in the form of sludge from Hias, organic waste fractions from Sirkula, as well as sand from a local sand and gravel provider. Furthermore, the production itself requires space and dedicated machinery (e.g., heavy machinery, grinder, sieve), as well as manpower for operation and administration, including a product specialist that was hired especially for the soil production.

Partners and partnerships: While Hias is a key supplier, regular consumers or citizens must also be considered as such, since they deliver the other organic waste streams used (mainly garden waste). A local biomass handler is also in the partnership, involved in the transport of biomass. Vea agricultural college is not a business actor or formal partner but contributes actively to the development and testing of the products. Opplandske Bioenergi, a producer of biochar, is also involved on this side. Furthermore, Felleskjøpet is mentioned, as there is dialogue with Nordic

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Garden about distribution and sale, and also with a view to upscaling the activity. Norgro wants to terminate their own production of soil containing peat. Since they are looking for alternatives, and since further expansion of the production at Sirkula will require more space and facilities for storing and processing, the potential for establishing a larger soil production factory near the region's main waste deposit is currently under study. Other intermunicipal companies in waste and wastewater management, most notably ØRAS, in an adjacent region, are involved, in terms of learning and experience-sharing. Furthermore, livestock farmers are included on the list since manure from livestock is highly relevant as additional source of biomass. Tree-clearing entrepreneurs may also provide feedstock in form of branches, tops, and roots, but are not included in the canvas, since manure would be preferred.

In terms of **intellectual property rights (IPR)**, the trademarks of the four soil products belong to Sirkula, while the specific biological treatment process for the wastewater is patented by Hias How2O, as the "Hias process".

The **cost structure** includes the following **operational expenditures (OpEx)**: procurement of raw material, production costs, land use cost of manpower (operators), quality assurance, marketing, distribution, and sales, in addition to depreciation. So far, biomass from Hias has been provided free of charge. Sand, which is another main ingredient, is bought at 56 NOK per ton. At the current site, seven operators and the product specialist have been added as a result of the soil production. Some of the operator positions at the recycling station are wholly or partly financed via state welfare, to provide work training or permanent employment for people in need of such support. Energy in form of fuel for machinery is also a substantial cost. Sales and marketing have so far been done at limited cost. In the future, Sirkula wants to focus on the production and leave these tasks to a partner, e.g., Nordic Garden. In terms of **CapEx,** the partners identified depreciation linked to past investments as the main element. This relates, amongst other, to the thermic hydrolysis for hygienisation and the anaerobic digestion unit at Hias, which are prerequisite for using the sludge as feedstock, as well as handling equipment at Sirkula, etc, which is needed for the soil production, but were already available before the start of production and serves multiple purposes. If the plans for a larger soil factory are realised this will require 15-20 000 m^2 of land, as well as a hall for more predictable process conditions.,

The current **revenue stream** is from cash sales by order.[2] In addition, reduced cost of sludge management must be taken into account. Currently, the soil products are slightly more expensive per ton than conventional ones. Thus, the present soil production does not make a very profitable business as such, but it does generate value from waste already and there are specific initiatives and plans for how this value may increase in future.

8.2.2 Environmental layer

As shown in Figure 13, the **functional value** in this case is estimated to be 7000 tons/year, based on previous production. However, Sirkula expects that for 2023, sales in big bags alone could

[2] Updated information on the prices of the different products can be found at Sirkula's website: https://www.sirkula.no/gjenvinningsstasjoner/kjop-jord/

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reach 6000-8000 tons. The estimated potential, based on biomass from sewage sludge and waste available in the region, is 30 000 tons/year. Still, there is no exact limit, since different source materials (such as manure, straw, etc.) could be added to increase the volume, if demand develops.



Figure 13: Environmental layer, TLBMC for Sirkula soil improvement products.

The environmental impacts associated with the **production** are difficult to estimate. Composting itself is associated with some climate gas emissions, primarily of CO_2. While some methane (CH_4) and nitrous oxide (N_2O) may be released, Sirkula turns the compost very often, therefore large anaerobic zones with methane emissions do not develop. Moreover, the decomposing feedstocks are from the short-term carbon cycle (Brown and Subler, 2007). Besides, other treatment methods are associated with CO_2 emissions in the same range. As long as fossil fuels are used, the energy consumption by machinery during production is associated with emissions, in the range of 2.75 kg of CO_2 per litre of fuel (ibid.) Packaging also requires energy, but this is in the form of electricity, which in Norway largely stems from renewable sources. Additionally, the stored product may emit some climate gases, which again form part of the short-term carbon cycle. At the present site, there may also be some surface runoff, which could be eliminated with the establishment of a production hall.

All the **materials** (30% organic and 70% sand) return to the natural cycle. The big bags are made of plastic, where the production has certain emissions. These may in principle be recycled and used for production of hard, industrial plastics, but so far this is not an option. Logistics, e.g., transport of biomass and soil products, is again associated with climate gas emissions, as long as the vehicles run on fossil fuels. Possible additives can be biochar or struvite based on phosphorus recovery, which both are associated largely with positive rather than negative environmental impacts.

The **use phase** is also associated with some emissions, to the extent that fossil fuelled machinery is used, e.g., mainly in the professional customer segment. Here it should be noted,

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however, that peat-free soil does not "shrink", i.e., there is no need to refill. It is also homogenous, without stones. Less machine work is therefore required than for peat-containing products. Generally, compost-based products are associated with improved carbon sequestration in the soil and preventing methane emissions through aerobic decomposition, as methane-producing microbes are not active in the presence of oxygen (US EPA, 2020).

Furthermore, the use of the soil products replaces use of non-renewable resources, e.g., peat. Peat takes around 1000 years to form. The degree of carbon sink it is associated with depends on many factors, such as type of peat, climate zone, etc. For Norway, the national Environment Protection Agency (2017) has estimated a conversion factor of 0,05t C/m^3. If we assume that 1 m^3 = 1.5 ton soil product, and consider the potential 30 000 tons, the production of soil improvement from Sirkula may thus save 20 000 tons of carbon per year. Moreover, the testing done with Vea agricultural school shows better cultivation results and less need for irrigation. According to Sirkula, their soil products are also without black-listed species, and microbial life is good for the biodiversity of the soil, at the same time as it improves resistance towards plant diseases and adds nutrients.

8.2.3 Social layer

As illustrated in Figure 14, the BM for the soil production at Sirkula is associated with multiple **social values**. In terms of job creation, the production presently employs 6-8 persons. Specific estimates have not yet been made, but a soil factory may increase this number, and there are also likely to be ripple effects along the value chain. Furthermore, social value is created in that the soil production helps promote circular economy in Innlandet, both in itself and through co-learning and potential synergies with other initiatives, such as the production of biochar at Opplandske Bioenergi. It also generates interest and may inspire similar initiatives in other regions. A sense of wellbeing and positive local identity, linked to the fact that valuable products are made from waste, must also be considered as social value, even if it is less concrete and difficult to quantify.

When it comes to **societal culture**, the BM is associated with increased environmental awareness and sense of environmental responsibility. The soil production is quite visible at the large recycling centre of Sirkula, where it is linked with education and demonstration activities targeting customers as well as other visitors, including school classes, students, and professionals from the private and public sectors. There are also significant impacts in terms of knowledge development, e.g., with Vea agricultural school, Opplandske Bioenergi and other companies in wastewater and waste management, both bilaterally and through the active roles Hias and Sirkula play in knowledge sharing and development in the national associations Norwegian Water and the National Waste Management and Recycling Association.

For the **end users**, the peat-free and waste-based product contributes to sustainable gardening and/or more circular production, e.g., of local vegetables and fruits. The satisfaction from using such products may contribute to a sense of wellbeing in the present, as already noted, but it may also be associated with bequest value, or sense of satisfaction at passing natural resources over to coming generations in a better state than when the current users inherited them. The availability of the offer and increased awareness of the circularity in question may also

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encourage more local food production, by strengthening the trend to grow one's own vegetables and design more edible gardens.

Especially for professional users, an important point is that the soil is shrink-free. Some customers report that this makes the soil cheaper to use than conventional products, even if the price is slightly higher at the outset. Moreover, the improved cultivation results and less need for irrigation saves time and costs. Because the product is sterilised, it is completely safe to use. The **scale of outreach**, as we have seen, is mainly local and regional right now. If the plans for a wider collaboration with Nordic Garden and/or a larger soil factory with Norgro are realised, the outreach will be to the whole south-eastern part of Norway.



Figure 14: Social layer, TLBMC for Sirkula soil improvement products.

For the **employees**, the soil production provides more meaningful work. This influences job satisfaction as well as the image or status of work at Sirkula and Hias positively. Moreover, the soil production involves increased opportunities for learning and innovation at work, as continuous process improvement and the development of new, designed-for-purpose soil products are integral parts of the activity. It is also noted that the jobs at the recycling stations are well suited for inclusive work and job training, which enriches the work environment and has benefits in terms of social cohesion.

When **governance** is concerned, all the key actors in this case are organised according to the Norwegian Model, e.g., with a close collaboration between owners/managers, employee organisations/unions and union representatives. Being owned by municipalities and providing public services, they follow the general guidelines for stakeholder involvement in their sectors. Hias uphold openness and honesty as key corporate values, and regularly issues a Climate, environment, and quality report.[3] Sirkula, likewise, issues an annual Sustainability report.[4] The

[3] https://www.hias.no/om-hias/miljo--og-klimarapport/

[4] https://nyweb.sirkula.no/siteassets/filer/miljorapport/barekraftsrapport2022_sirkula.pdf

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BM considered for the soil product/s may further stakeholder engagement and transparency, as a visible CE initiative, and it may contribute to more integrated natural resource management.

For local communities, the BM is a step towards realisation of the strategy for regional development in Innlandet, which has a strong emphasis on bioeconomy.[5] It may help attract new inhabitants, by contributing to an environment-friendly and innovative image and local identity. E.g., a new eco-friendly house and living concept, based on CE and zero emission solutions has recently attracted more than 300 applicants.[6] Thus, there may also be long-term lifestyle and health benefits. Another point that came up in the group discussions is that the value creation from sludge and other waste fractions over time could lead to a reduction of wastewater and waste management fees, as these are based on a full cost coverage principle.

Overall **social benefits**, as summarised in Figure 14, are increased value creation and employment without increased resource use, freedom to choose and sense of well-being gained from production and use of sustainable products, and social inclusion. While there are no negative social impacts as such, the products still cost more than peat-based alternatives. If the price remains high, the products may not be equally affordable to all.

8.2.4 Alignment across the TLBMC layers

As we have seen, this BM is associated with substantial environmental and social benefits. There seems to be a growing market, and the production is already generating economic value. In pure business terms, the revenues are less than the costs. At first sight, this might seem like a misalignment. However, the alternative would be conventional management of the sludge and organic waste fractions, without any revenues. In this sense, and given the cost price principle involved, the model seems to be sustainable also in economic terms. Moreover, the BM is still under development, in that there are efforts to expand the collaboration and establish a larger production facility. This will require substantial investment but may also increase the market and pave the way for new products that can be sold at a higher price.

Another issue is the risk associated with the end of life. This risk is, however, seen as very small, and the dialogue with the Norwegian Food Safety Authority and strict adherence to their rules and guidelines ensures that this is handled responsibly. As there are ongoing efforts to reduce the contaminants entering into the wastewater system, this risk is also likely to decrease over time. Finally, we note that the slightly higher price compared to peat-based products in principle may contribute to inequality. This would apply mainly to individual houseowners and not for professionals, where the use properties balance the higher price. Given the need to limit harvesting of peat, it is also likely that the price balance may change in future.

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5 https://www.regjeringen.no/contentassets/afc36304247d48f0a3546f992e0e0305/biookonomistrategi-for-innlandet_feb18.pdf

6 https://www.nrk.no/innlandet/over-300-interesserte-i-miljovennlige-boliger-pa-moystad-okotun-i-hamar-1.16319787

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8.3 Maturity, dependencies, and uncertainties

The BM for the soil products is based on mature technology, in that co-composting is a well-known process. At the same time there is a wide scope for improvement and development of new designed-for-purpose products. In this case, we also find a working BM in place, whereby the products already are being made and bought by a range of customers. In this sense, the BM has all elements in place and is clearly defined. However, as we have seen there are also efforts to develop it further. For this, all details are not yet defined. The outcome of the ongoing dialogue between the present and prospective new partners will affect the activities and key resources, as well as customer relationships, in turn influencing the scale of outreach and wider benefits for society. However, this dialogue is quite concrete, and according to some of the partners the new larger-scale facility will be realised within the next couple of years. Still, as the following section will show, the actors do not necessarily share exactly the same view on the future direction, as there are multiple options both for the use of struvite and the mix of organic waste fractions used in the soil production.

In terms of **dependency on existing infrastructures or building new 'sustainable infrastructures'**, there are no critical factors, but as we have seen a new site and production hall are considered as important for further upscaling. Some of the partners also emphasized the potential for cleaning polluted masses from the waste deposit next to the considered site and reusing them in the soil production, thereby limiting the use of sand. Others saw less potential for this, as such cleaning facilities already are planned in two nearby regions. However, they did not rule it out completely.

Dependency on other resources is a relevant issue, since the digestate from Hias currently is an important ingredient in the soil products. Given the close collaboration between the two intermunicipal companies, Sirkula does not see any immediate threat in this. The availability of sludge is not necessarily a limitation, since other organic waste fractions, e.g., manure from chicken and livestock, straw, sludge from fish farming etc. also may be used in future products. Should the requirements concerning use of sludge for soil and fertiliser production become stricter, it is possible to use some biochar and less sand.

When it comes to **dependency on existing value chains**, the BM at present is highly dependent on receiving free or very cheap digestate from Hias. That the key activities are carried out alongside other CE activities at Sirkula's recycling station has also been a key enabling factor. Entering a more formal and long-term collaboration with established actors could lead to new dependencies, including lock-in to soil production whereas other, higher-value uses of the biomass could become available in future. The latter is a matter of concern for Sirkula, who emphasize that they want to stay in control and ensure that the resources are used to create value locally and in the most sustainable way. Hias is less explicit about this and more focused on the opportunities involved in collaborating with a national distributor.

Interaction with existing business models: The soil products from Sirkula compete with soil products from established national chains, where the mentioned Felleskjøpet and another called Plantasjen are major ones, providing both peat- and compost-based alternatives. Soil improvement products are also offered by other wastewater and waste management

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companies, such as ØRAS, which covers a district in the county south of Innlandet. In these cases, there is more of a symbiotic relationship, where the actors share experiences and make joint efforts to promote sludge-based products. This is related to the geographical distribution and bulky nature of the resources, which lead to high transport costs. In the case of Felleskjøpet and Norgro, we have noted, a relationship of mutualism is foreseen, as growing interest in peat free products makes Norgro interested in new alternatives, and increasing volume and range of products from Sirkula makes it relevant for the local partners to collaborate with a professional market actor.

As regards **social acceptance**, Norway has a long tradition for and still allows spreading sludge on farmland under certain, regulated conditions. Hence, the use of sludge-based soil improvement is not surrounded with scepticism. The experience so far is that once the target users know or experience the quality of the products, in terms usability as well as improved soil and plant growth, they are easily convinced. Public acceptance has not been examined in detail, but there is a strong public interest in the CE activities at both Hias and Sirkula.

In terms of **political support**, the initiative is fully backed by the owning municipalities. It is also contributing towards realisation of the bioeconomy strategy for Innlandet county, and Norway's strategy for circular economy. In terms of regulations, the national fertiliser regulation has been under revision for several years. It is anticipated that there will be stricter requirements concerning quality and seed content, which could strengthen the competitiveness of products with sterilised content like those of Sirkula. The Norwegian Government is also considering a range of measures to limit the harvesting and use of peat, which would be conducive.[7] On the other hand, the EU has strict policies and requirements concerning use of sewage sludge. In a longer-term perspective, this could exert a stronger influence on Norwegian regulations and/or the social acceptance of the products. This uncertainty is acknowledged, and it is amongst other with this in mind that biochar is considered as an ingredient with a strong future potential.

8.4 Business model for struvite from Hias

As noted above, the struvite from Hias may be added to the soil improvement and organic fertilisers. However, it may also be sold as a fertilised on its own. Internationally, struvite is also used as input for conventional fertiliser production. Under the initial agreement with Ostara, which was organised through Enwa for the first five years, Ostara would buy 50% of the harvested struvite and distribute it on the international market. Hias would aim to sell the other 50%, and whatever part they could not sell would also be taken by Ostara. As noted above, this has changed, and Hias is more confident about the opportunities in the Norwegian market.

8.4.1 Economic layer

The main **value proposition** for the struvite is the provision of a fertiliser product, which is based on renewable P, thus contributing to sustainability and CE by limiting the use of virgin phosphate rock. Moreover, struvite works as a longer-time slow-release fertiliser, in comparison with conventional fertiliser products. Hence, its use will involve less run-off.

[7] Torv-forbud kan kutte utslipp av klimagasser - Miljødirektoratet (miljodirektoratet.no)

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Struvite is certified as green and recycled, and also very homogenous, both in terms of content and physical properties, and has been approved as a fertiliser for ecological farming in the EU. Moreover, it is locally produced and therefore easily available, also in times of uncertainty. Another important point is that the biological P removal process, which the struvite extraction is linked to, leads to less use of chemicals in wastewater treatment compared to chemical precipitation of P.

In terms of **customers**, a mapping of potential applications and customer segments was carried out by Hias in 2020.[8] This study identified green keepers as a key target customer group, since struvite provides strong root growth. In addition to ecological farmers, actors in horticulture are an important customer segment since their production requires intensive root growth. Intensively irrigated areas, and production of higher value crops (in the Norwegian context; carrots, salad, potatoes, swedes, onion, etc.), as well as private niche markets are also associated with considerable potential. Internationally, e.g., in the Netherlands, recovered struvite is also sold to conventional fertiliser producers. In Norway, Yara is partnering in research and development on nutrient recovery but given the limited volume of struvite and type of process technology applied in their Norwegian plant, they are not a likely user right now.

As regards **customer relationships**, the focus currently is on developing a key partnership with Norgro. According to Hias this is so far expected to be exclusive. However, as noted above, struvite is also attractive as an additive to the soil improvement production with Sirkula. Currently, the thinking is that Norgro will sell the product through Felleskjøpet, both directly to professional buyers, and through their national network of shops.

As shown in Figure 15, the **key activities** include biological wastewater treatment and the struvite extraction itself, in addition to input and quality control, packaging, marketing and distribution. The **key resources** are wastewater (running through an effective bio-P process), the struvite reactor and precipitation chemicals, including polymer, magnesium chloride and lye. Human resources in form of operators and management/QA must also be counted in, although the operation of the struvite plant is closely integrated with the other activities in the wastewater resource recovery facility.

The **key partners** identified are Hias, working hand-in-hand with Hias How2O to promote biological treatment and nutrient recovery, Enwa with Ostara as their international partner, and Norgro, which as we have seen is closely linked with Felleskjøpet. Sirkula has also been part of the initiative, and all three are in dialogue about the establishment of a larger soil factory. The Norwegian Centre for Ecological Farming and the research institute Nibio have been important in documenting the properties and promoting struvite as a fertiliser product. The national Association of Agriculture Advisors (NLR) is also seen to have a key role but has so far not been much involved. At the local level, a knowledgeable greenkeeper at Atlungstad Golf Club has contributed to the testing and played a key role as early adapter. In terms of **IPR**, the product was trademarked in 2023, as "Hias Struvitt", owned by Hias, and to be distributed by

[8] https://www.vanytt.no/?p=16234 (the report itself is internal and in Norwegian language).

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Norgro, where a formal intention agreement has been signed. As noted above, the biological treatment process that the struvite is harvested from is patented by Hias How2O.

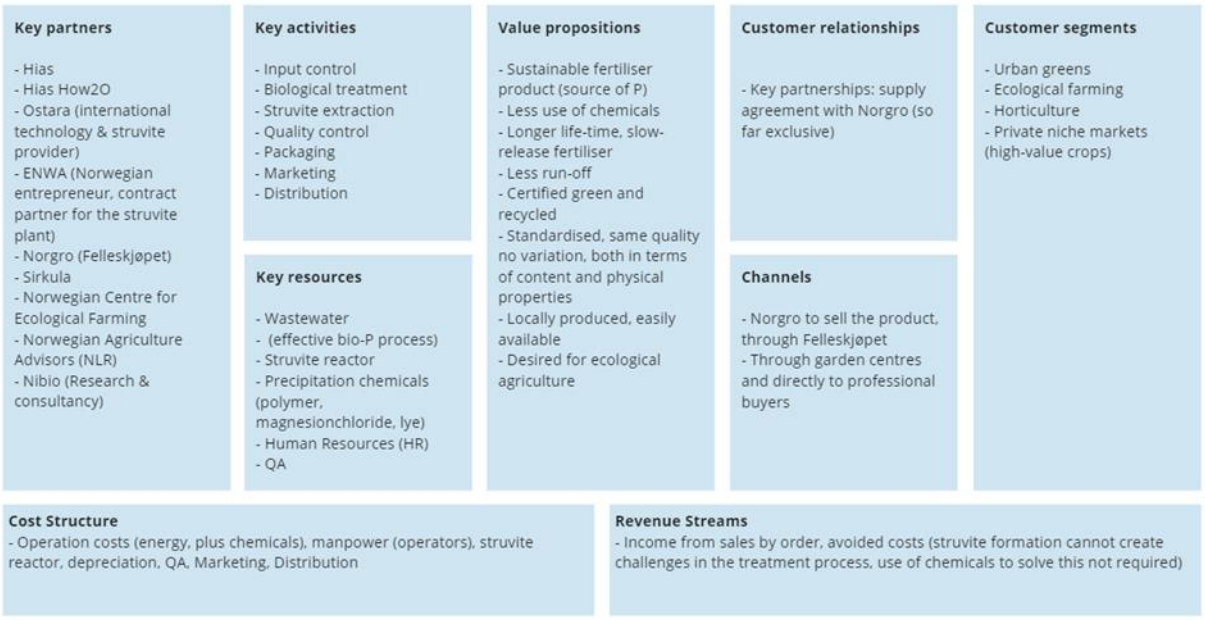


Figure 15: Economic layer, TLBMC for struvite from Hias.

Considering **cost structure**, the main **operational expenditures (OpEx)** are for energy and chemicals. Currently biogas is produced in the sludge treatment, but not used as energy source for the aeration of the biological phosphorus (Bio-P) recovery. However, Hias will soon install equipment to use the biogas as energy for its own processes.[9] The energy cost will therefore be relatively low and stable, as compared with hydropower-based electricity from the grid. However, the costs of chemicals have increased sharply during the last year – around 30% for magnesium and 300% for lye, which is noted as a challenge. The **capital expenditures (CapEx)** did not include any expansion of the wastewater treatment plant since space was available within the existing facility. Hence, the main element here, in addition to depreciation, is the acquisition of a Pearl K2 struvite reactor system from Ostara. The **revenues** will come from sales by order, and by March 2023 the approximate price for struvite was 10 NOK (or around 1 EUR) per kg. The most important element here is avoided cost, from not getting problems with struvite formation in the sludge treatment process, as struvite precipitation in anaerobic digesters is a known operational problem. During the initial workshop, this was roughly estimated to be around 2 million NOK per year. Also, the bio-P process preceding the struvite extraction leads to cost reductions due to significantly reduced consumption of phosphorus precipitation chemicals. A further consequence is less chemical sludge that is inert and takes up capacity during sludge treatment.

8.4.2 Environmental layer

The capacity of the biological treatment at Hias is 160 000 PE, and it is estimated that the production of struvite will reach 250 tons per year. The key **material**, as we have seen, is

[99] Up to now, there has been an agreement with the Nordic biogas provider Gasum, to provide liquid biogas (LBG) as fuel, amongst other for waste collection trucks at Sirkula.

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biologically treated wastewater, while supplies and outsourcing include the above-mentioned precipitation chemicals, energy, packaging materials and logistics. As shown in Figure 16, the **production** involves biological treatment of wastewater, harvesting and drying of struvite, and finally packaging. The environmental impacts from **distribution** will be linked to fossil fuels for trucks.

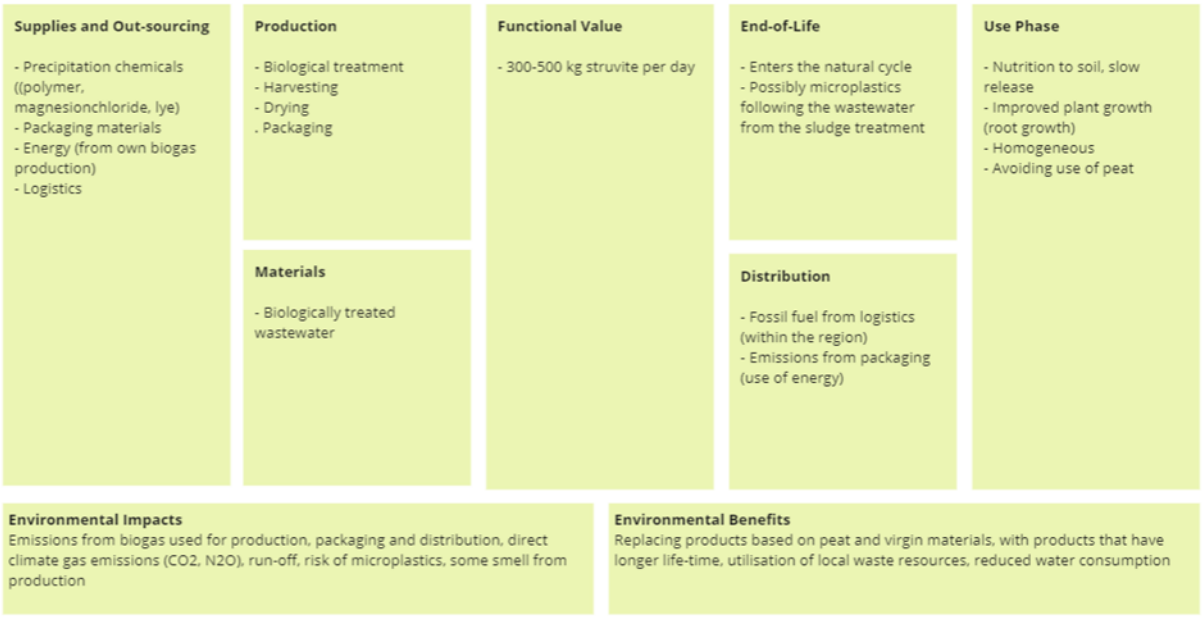


Figure 16: Environmental layer, TLBMC for struvite from Hias.

While the combustion of biogas produces CO_2, the carbon in this case comes from plant matter that fixed this carbon from atmospheric CO_2. Thus, biogas production is carbon-neutral and does not add to greenhouse gas emissions. However, the struvite production itself is associated with some emissions. Likewise, some climate gas emissions from distribution can be expected. When it comes to **use**, some run-off is anticipated, but since struvite is a slow-release fertiliser this may be less than for conventional products. Since treated wastewater may contain some micro-plastics one cannot rule out its presence but given that struvite is crystalline this would be very limited, if any.

The **environmental benefits** seem to outweigh the **negative impacts** by far. The product adds nutrients to the soil and helps improve plant growth. A study by the Norwegian Centre for Ecological Farming (NORSØK, 2019), documents the positive effects of struvite application on the P budget and ley yield, and supports struvite as input in organic farming systems. Struvite may also impact positively on soil structure, thereby leading to reduced water consumption. According to the above-mentioned estimate, struvite production at Hias may eventually replace 250 tons of phosphorus extracted from phosphate rock per year. In Norway, the traded volume of phosphorus from mineral fertilisers was 9000 tons, and 12 000 tons were available from livestock manure. Thus, what Hias can provide is a small, but important share.

The degree of material circularity in this case is very high, since the product itself consists only of recovered resources, the process produces no waste, and the energy used for production is circular as well. Moreover, a substantial amount of chemicals is saved from not having struvite

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precipitation in the wastewater treatment process. Before the bio-P process was installed, Hias used about 1440 tons precipitation chemicals per year, but in 2022 the figure was 530 tons, and the aim to reduce it to zero.

8.4.3 Social layer

The social impact layer of the BM for struvite, shown in Figure 17, has many parallels with that for soil products from Sirkula. So far, the benefits are difficult to specify in quantitative terms, but the **social value** includes value creation and employment effects, related to new soil production with Norgro and/or Sirkula. Norgro has forty-four employees at Degernes, where their soil and peat products currently are made.

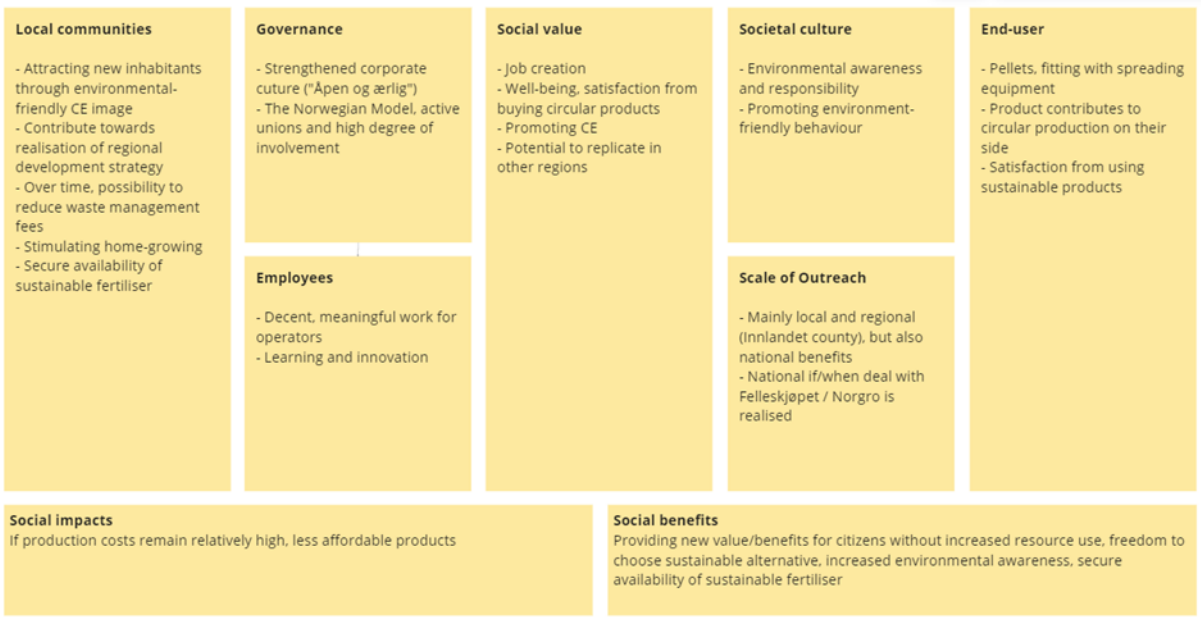


Figure 17: Social layer, TLBMC for struvite from Hias.

A sense of well-being at contributing to sustainability transition must also be counted in. Furthermore, the struvite production at Hias is the first of its kind in Norway and the BM has benefits in terms of knowledge development and potential for replication in other regions. The struvite production is less visible to the ordinary citizen than the soil production, but the innovation has received due attention in the media. Also, if/when a larger soil factory is realised, the **scale of outreach** would be along the same lines as for the soil improvement products. However, in the case of struvite, target **end-users** initially are not citizens, but professional users and producers of soil and fertiliser products who may benefit from a more environment-friendly image. As the struvite is homogenous and may be provided as pellets in different sizes, it fits well with existing spreading equipment.

When it comes to health and safety, the EU defines struvite as a product that is safe to use and handle. While it has been observed that pharmaceuticals may be adsorbed to sorbent-amended struvite fertilisers recovered from urine and passed on to plants, the risk to human health from eating the crops is seen as insignificant (de Boer et al., 2018), and the concentration of pharmaceuticals in biologically treated wastewater is likely way less than in urine. For **employees** and in terms of **governance** the social impacts of the BM for struvite parallel those

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identified for the soil improvement, except the struvite production itself is not connected with inclusive workplaces. For the **local community**, the new BM bolsters Innlandet' as a bioeconomy stronghold. It may thus also increase the attractiveness of Hamar and the other owner municipalities as places to live and work. The BM also helps ensure availability of sustainable soil improvement products and fertilisers locally.

8.4.4 Alignment across the TLBMC layers

The BM for the struvite is associated with substantial environmental and social benefits. At present, it is not lucrative from a purely business point of view, but the avoided cost related to eliminating problems with struvite precipitation in the sludge treatment process is a substantial saving for Hias. Wastewater treatment fees based on the cost price principle cover the wastewater treatment, which should be based on best available technology, and the demand for struvite worldwide is expected to increase.

8.5 Maturity, dependencies, and uncertainties

The BM for struvite is related to a full-scale struvite facility that entered into operation in 2023. The necessary agreement between Hias and Norgro is in place, and the partners are currently testing the struvite in different soil and fertiliser products. The plan is to spend a few months on this, and then start commercial production targeting selected products and customer groups. The full-scale implementation was preceded by pilot production and research and development activities documenting product properties and compliance with the relevant regulations. So far, there seems to be a high degree of agreement between the partners, although there is a potential tension between having an exclusive agreement with Norgro and the desire to further explore struvite as an additive in the local soil production at Sirkula.

In terms of **resource dependency**, the production at Hias is limited by the volumes of treated wastewater. Struvite can also be produced from other sources, but this is not within the mandate of Hias. In future, the demand for biogas for other uses may increase. Then, renewable electricity will be the alternative. Historically this has been cheap, but future energy demand in Norway is expected to increase and prices are associated with uncertainty. A more immediate concern is the recent increase in cost of precipitation chemicals, which at the outset is seen as a main bottleneck of struvite production (Zin and Kim, 2021).

Interaction with other business models: When it comes to other BMs, we will find some mutual dependency between the partners, once a new and larger soil factory is established. Organic material may be drawn from multiple sources, but transportation costs and distance will be a limitation. Moreover, Hias and Sirkula will need Norgro and/or Nordic Garden and their existing networks for distribution and sales. Competition from other water resource recovery facilities is not likely to be problem since the volume of struvite from each plant will be limited. Implementation by other wastewater companies may rather increase the potential for sale of struvite as a fertiliser on its own and eventual collaboration with conventional fertiliser companies.

When it comes to **social acceptance**, the feedback from initial users is positive, and as noted ecological farmers in Norway have high expectations when it comes to struvite. While the use

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of phosphorus in Norwegian agriculture has been rather stable in recent years, demand could increase with increasing focus on local food security. Concern with climate change may lead to less livestock and more vegetable production, with the implication that less phosphorus will be available from manure. Still, cost tends to be a concern for customers. While the price for struvite is low, as compared to the production costs, the approximate price noted by the local partners is 20-40% higher than the price of phosphate on the international market, which was 7-8 NOK per kg in 2022. At the same time, most of the phosphate comes from China and Morocco,[10] and the price is expected to increase with increasing scarcity.

The BM for struvite has **political support**, e.g., in the regional and national strategies for circular bioeconomy. In the beginning of 2023, the Norwegian government launched a new programme (Bionova) that offers support for sustainable bioeconomy innovations and can be of relevance. On the other hand, a revision of the national fertiliser regulation has been prolonged over several years and is creating uncertainty about the future framework conditions. The proposal for a revised Urban Wastewater Treatment Directive in the EU will have huge implications for Norwegian water utilities, in terms of requiring new investments in secondary and tertiary treatment. While some are more focused on how to meet or seek exemption from the proposed requirements given the geographical conditions in Norway, this could also stimulate more nutrient recovery and synergies in terms of pooling knowledge and resources.

[10] https://www.statista.com/statistics/681617/phosphate-rock-production-by-country/ [27.04.2023]

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9 BM exploration in the Stavanger case

9.1 Solution in focus

The focal BM in the Stavanger case is related to the production and sales of the organic fertilizer called Minorga. Minorga represents a portfolio of mineral organic fertilizers developed for use as fertilizers in Norwegian cereal production. This fertilizer is strengthened by adding nitrogen (Urea) and potassium (KCl) and provides plant nutrition by contributing to soil amendment composed of the optimal amount of nitrogen and phosphorus. Today the fertilisers (NOR) have turned more organic, adjusted to the Vietnamese markets focusing on organic matter.

9.2 BM for organic fertilizer

Minorga is based on wastewater sludge and biomass from aquaculture side streams and produced by IVAR, while Terramarine is responsible for the design, marketing, and sale of the fertilizer on the international market, mainly Vietnam.

9.2.1 Economic layer

The main **value proposition** is that the product contributes to soil amendment, where key features include a high level of dissolubility and the optimal amount of nitrogen and phosphorus. Other value propositions, shown in Figure 18, are linked to the high portion of organic material and ability to design different versions of the product for specific purposes and conditions. Moreover, the product is delivered with a consistent quality and secure supply. The key partners further highlight that the product enhances nutrivity and plant growth and limits the prevalence of plant diseases.

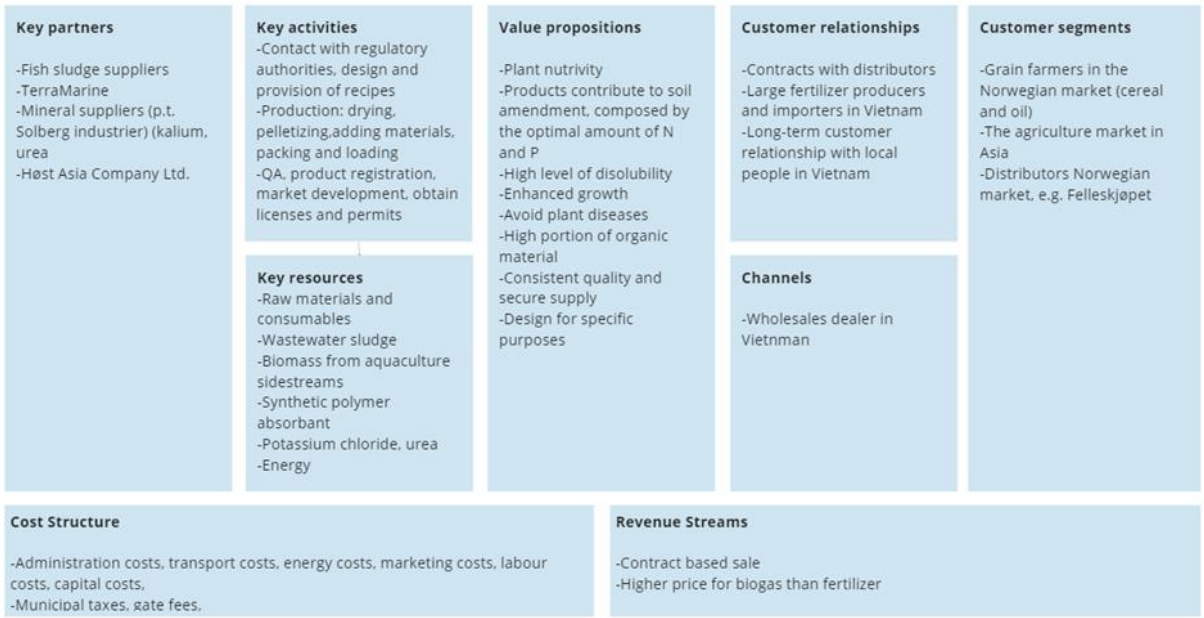


Figure 18: Economic layer, TLBMC organic fertiliser from IVAR and Terramarine.

Key partners besides IVAR and Terramarine are Bioretur, which are fish sludge suppliers, Solberg Industrier, a mineral supplier of kalium and urea, and Høst Ltd., the local wholesales dealer in Vietnam, which Terramarine is a shareholder in. In terms of intellectual property rights

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(IPR), there are two trademarks for Minorga, one issued to Norsk Jordforbedring (a predecessor of Grønn Vekst and Terramarine), in 2009, and one to IVAR, in 2013. Currently, the main **customers** are the agriculture market in Vietnam, where the product is sold to large fertilizer providers. The marketing, sales and field tests related to registration take place through Høst Ltd. Vietnam. This **customer relationships** are based on a long-term contact with the wholesaler and long-term contracts with local actors in Vietnam, which are for specified time periods where the buyer is obliged to buy all the fertilisers they sell from the same supplier. Another target customer segment is grain farmers in the south-eastern part of Norway. On the Norwegian market, Minorga has previously been distributed through contracts with Felleskjøpet, as the main retailer for agriculture in Norway. Using the product on green fields is also considered as an option in Norway. That the product is legal and approved for sale in Norway is a precondition for the Free Sale Certificate (FSC) issued by the Norwegian Food Safety Authority, which is required for the export of organic products, e.g., by most Asian countries.

Key activities, as listed in Figure 18, involve contact with regulatory authorities, first and foremost The Norwegian Food Safety Authority. The next step in the process involves the design and provision of recipes for the product. The production of organic fertilizer includes four main stages: 1) drying, 2) adding materials, 3) pelletizing, 4) packing and loading. The product is then checked through quality assurance and product registration before further market development and licenses and permits are obtained. In terms of key resources, a distinction has been made between raw materials and consumables and infrastructure/hardware. The **key raw materials** and consumables are energy, wastewater sludge, biomass from aquaculture side streams, synthetic polymer absorbent. For the fertiliser intended for the Norwegian market, potassium chloride, and urea was added, but these components are not included for the Vietnamese market. The key infrastructure and hardware include laboratory facilities, ERP business systems, a warehouse, and the fertilizer plant at IVAR.

The **cost structure** includes **operation expenditures (OpEx)** in terms of administration costs, transportation costs, energy costs, marketing costs, labour costs. In addition, municipal taxes, and gate fees. **Capital expenditures (CapEx)** are difficult to specify, in this as in the other cases, but it can be noted that the fertiliser production facility established at IVAR originally had a cost of 55 million NOK, of which 5 million NOK were provided through a public grant scheme (Innovation Norway). The **revenues** are from sale of the product, where the price is determined based on the proportion of organic material in the end product: High organic matter, in this case a proper C/N ratio, results in higher prices. Another factor driving the price is immediate solubilization in the water phase, which is a much-desired property.

9.2.2 Environmental layer

For the environmental layer, shown in Figure 19, numbers are based on input and output volumes for Minorga/NOR in 2021. The **functional value** of pellets produced that year was 4400 tonnes. Key materials are wastewater sludge, sludge from aquaculture, digestate, and other organic waste. For Minorga/NOR in 2021, the **materials** used to produce pellets originate from 13748 tonnes of digestate. After drying and adding of 400 tonnes of dried fish sludge this produces 4400 tonnes of bio-pellets. The fish sludge is surplus biomass from closed production of salmon fingerlings and smolt, which are naturally rich in nitrogen and phosphorus, and has

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a modest oil fraction believed to ease the pelletizing process and increase the solubility of the final pellets.

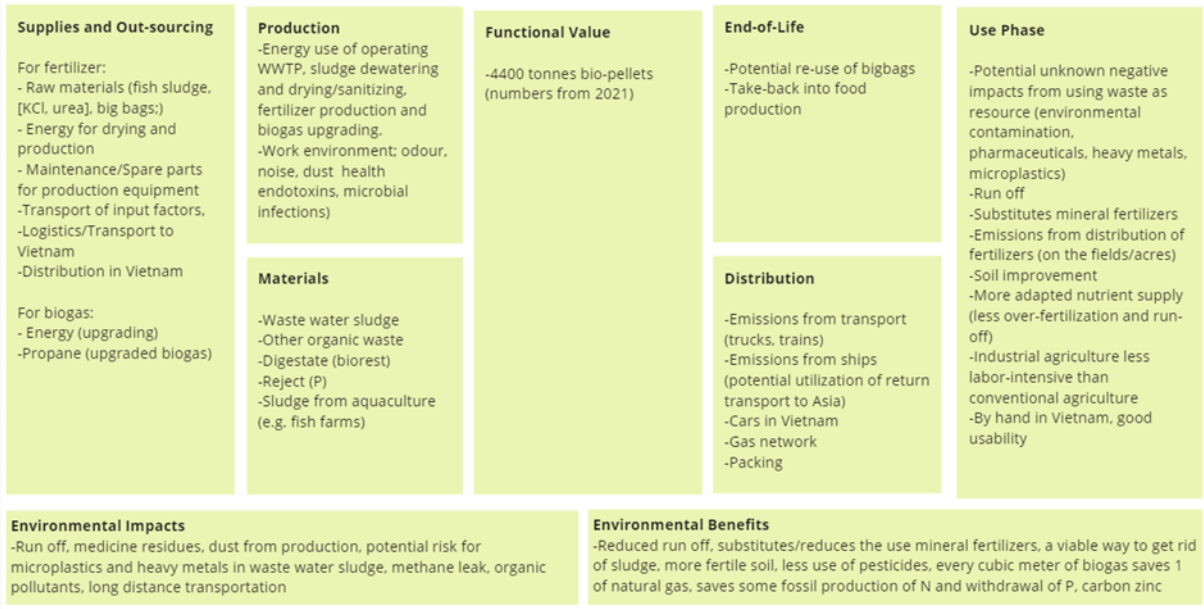


Figure 19: Environmental layer, TLBMC for organic fertiliser from IVAR and Terramarine.

The activities related to the environmental impact of the **production** are energy use related to operating the wastewater treatment plant and key activities involving sludge dewatering and drying/sanitizing, fertilizer production, and biogas upgrading.

Further activities include the transport of input factors, as well as the logistics involved in transporting and distributing products to Vietnam. With regards to emissions from the **distribution** of the organic fertilizer, emissions are produced from transportation by trucks, trains, and ships. The utilization of return transport related to the shipping of consumer goods from Asia to Europe limits costs and lowers the environmental impact of the long-distance transport leg. In Vietnam, cars are used to distribute further to farmers.

The **use phase** provides soil improvement, with more adapted nutrient supply and positive effect on growth and plant health. Run-off is reduced, as compared with conventional fertilizers. On the negative side there may be emissions from fuels used for distribution and potential unknown impacts from micro-pollutants. Considering the **end-of-life**, the product itself returns back into the natural cycle. Big bags used in distribution can be reused or in principle be recycled, but this is not an available option in this case now. The **main negative and positive impacts** are summarised in the bottom cells of the canvas (Figure 19). IVAR and Terramarine further emphasize the global perspective: Globally, the loss of soil is formidable and soil fatigue threatens efficient utilization of nutrients. The use of high-quality organic fertilizers is one of many tools for turning this development. Utilizing sludge from aquaculture provides a viable way of valorising this resource, while creating regenerative positive ecological value.

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9.2.3 Social layer

The social layer for the BM for the organic fertilizer, as shown in Figure 20, entails several **social values**. Firstly, the product is seen to create increased environmental awareness in the local community, and this work has been acknowledged with a sustainability award. In terms of jobs, three man-years at IVAR, three man-years at Terramarine and four people in Vietnam are currently working to develop, market and deliver the product. In addition, the BM is reducing the cost of waste management for society, by making business and creating value out of resources previously classified as waste.

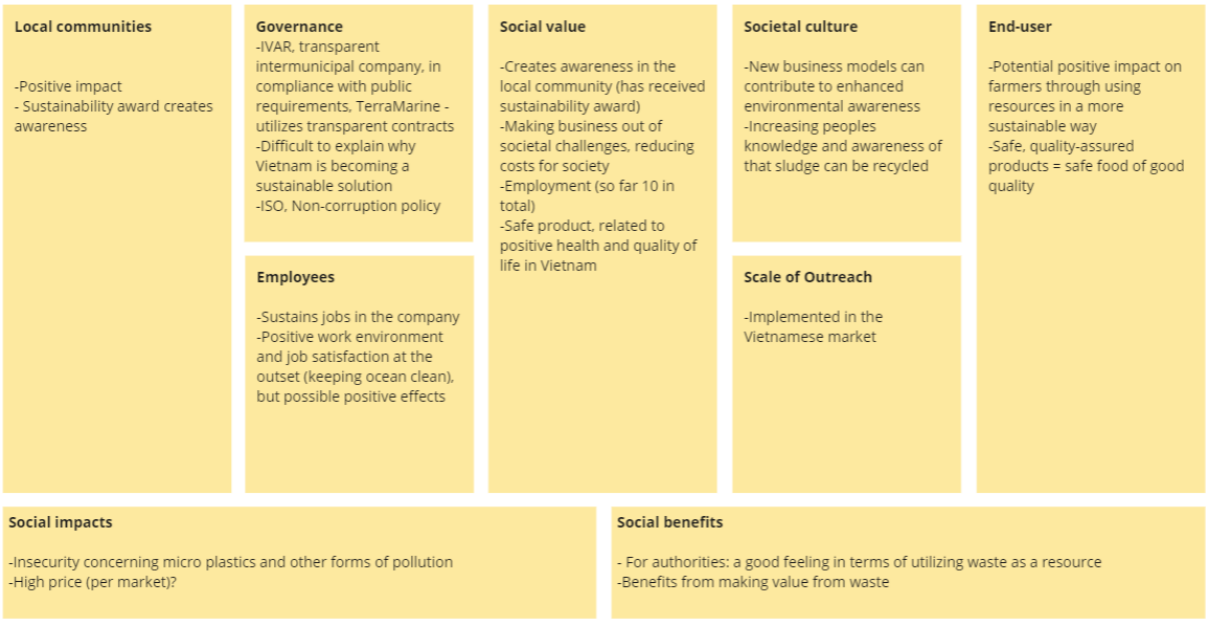


Figure 20: Social layer, TLBMC for organic fertiliser from IVAR and Terramarine.

In terms of impact on **employees**, the BM is associated with a positive working environment and high scores for job satisfaction. As regards **governance**, IVAR, as an intermunicipal company, is transparent as an organization and working in compliance with existing public requirements. Terramarine utilizes transparent contracts in their operations and is working according to a non-corruption policy.

In terms of **scale of outreach**, the BM relates to an international market, and sees further potential in regions with high need of phosphorus. The partners are also active and influential in terms of experience-sharing and knowledge development at the national and regional level, and the BM helps generate value also from other waste streams, e.g., aquaculture and fishery, as well as wastewater. Overall, the organic fertilizer contributes to a range of **social benefits** by making value from waste as a resource, which may raise awareness and positive impacts for both users, public authorities, the companies, and other stakeholders. It is noted, however, insecurity concerning micro plastics and other forms of pollution, which needs to be addressed in further work.

9.2.4 Alignment across the TLBMC layers

As shown in the TLBMC canvases, the BM is contributing to both economic, environmental, and social benefits for public and private actors involved, and for society in general. The product

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can be customized to a wide range of purposes and conditions, which also contributes to alignment between the three layers. At this point, the application is more uncertain in the Norwegian market. It is legally approved, but the more recent versions are tailored to the conditions in Vietnam, where zinc is needed for soil amendment, and have challenges meeting current Norwegian requirements concerning levels of zinc. Thus, the BM is more aligned with the Asian market at present. Further development for the Norwegian market would likely increase the environmental benefits, as more local value chains would lead to lower transport emissions.

9.3 Maturity, dependencies, and uncertainties

The technological maturity of the solution is high. The design is continuously developed, and a stable market has been found in Asia where fertilisers are needed throughout the year. In terms of **infrastructure dependencies**, the current production is limited by the capacity of the production unit at IVAR. Sludge from IVAR is also the main feedstock. However, Terramarine is experimenting with different designs and feedstock mixes, e.g, from aquaculture, thus reducing the **resource dependency** on one specific type of biomass.

When it comes to **dependency on existing value chains**, the BM at present is highly dependent on Terramarine's getting free biomass and cheap production services from IVAR.

In terms of **dependency on other business models**, there is a symbiotic relationship between IVAR and Terramarine, although it may be argued that IVAR has a dominant role, controlling the main resource and production assets. At the same time, there are potential synergies linked to waste streams from other sectors (e.g., aquaculture, poultry). In relation to conventional fertiliser producers, there is a competitive relationship in Vietnam, potentially also in other regions where high amounts of phosphorous are needed, and as the price of conventional fertilisers are increasing. As noted in the preceding chapter, there can also be a certain mutualism between wastewater and waste management utilities, in terms of promoting sludge-based products.

As regards **social acceptance**, Terramarine (then Grønn Vekst) did a willingness to pay survey among 400 Norwegian farmers, before WIDER UPTAKE.[11] Here, 68% said they would be willing to test a new organic fertilizer. Only 33% were willing, if the fertilizer was based on sludge, but if the price was lower than for conventional fertilizers, 76% responded that they were positive also to sludge-based fertilisers. Measuring of heavy metals and field experiments to address these challenges are ongoing. Current results indicate that Minorga performs at the same level as full fertilizers from Yara (same yield), and that it works better than calcium nitrate as maintenance fertilizer. However, the full-scale trials in the first phase also identified some challenges in terms of compatibility with the spreading equipment used by Norwegian farmers. As double dosages were needed, the product also required more labour and prices made it hard to penetrate the cereal market in Norway. Thus, Vietnam has so far remained the main market.

[11] In Norwegian. Presented at local CoP workshop in Stavanger, 06.10.2020.

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Dependency on policy and regulations. Initially, the product faced regulatory barriers in Norway, but after several years of lobbying, the Norwegian legislation was amended, opening up for using Minorga as a fertilizer commodity. While organic fertilisers are encouraged, EU's Fertiliser Regulation does not allow organic fertilisers based on sewage sludge. Norway has more liberal legislation, but the national Fertiliser Regulation has been under revision for more than ten years. While circular bioeconomy is strongly encouraged, this prolonged legislative process is creating some uncertainty about the future framework conditions for organic fertilisers in Norway. Another uncertainty, highlighted by Terramarine, is related to being a supplier of organic material, which is dependent on a product with the right proportion between carbon and nitrogen. With respect to this, receiving sludge from IVAR, where this composition is well documented, is a strong advantage.

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10 BM exploration in the Dutch case

10.1 BM for biocomposite riverbank protection from NPSP

The demonstration case in Amsterdam is focused on the development and production of biocomposites, e.g., for application in canal riverbank protection structures as a substitute of tropical hardwood. The biocomposite is made with raw materials recovered from wastewater treatment, from the drinking water production process and of biomass from surface water management by the water board. The calcite used is a residual from the softening process in drinking water treatment and used as a filler in the biocomposite. Cellulose is recovered from wastewater treatment (recovered and upcycled toilet paper) and used as a fibre in the biocomposites, just as water reed and dike grass from the water board.

Waternet provides calcite recovered from drinking water treatment and cellulose in the form of gras or reed for the demonstration and production of biocomposites in the WU project. Cellulose from toilet paper is currently sourced from the start-up company Recell. Calcite recovery is a common process in drinking water treatment in the Netherlands, as the water is characterised by its hardness, and is recovered by Waternet before it is sent to the Calcite Factory where it is treated before it is sold to users, such as NPSP, by Aquaminerals (a resource intermediary). The calcite is also used in other industries, such as food and cosmetics.

NPSP has used the material for several application areas; e.g., road signs, façade panels, and interior design products. For the work done on BMs in WU we conducted two workshops. The initial workshop was held in Amsterdam, while the follow-up workshop where findings from the first workshops were verified and discussed was done online using Teams. Ahead of the first workshop, in agreement with the partners from NPSP and Waternet, we decided to focus on the BM for riverbank protection panels made from biocomposites. As the project has developed and NPSP has continued to develop their biocomposite material, they now have four different types of biocomposites which have a varying percentage of biomaterial input, ranging from only water reeds in the Nabasco 8010 (Nabasco being the name of the material produced by NPSP, and an acronym for Nature Based Composites) to the 100% biobased Nabasco 8040 (with grass fibres and with resin and filler both based on agricultural waste, e.g. a furan resin made from sugarcane bagasse).

10.1.1 Economic layer

As illustrated in Figure 21, several **value propositions** are identified. The main one is that biocomposites is a sustainable alternative to tropical hardwood and other plastic alternatives. Second, NPSP highlights the potential for using locally recovered resources (from drinking water and wastewater, and biomass from surface water management) in the production of riverbank protection that are used in the same area.

The **customers** for the riverbank protection panels are mainly water boards (in the Netherlands there are twenty-one water boards (authorities) that are responsible for surface water management), municipalities and private actors (i.e., individuals). These customers are mostly reached through NPSP's contact with contractors who are in charge of installing and maintaining riverbank protections. As such, the main **channels** are contractors and the

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wholesalers who provide these contractors, while they also see potential for selling the product through distributors. In the second workshop on BMs, NPSP argued that they have to target water boards as end-users to promote their biocomposite solutions. The reasons for this being that to the **end-users**, water boards are front runners in terms of prescribing environmentally friendly solutions in their tenders, and NPSP has to convince contractors and distributors to include the biocomposite riverbank protection solution in their proposals to tenders. In terms of reaching the municipalities, NPSP foresee that they will largely work through resellers.

Regarding **IPR,** the intellectual property rights of the different Nabasco materials are owned by NPSP

The **key activities**, as listed in Figure 21, include product development, material development, R&D, licencing, and engineering. The **key resources** are raw materials used in biocomposite production, engineering and consultancy, and the **key personnel** working in NPSP. The key partners for NPSP are raw material providers, such as Waternet (calcite), cellulose (Recell – toilet paper, Waternet - gras and water reeds). Furthermore, contractors who install and maintain riverbank protection, distributors of materials, and manufacturers who manufacture end-products from biocomposites constitute **important partners** for NPSP.

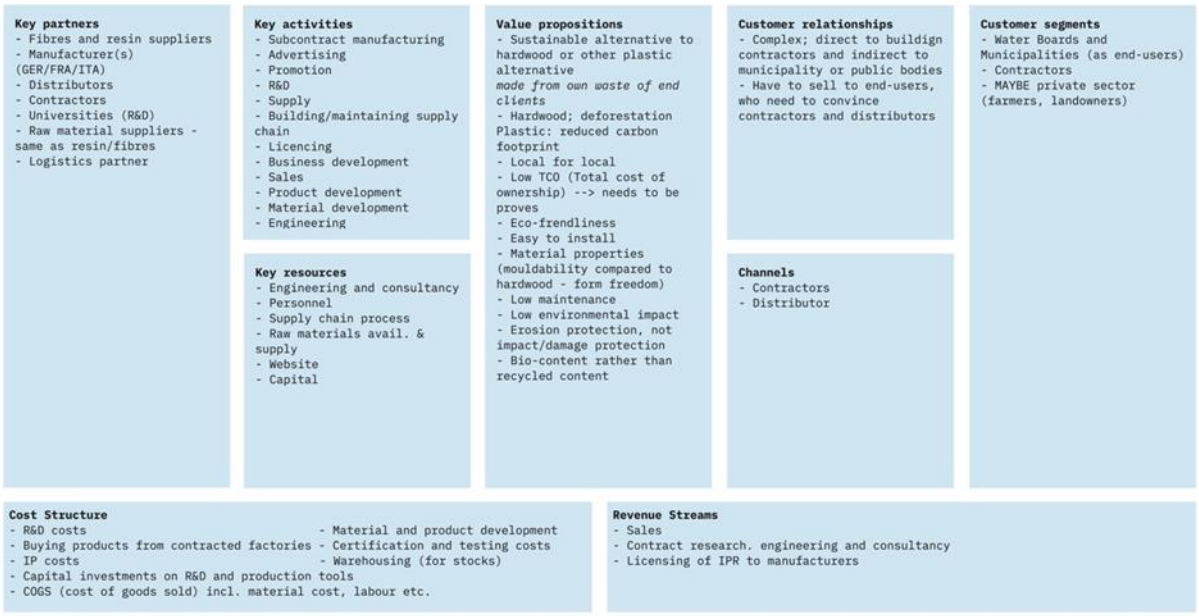


Figure 21: Economic layer, TLBMC for biocomposite for riverbank protection, NPSP.

The **cost structure** includes R&D and IP costs, costs for certification and testing, costs related to storage infrastructure such as warehouse. Additionally, the cost structure includes costs of goods sold (COGS), which include the cost of materials (ca. 35%), labour (ca. 50%), energy (ca. 10%) and machinery (ca. 5%). Based on NPSP's preliminary calculations, **the total cost** of the biocomposite solutions is up to 50% higher than that of the conventional hardwood solution. Specific information on **OpEx** and **CapEx** is not possible to render in this report as it is not readily available due to uncertainties regarding the BM. In addition, NPSP as a commercial actor, does not want to share estimates in public reports, due to competitive considerations. The **revenue stream** for biocomposite riverbank protection panels comes from direct sales to contractors.

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10.1.2 Environmental layer

As shown in Figure 22, **the functional value** in this case is estimated to be 129km of riverbank protection panels per year. This is a rough estimate for a full substitute of wooden bank protection provided by NPSP based on the calculations of average replacement rate and number of kilometres replaced per year by four water boards, then extrapolated for all water boards (a total of 21). However, this depends on production capacity of a third-party manufacturer and market uptake. For the Dutch case, a LCA analysis has been conducted for the different types of Nabasco material qualities. The functional unit used in the LCA analysis is 1kg of biocomposite material.

The **environmental impacts** of the riverbank protection panels are associated with **production**, **transport** of end-products by trucks, and potential leaching of materials into the waterbodies where the riverbank protection panels are installed. The leaching potential is to be further evaluated in WP2 in the WU project is currently not known. The calculated energy consumption and global warming potential (kg CO_2 equivalents emitted into the air) per kg of material varies depending on the composition of the material, i.e., percentage of biological input material and raw material source. WP3 of WIDER UPTAKE has provided preliminary results on this, summarised in Table 9.[12]

Table 9: Material composition, global warming potential, and energy consumption for different Nabasco materials (adapted from MS15 report)

Nabasco	Fiber	Filler	Resin	CO2 eq./ kg	kWh/kg
8010	Water reed	Mined calcite	Unsaturated polyester	1.83	9.52
8012	Water reed	50% Recovered + 50% mined calcite	50% bio-unsaturated polyester	1.64	8.90
8020	Recovered cellulose	50% Recovered + 50% mined calcite	50% bio-unsaturated polyester	1.67	9.04
8040	Grass	Bio-filler from agricultural waste	Polyfurfuryl alcohol	0.62	15.60

In the **production**, there are several elements that influence the environmental impact of the BM; chemical waste from cleaning of production tools, dust from cutting of materials at installation sites, treatment of raw materials (e.g., related to hygiene), and energy consumption (which has been accounted for above, in Table 9. According to NPSP, at the end-of-life, 100% of the Nabasco panes can be ground and reused as filler in new biocomposite materials and can replace 30–50% of the chalk that is being used as filler.

[12] For further detail, see MS15: Adrian Werner, Anurag Bhambhani & Zoran Kapelan (2023). Application of the circularity/efficiency assessment framework – key results

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The **environmental benefits** are closely connected to the replacement of tropical hardwood in the sense that riverbank protection panels made from biocomposites can be used as a substitute for the conventional tropical hardwood. This can contribute towards reducing deforestation in the areas where tropical hardwood is sourced from. That being said, the LCA analysis shows that 1kg of tropical hardwood has a lower global warming potential (0.31 CO2 per kilo material) than 1kg of biocomposite material. However, by using biocomposites for riverbank protection you reduce the number of kilos of material needed compared to that of hardwood. That means that you can use 3-4 kgs of biocomposites to replace 10 kgs of tropical hardwood, which in sum makes the carbon footprint of biocomposite riverbank protection lower than that of those made from tropical hardwood.



Figure 22: Environmental layer, TLBMC for biocomposite for riverbank protection, NPSP.

10.1.3 Social layer

As illustrated by Figure 23, biocomposite production based on recovered resources from surface water management, drinking water and wastewater treatment is associated with relatively few social benefits. For the **end-users**, NPSP see that the use of biocomposite materials can provide a good narrative on sustainability for the water boards and municipalities that replace hardwood with biocomposite riverbank protection. **Building awareness** through sensitization is also mentioned as a potential positive contribution from attention around the utilization of biocomposite protection panels in different localities. Additionally, enhancing young people's knowledge on the potential benefits and use of biocomposites through internships at NPSP can help spread the knowledge further and among the younger generation.

Although there is no immediate increase in number of jobs related to the BM, NPSP believe that the **employees** who work with biocomposite production can experience more meaning as they contribute to making products from what has been considered waste. As seen in Figure 23, the **social benefits** of the BM on riverbank protection from biocomposites can be enhanced aesthetics for the N8040, which is dark brown, in the local environment (compared to hardwood, which turns grey), the protection panels enable better water drainage which in turn

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can increase the value and usage potential for private persons' lands, and the creation of valuable products from waste can inspire future generations.



Figure 23: Social layer, TLBMC for biocomposite for riverbank protection, NPSP.

10.1.4 Alignment across the TLBMC layers

As we have seen, there are potential environmental and economic benefits to the BM for riverbank protection made from biocomposites. Yet, there seems to be a lack of alignment between the potential environmental benefits and potential economic benefits proposed. The environmental benefits presuppose more market uptake to be economically sustainable. Pending testing in aquatic environments, the higher price on biocomposites could be compensated for by a longer life span compared to that of hardwood, which could contribute to aligning the economic and environmental aspects of the BM. There is also a misalignment between the environmental and social layer, as there are limited social benefits compared to potential environmental benefits. In sum, the BM is currently more focused on environmental sustainability, whereas the economic and social sustainability aspects seem to be less prominent.

10.2 Maturity, dependencies, and uncertainties

NPSP's BM on biocomposite riverbank protection panels is subject to **external pressure and dependencies** that can both promote and hinder the success of the BM. A general condition that NPSP regards as a potential driver is an increased interest in circular economy and circular products among water boards. This is in line with the water boards' ambitions as represented in the Wastewater management roadmap towards 2030 where recovery of resources is one of three pillars.

The BM is based on a new, less mature material, where the material composition is in need for testing in the types of wet environments where it is to be applied. One main issue concerning the BM is **market adoption and acceptance**. To this end, NPSP has been successful in getting

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funding from the Small Business Innovation Research Programme (SBIR)[13] to conduct a feasibility study to uncover the business potential of the biocomposite riverbank protection solution. As such, we see that the BM at the moment is rather immature and in need of further work, both regarding testing and verification of material qualities and market development, to be successful.

In terms of **dependency on existing infrastructure**, NPSP has invested in new production equipment enabling them to produce smaller (H)40x(W)60x(T)0,6 cm biocomposite tiles. The equipment is suitable for production of demonstration panes for the riverbank protection demonstration, but there is a need to establish agreements with manufacturers that have equipment to manufacture full scale products that are 120cm wide. As such, there is currently a lack of a manufacturer that can contribute to realizing the BM.

Resource dependency is also a relevant issue for the BM, as it depends on access to renewable raw materials such as water reeds and recovered toilet paper from wastewater treatment. Water reeds are in demand also for other applications such as roofing, where NPSP receives what is left after the production process. If water reeds would become more in demand for roofing production, NPSP could potentially face problems with supply of reeds. However, the different types of Nabasco materials depend on different types of fibre materials. As such, water reeds could be substituted with grass clippings, which currently is abundant. It could also be replaced by recovered toilet paper, although this is an abundant resource in wastewater, it requires technology investments in order to recover it, which at this point in time is done by relatively few actors, Recell being one of them.

NPSP also identify other potential risks that can influence the success of the BM. One of them being the **competition** their biocomposite product is in with other circular products. It is argued that the circular economy has received vast attention in society and NSPS therefore suspect that products that deemed circular will be more competitive in the market than bio-products, a trend they foresee that this trend will last at least until 2030. Second, NPSP is a small company with limited financial resources. If a larger actor with more resources were to move into the market and provide biocomposites or other biomaterial NPSP believe that they will be ousted, leading to **parasitism**. Third, although tropical hardwood is associated with negative impacts on the environment such as deforestation, loss of biodiversity, and the end-of-life solution being incineration for energy, the material is still competitive in terms of Environmental Cost Indicator (ECI), which is a single score indicator that provide one monetary number based on a range of environmental data points.[14] As ECI is being used in public procurement processes, and contractors are deemed to be more willing to work with known riverbank protection solutions (i.e. hardwood), the uptake of biocomposite riverbank protection as a solution is challenging, and in **competition** with tropical hardwood. Therefore, NPSP believe that scarcity of, or a price increase for, hardwood, e.g., due to political decisions to reduce the use of hardwood, could contribute to promote the biocomposite solution.

[13] https://business.gov.nl/subsidy/small-business-innovation-research/

[14] Ecochain: https://ecochain.com/knowledge/environmental-cost-indicator-eci/

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11 BM exploration in the Sicilian case

11.1 Solution in focus

In the Sicilian case, reuse of treated wastewater for irrigation, demonstrated in relation to the wastewater treatment plant at the town of Corleone was selected as a case for the work on BMs. The wastewater treatment plant has established an ultrafiltration plant. Even if this is not in operation for technical/administrative issues,[15] this is the most mature of the three solutions.

Over the last 20 years, 40 billion EUR have been invested in wastewater reuse facilities in Sicily. According to the local stakeholders, there are currently fourteen existing plants that could be put into operation. Four of these, including the one at Corleone, have been formally approved. However, none are yet in operation (data source: Department of Water and Waste of Sicily Region). Given the impending water challenges in the region, it is therefore crucial to address institutional barriers, and exploration and eventual identification of sustainable BMs may have a huge impact in terms of implementing and upscaling water reuse.

11.2 Business model for water reuse in Sicily

The facility for wastewater reuse in Corleone was established in 2003 through financing from the Sicilian Region (POR and POFERS funds). The facility is composed of an ultrafiltration system (UF) and a series of reservoirs. The facility is partially in place (i.e., pipeline and reservoirs (two tanks), but needs revamping since it has not been in operations for many years and lacks distribution network to fields/farmers.

11.2.1 Economic layer

As highlighted in Figure 24, the main **value proposition** linked to reuse of treated wastewater in Corleone is more water of good quality, to secure sustainable agriculture. Availability of reuse water will ensure stable water supply for irrigation, which also may result in improved existing crop yields and be convenient for the farmers in practice. By providing an alternative when other sources dry out, water reuse is also associated with increased economic stability, both for farmers and for the agro business/food processing industry in the area. Indeed, stable water supply will also give the farmers opportunity to variegate the cultivar of the same crop to produce high-quality products (e.g., DOP – "Di Origine Protetta" olive oil). For the municipality, reuse of wastewater may save potable water for domestic use. As 60% of the water budget currently is used for agriculture, this could increase the system resilience considerably. Availability of reuse water may also provide opportunity to take up agriculture on land previously abandoned.

[15] Which are partly solved and will be further addressed in WIDER UPTAKE.

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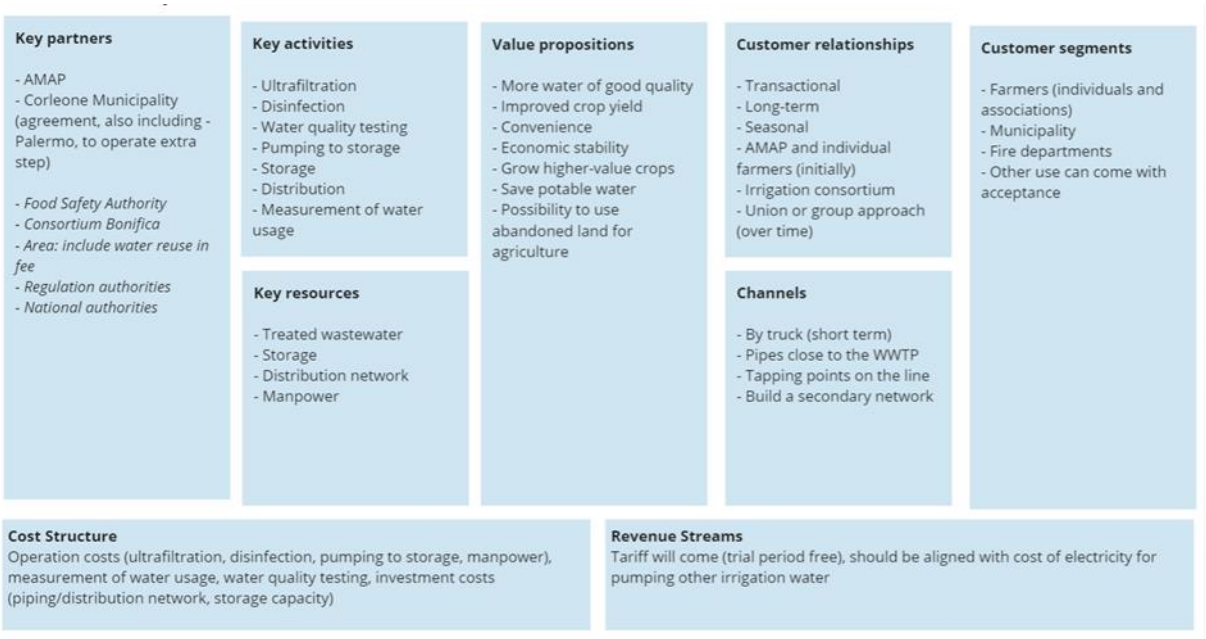


Figure 24: Economic layer, TLBMC for treated wastewater reuse, Corleone.

The target **customers** are mainly local farmers. Elsewhere in Italy, e.g., Lombardi region, strong farmers associations have been a key to successful implementation of water reuse. However, farmers near Corleone are currently not organised to the same degree, so individual farmers must be targeted, at least initially. The municipality itself is also seen as a customer, in that reuse water may be used for irrigating their public garden, and the regional fire department, which may need large volumes to extinguish wildfires which occur during dry periods, is another potential early adapter. With time and experience, other uses may also come into consideration. (e.g., street washing; industrial, etc.).

The **key partners** are AMAP and Corleone Municipality (as plant owner), where an agreement formally has been made to assign the ultrafiltration plant operation to AMAP. To create an institutional setting that is favourable, the regional Consortium of Reclamation (Consortium Bonifica) or irrigation board, and relevant authorities in Sicily are also considered as important partners. The **IPR** cannot be definitively identified, since the plant owner is the Municipality of Corleone, but the operator (both for the WWTP and UF plant) is AMAP. Indeed, in 2023 in view of boosting the water reuse process, and agreement between the Municipality of Corleone and AMAP was established to assign to AMAP the UF plant operation. The ideal **customer relationships** would be long-term, with a union or group agreement between AMAP and farmers, or with the current irrigation consortium, where individual farmers may draw water when needed. However, in an initial phase the relationships will be with individual farmers. During a first trial period, interested farmers will receive the water for free, sponsored by the municipality, to build experience and acceptance. In the shorter term, reuse water will be **distributed** by truck. In a medium to longer term, the key partners foresee a pipe distribution system, with different tapping points on the line, and eventually a whole secondary network. The medium/long perspectives depend on external funding, which is sought through two project applications that recently have been submitted.

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The **key resources**, apart from treated wastewater, are storage, distribution system and manpower. The **key activities**, as listed in Figure 24, are ultrafiltration, disinfection, water quality testing, pumping to storage, storage, distribution, and measurement of water usage.

The **cost structure** includes both **OpEx and** CapEx. The **OpEx** includes personnel, energy (for pumping), chemicals for disinfections and membrane chemical cleanings, and membrane maintenance, laboratory for water quality testing and monitoring of use, and equipment maintenance. Among these costs, the energy costs are relevant due to the need to heat the Clean In Place (CIP) Reactor and to the pressure behaviour of UF membrane. From previous estimation, the energy demand could achieve up to 1kWh/treated m^3 which could represent up to 40% of the **OpEx**. The water quantity testing will be done via flow rate devices for water usage measurement installed by Corleone Municipality. The **CapEx** includes: the investment costs related to distribution. Further, investments have to be done in future long the term applications, after the project ends. Specifically, the investments costs for recovering the existing artificial reservoirs and their connections with UF plant have to be taken into account. According to preliminary calculations made by TU Delft in WP3, economic efficiency for distribution by gravity to users in agriculture would be 5.79 m^2/EUR for the resource recovery stage, as compared to 8.63 m^2/EUR for the base case (when treated effluent is discharged into the river and groundwater is pumped for irrigation). For the use side, economic efficiency would be 0.76 m^2/EUR, compared to 9.81 m^2/EUR for the base case.[16] This would indicate that with the current cost of pumping groundwater, for the farmers reuse is not competitive from a purely economic point of view.

In a full-scale operation phase, the **revenues** will come from tariffs paid by water users. The tariff will depend on regulatory requirements and related costs (e.g., of water quality testing). Currently, national regulations place the additional costs (including operation and maintenance costs) related to reuse of wastewater treatment on citizens, while distribution and monitoring costs fall on the final users (Ventura et al., 2019).

11.2.2 Environmental layer

As shown in Figure 25, the **functional value** in this case could be 220 000 m^3 of treated wastewater per year (depending on suited reservoirs). The environmental impacts linked to **production** will mainly be climate gas emissions linked to energy use for the different steps: UV filtration, disinfection, water quality testing, pumping and storage.

[16] For details, see MS15: Adrian Werner, Anurag Bhambhani & Zoran Kapelan (2023). Application of the circularity/efficiency assessment framework – key results.

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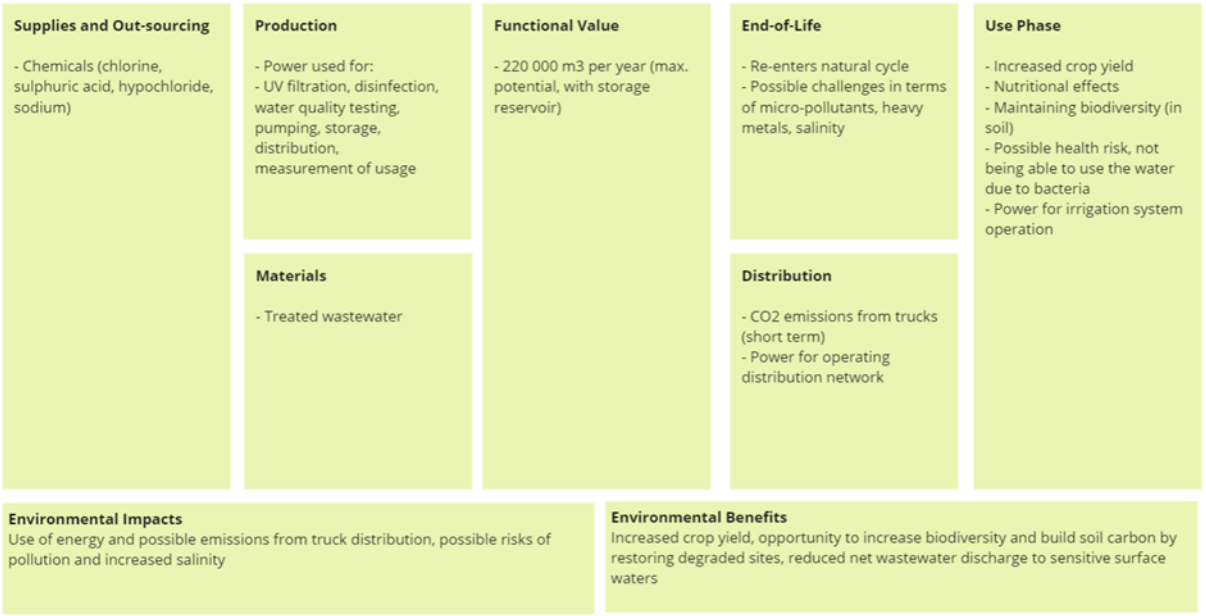


Figure 25: Environmental layer, TLBMC for reuse of treated wastewater, Corleone.

The key **material**, obviously, is the wastewater being treated. Currently, the plant at Corleone is treating about 3800 m^3 per day. The UF system has a maximum rate of 55-60 m^3/h, with two membrane modules under operation. Nevertheless, only one module could be reactivated, and the net flow rate is about 25 m^3/h. The current reservoir volume refers to two tanks, one close the wastewater treatment plant having a volume of 185 m^3 and another within the municipal garden having a volume of 25 m^3. In addition, chemicals for membrane maintenance (chemical cleaning) in the form of chlorine, sulphuric acid, hypochlorite, and sodium are brought in from other sources. These chemicals do not modify the water quality, and therefore no analysis of their impact has been made.

As noted above, trucks will be used for **distribution** initially. As long as these are fossil fuelled there will be CO_2 emissions. These have not been calculated in detail but may roughly be estimated to be around 105 g/km, for diesel trucks. In a medium to longer-term perspective, pipe distribution seems more feasible. The wastewater treatment plant is located on a steep hill, below the main settlement of Corleone but above a large, agricultural plain. Thus, the intention is to establish a distribution network, where gravity for a large part will be used to distribute water to farms in this area.

WP3, led by TU Delft, has made initial calculations of water circularity and nutrient circularity in the Sicilian case, comparing a base case with pumped ground water and use of treated wastewater for irrigating the public garden and/or distribution for agriculture under certain, specified conditions. These indicate that for the water recovery stage, water circularity will be 96%, and nutrient circularity will be around 70%. For the **use phase**, energy efficiency is significantly improved, especially for agricultural use, since pumping of ground water requires a lot of energy. Water and nutrient circularity for the alternatives involving agriculture were 23% and 65%, respectively, and slightly less for the garden.

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In addition to the water itself, the provision of nutrients may increase crop yield, and possibly reduce the need for fertilizers in some cases. Moreover, the organic content may influence microbial life and soil structure positively. The use phase is also associated with substantial energy consumption. WP3 found that energy efficiency increases manifold, when reuse for agriculture is compared with pumping groundwater. To assess the environmental impact of the BM as such, we must also consider the type and amount of energy used for the irrigation itself. Diesel engines are often used as energy source for irrigating (related emissions per litre 2.7 kgCO2eq/L) in this case irrigation requires 1.7 L of diesel/h. Under the hypothesis of considering only 28 days of irrigation (8 irrigation per month in 3 months) and 0.5 hrs/ha per irrigation the CO^2 emission can be quantified as 110 kgCO2eq/ha.

Thus, **overall environmental impacts** are linked to energy use and the potential for micro-pollutants and increased salinity. The latter have not been studied in detail in WIDER UPTAKE but are noted in other studies from Sicily (Ventura et al., 2019). On the positive side are increased crop yields, opportunity to increase biodiversity and build soil carbon, and reduced discharge to sensitive surface waters.

11.2.3 Social layer

The **social value** of this BM is related to the increased security of water supply for irrigation, which can help maintain and increase value creation and employment linked to agriculture. Thus, it may also limit or reverse land abandonment and improve the prospects for future generations. As noted in Figure 26, the intended implementation of water reuse is also likely to have positive effects on community cohesion and identity in Corleone, which is trying to improve the uniqueness of the Sicilian territory to contrast and manage the historical association with illegal activities. Being a frontrunner in CE will add a new dimension. Furthermore, water reuse is likely to increase environmental awareness and stimulate environmental behaviour in other areas, e.g., in terms of what is thrown into the sewage system, private householding with water, and recycling and reuse of other resources.

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Figure 26: Social layer, TLBMC for reuse of treated wastewater, Corleone.

For **end users** in agriculture, the model will increase water security, thereby also increasing farmers' economic security. Well-water may be saved for domestic use, and farm yields may improve. Since the use will be monitored, there will also be impacts in the form of increased knowledge, which could lead to more sustainable farming practices, and in turn may increase well-being and socio-economic status. Social impacts linked to end-use for public greens will be improved quality of life for citizens, in terms of local climate effect and aesthetics. On the negative side, previous trials in Sicily have shown contents of bacteria and pollutants below, but also at times above, required limits (Ventura et al., 2019). This underscores the need for careful monitoring.

While the **scale of outreach** will be local in the short and medium term, the BM can be an example for other towns in Sicily and thus have a much wider scale of outreach in the longer term. **Employees** may experience more meaningful work, and also benefit in terms of knowledge and capacity-building.

11.2.4 Alignment across the TLBMC layers

In this case, the economic layer has many elements clearly worked out, but the revenue stream is uncertain, and the economic efficiency of the solution, compared to present irrigation practice, is also not well documented yet. This is in contrast to the considerable environmental and social benefits that may ensue. To detail out the TLBMC further, more stakeholder dialogue is needed, and further documentation of the overall benefits for both community and end users, may be useful.

11.3 Maturity, dependencies, and uncertainties

Significant steps towards realisation of water reuse in Corleone have been taken. The most important are a) identification of technical issues and solutions limiting the re-starting of the UF system, b) signed agreement to clearly identify the UF operator, and c) identification of

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possible funds to realise/restore infrastructure. Specifically, AMAP Spa will re-start and operate the UF system. Corleone municipality will provide administrative and technical support for starting up managing the water distribution (coupled with the local farmers associations). All the key actors (Consorzio di Bonifica, GAL Terre Normanne, ATI) will provide their support in view to making the water distribution realistic.

In this case, there is a high **dependency on infrastructure development**: Long-term prospects and potentials are strictly dependent on securing the funds that are needed to restore the existing infrastructure (reservoirs) and/or realise network for water distribution. The BM is also **resource dependent**, in the sense that it depends on the quantity of quality of wastewater received for treatment.

Dependency on other value chains: The cost will be strongly influenced by the cost of treatment chemicals, and the demand will depend on how the local farming practices and framework conditions for agriculture in Sicily develop in the years to come, as well as on the energy costs associated with other sources of irrigation water.

Social acceptance: Farmers have a good awareness of the importance of having stable water availability for their crops and economy; therefore, they are also willing to collaborate to identify the best solution for their needs. Moreover, they are aware that the cost of current irrigation practices (often using diesel or electricity) will increase in the future and therefore they would be willing to pay water, but they require a quality certificate. Future demand, willingness/ability to pay will also depend on other framework conditions, such as the competition with other countries in terms of agricultural productivity and the Italian Country Agricultural incentive policies.

In terms of policies and regulations, a critical factor is the cost of the required water quality testing.

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12 BM exploration in the Ghana case

12.1 Solutions in focus

In Ghana, there are two areas of focus for the solutions: a) production of biochar from recovered sludge to replace wood-based charcoal, and b) water reuse for urban agriculture.

12.2 Business model for biochar from SSGL

Energy access and use in developing countries continue to remain low, and often to the detriment of its low-income earners. In Ghana, energy from wood sources has been used for cooking and heating at the domestic and industrial levels and wood fuel energy consumption consistently remained the largest household energy consumed according to the Energy Commission (2018). Thus, the biochar solution is intended to contribute to a reduction in the consumption of wood-based fuel.

12.2.1 Economic layer

The main **value proposition** in this case is the provision of a sustainable biofuel (biochar) as an alternative to wood charcoal from the forest. For the biochar to constitute a real alternative, competitive pricing is essential. Other key aspects are that the biochar is circular and local and will distributed regularly as demand grows. SSGL notes that the biochar can be made into different sizes, including extra small size, which could be suited other industrial use. Regarding **intellectual property rights (IPR)**, SSGL have not patented any aspect of the innovation as yet; although they have plans to acquire intellectual property rights on the biochar.

The targeted **customers** are micro, small, and medium-sized enterprises who wholly or partly use charcoal as an energy source. Initially there was a strong focus on food vendors and operators of small roadside cafes and restaurants, but the focus has gradually shifted to enterprises that do not use the charcoal directly for cooking, such as producers of cosmetics, detergents, and shea butter processors. Once all the formal documentation requirements are met, those preparing and selling food could also be targeted. **The customer relationships** are transactional. SSGL has an agreement with a wholesaler who already is a professional with a large network for distribution of regular charcoal. They do not foresee long-term agreements. In terms of **channels**, the noted wholesaler is using the same marketplaces and trucks for distribution as for regular charcoal, ensuring that customers find the biochar readily available. The product is also provided for sale in local supermarkets, where it is nicely packaged.

One possibility discussed is to promote the biochar together with more energy efficient charcoal stoves (where a small, electric fan makes the charcoal light and burn more efficiently). However, there are mixed views on the sustainability and popularity of such stoves, which are more expensive than regular ones. On the other hand, the testing done with CSIR's Institute of Industrial Research show that the biochar indeed burns more easily and efficiently with the energy-efficient version of the charcoal stove.

As listed in Figure 27, the **key activities** include wastewater treatment, sludge management, production of the biochar itself, certification, packaging, and distribution. Presently, the

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partners also see advocacy as a key activity, given that the use of biochar for energy in Ghana is neither addressed in energy transition policies nor directly regulated.

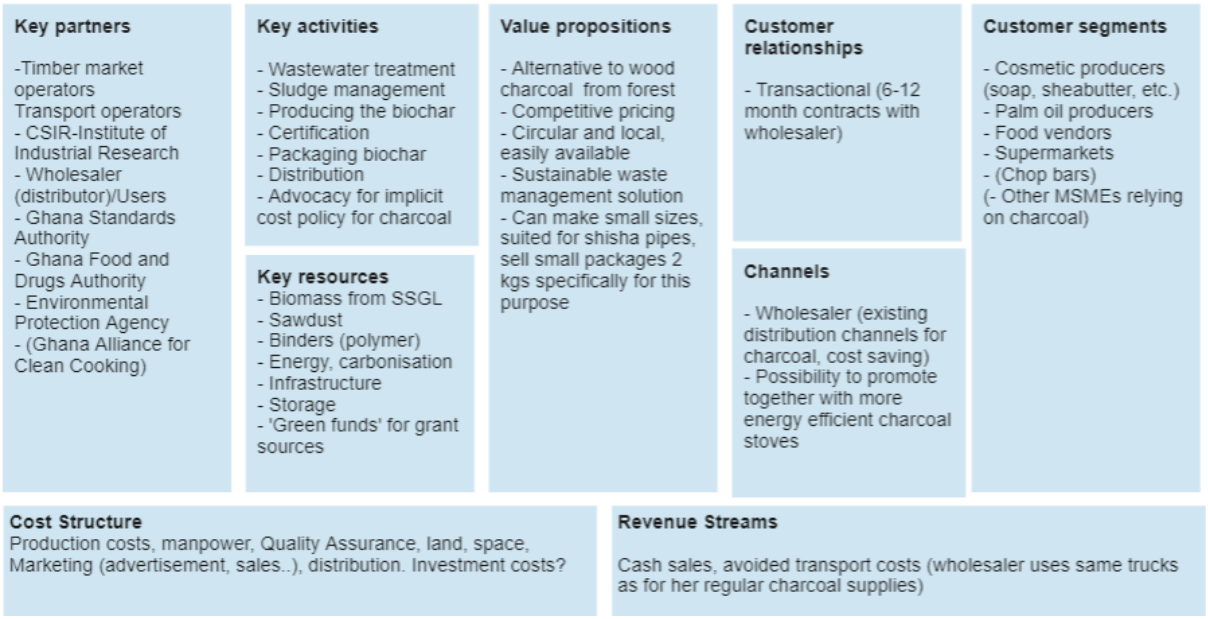


Figure 27: Economic layer, TLBMC for biochar from SSGL.

The key resources, apart from biomass from SSGL, include sawdust (2 parts sawdust to one part biomass), binders, and energy for the carbonisation of the biochar. In addition, infrastructure and storage space must be counted in. 'Green funds' to finance upscaling was also mentioned, although the biochar is quite a small activity for SSGL.

The key partners for the activity itself are SSGL, the wholesaler and CSIR-Industrial Research, which helps document, test and further develop the product. In addition, operators at the nearby timber market are crucial, considering the volume of sawdust that is included, and transport operators are also counted in. In terms of permitting, certification and advocacy, Ghana Food and Drugs Authority, the Environment Protection Agency and Ghana Standards Authority are also seen to play important roles. Ghana Alliance for Clean Cooking is also on the list, important if/when use for food production gets more central.

The cost structure includes the investment cost of the carbonisation unit, production costs (most notably supplies (sawdust, additives (polymer), energy and manpower), as well as advocacy and marketing. Thus, the cost structure may be broken into capital expenditure (CapEx) and operation expenditures (OpEx). CapEx includes the expenditure SSGL made to construct the biochar making shed, the procurement and installation of the biochar making plant that includes components such as a crusher, kiln, conveyor system, mixer, and a mould. The OpEx, on the other hand, emanate from paying labour (workers), cost of binder, transportation of saw dust to biochar plant, and electricity. In respect of revenue streams, SSGL expects revenue mainly from cash sales. In addition, avoided transport costs may be counted in, since SSGL does not need to transport dried sludge away and the wholesaler can use the same trucks as for her regular charcoal supplies.

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12.2.2 Environmental layer

An overview of the environmental impacts identified is provided in Figure 28. As noted, the main **material** used in the production is sewage sludge. This is combined with sawdust from the nearby timber market in Accra central (two parts sawdust to one part biomass). Another production input is a polymer binder that is used to bind the carbonised crushed sludge. Then the biochar **production** facility produces a limited level of smoke, soot, and CO_2 from the kiln used to carbonise the sludge. There could also be fugitive emissions. However, these are yet to be determined as the type and amount of emissions are also not quantified in detail for this particular plant.

By way of **supplies and outsourcing**, the initial source of energy for firing the kiln was wood based until there was enough biochar produced to fire the kiln. In the near future SSGL intends to fire the kiln with biogas that is produced by the wastewater treatment plant. Additionally, the supplies for the biochar production includes packaging materials such as paper, sacks, and plastic sheets. The biochar production facility presently produces 10% of its operating capacity, mainly due to limited storage and the fact that the product is still being tried and tested. So far, it emerged that in the **end-of-life** of the biochar, it leaves more ash than regular charcoal produces and although this may be a draw-back, the ash could be used for soil amendment after tests have been done to ensure that it is safe.

The biochar is presently being tested by selected SMEs. Feedback so far has been that there is no unpleasant smell when the biochar is burning except a whiff when it is being extinguished with water. The studies conducted show that the biochar works better with energy-efficient stoves that have tiny fans that give more air (oxygen) to the fire. At the moment, the biochar produced is **transported** by pickup trucks for delivery to the SMEs using them. These are associated with climate gas emissions, as they run on petrol or diesel.

Overall, a major **environmental benefit** of the biochar product is thought to be as a solution that can reduce deforestation associated with production of wood-based charcoal. According to the Energy Commission of Ghana, although biomass consumption in the fuel mix is declining (from 3,432 ktoe in 2000 to 3,162 ktoe in 2021), it still remains the major source of fuel for most Ghanaians (Energy Commission, 2022). Thus, finding an alternative in biochar will contribute to reducing deforestation in Ghana. On the other hand, the biochar production process and use can have adverse **environmental impacts** such as emission of sulphur and other gases, and inappropriate disposal of ash and packaging materials.

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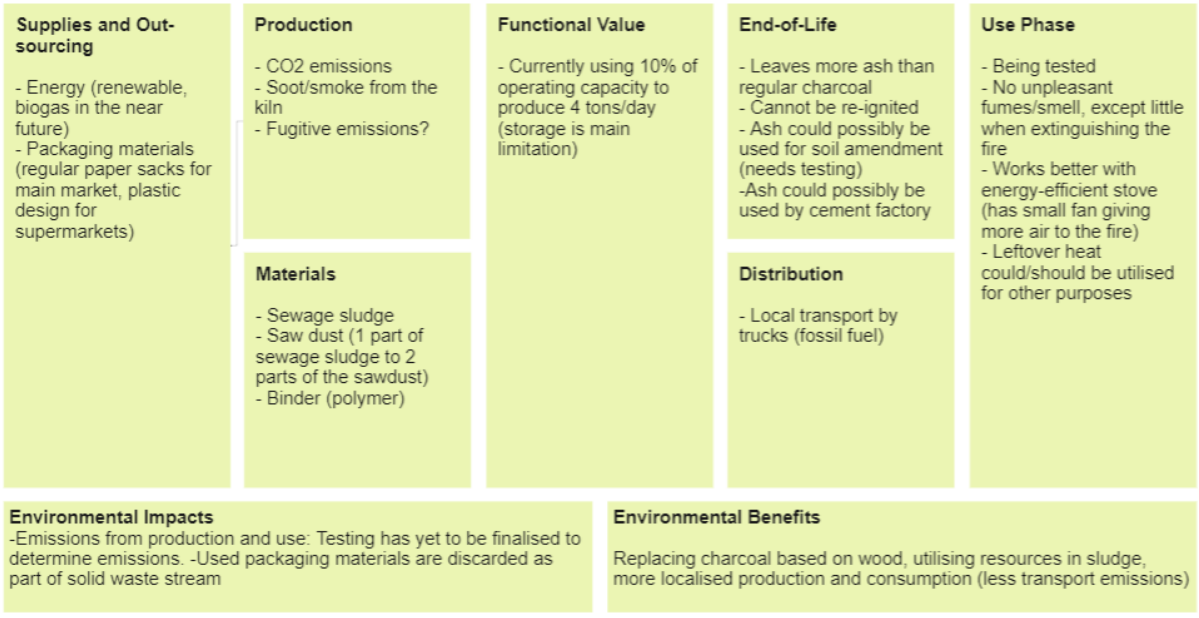


Figure 28: Environmental layer, TLBMC for biochar from SSGL.

12.2.3 Social layer

The biochar BM is associated with strongly positive **social value** creation. From the **local communities'** perspective, there is no complaint about noise, smoke, or any other nuisance beyond the SSGL's facility. As shown in Figure 29, the operation of the biochar production line has prospects for jobs creation. At present the demonstration facility employs six people on full-time and that is expected to increase to eighteen when the plant begins full operation to its capacity. Hence, the social value of job creation for the local communities.

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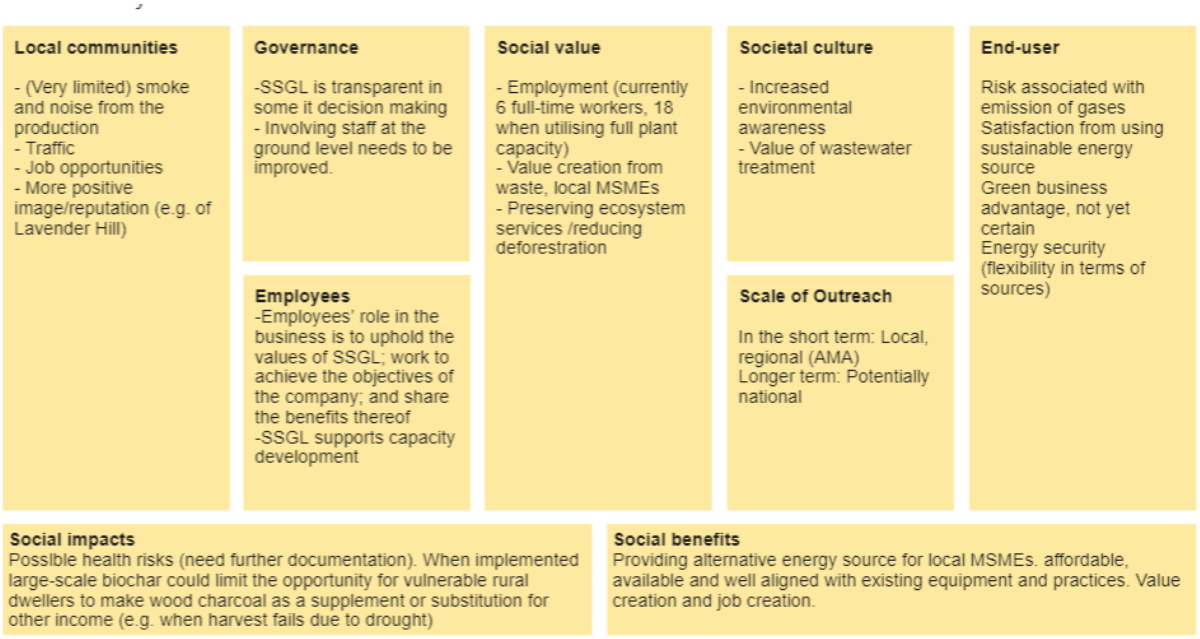


Figure 29: Social layer, TLBMC for biochar from SSGL.

Furthermore, the conversion of 'waste' to charcoal (resource) is expected to influence **societal culture** by increasing environmental awareness and to improve on the image and reputation of Lavender Hill, a name given to the current location of the wastewater treatment plan as an irony to the sight and smell of the place a few decades ago. As far as the **scale of outreach** is concerned, the biochar will be available in Accra, the capital city of Ghana in the short term and potentially going national in the long term.

Employees of SSGL are to uphold the values of the company teamwork, integrity, accountability, and safe operations. In terms of **governance**, we do not, at present, have detailed knowledge on transparency and involvement in decision making. As regards **social impacts**, there is also the need to document possible human risks related to the use of biochar. Furthermore, when implemented on large scale, biochar could limit the opportunity for vulnerable rural dwellers from making charcoal as a supplement or substitution for other income. Hence, there is need to begin to think about ensuring just transitions while the solution is being further developed. Nonetheless, the solution will provide important **social benefits**, in terms of affordable and available alternative energy source for local micro, small and medium enterprises.

12.2.4 Alignment across the TLBMC layers

The economic layer of the BM for the biochar is quite well worked out, with strong value propositions, a clear target customer group, as well as a good overview of costs and potential revenues. On the environmental side, the material circularity is close to 100%, e.g., almost all of the content is from biomass waste streams and returns back into the natural cycle. Since they form part of the short-term carbon cycle, the emissions from the use of biogas are not regarded as climate gas emissions. However, more documentation is needed as to whether the ashes can be used as soil improvement. The social layer, likewise, lists a range of positive benefits, but the safety of use must be better documented before the product can be certified.

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Thus, the three layers seem well aligned, but the BM will be strengthened when the risk and safety aspects are fully documented.

12.3 Maturity, dependencies, and uncertainties

The core technology in this case is pyrolysis, which is a quite mature technology. However, the intended use of the biochar is new. As we have seen, the economic layer of the BM has all elements in place, and the product is near competitive with regular charcoal in terms of price. Indeed, SSGL sees a huge potential, since the company is running several wastewater treatment plants, and more are planned. Charcoal, and the simple stoves it commonly is used in will also continue to have a role in Ghana's future energy mix (Energy Commission of Ghana, 2019). Still, while the public stakeholders consulted are positive, biochar is not included. The process to get permission for full-scale production could take long, due to lack of capacity in the public administration system, among others. Thus, there are **policy/regulatory dependencies** with respect to the biochar uptake. On the other hand, SSGL is an influential actor and there is a drive for CE in Ghana.

Dependency on infrastructure is not a major issue for now and with proper planning, the production capacity can be increased as demand rises; however, more space for storage of produced biochar will be needed. Furthermore, SSGL will have to add a dehydration system for drying the biochar indoors so as to ensure continuous production during the rainy reason.

Dependency on other resources is also not seen as a challenge, since the wood processors produce huge amounts of sawdust. However, SSGL needs to establish an agreement with the wood processing companies for guaranteed supply of saw dust. When it comes to **dependency on other value chains**, liaising with established charcoal dealer will be of mutual benefit. On the other hand, there will be a **competitive relationship** with the producers of charcoal, as well as with providers of LPG. Gradually, one may also foresee a development when charcoal stoves are replaced by renewable electricity, for convenience and in order to save biomass for other uses. Indeed, SSGL considers charcoal for this use as a solution for the short and medium-term, seeing more options for the use of sludge and biochar in future.

12.4 BM for reuse of treated wastewater

Wastewater use for irrigated agriculture has been in existence for many years because of increasing scarcity of freshwater resources in inner-city areas. As a result, the use of treated wastewater is increasingly being promoted by governments and development practitioners (Becerra-Castro, et al., 2015). In the second arm of Ghana's demonstration case, treated wastewater is therefore polished for supply to urban farmers to use for irrigating vegetables.

12.4.1 Economic layer

The **value proposition** for the reuse of treated water for urban farming is the reliability and timely supply of water for farming in the city, which will replace the use untreated "gutter" (drain) water presently being used for vegetables production. The treated wastewater is nutrient-rich and will help improve soil and plant quality. Moreover, using the treated wastewater for irrigation will make the vegetable cultivation safe for the farmers and ensure

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that the vegetables produced are good quality and safe to sell, buy and eat. This value proposition will be delivered through urban farmers in the city of Accra for a start.

As shown in Figure 30, **key resources** required in the production and delivery of water to farmers include the wastewater, treated wastewater, wastewater treatment plant (regular operation), water tankers for regular supply of water to farmers, development of a BM, road infrastructure for transportation of water to farmers, water storage facilities, and water-efficient irrigation systems. **Key activities** towards the supply of the wastewater include polishing of the wastewater treated at the Mudor Wastewater Treatment Plant, transportation of the treated water for delivery to storage tanks where the water is released into shallow reservoirs for final treatment before farmers use to irrigate their crops. Several key partners are involved in the product development. They include the Accra Metropolitan Authority, Council for Scientific and Industrial Research, Environmental Protection Agency, Farmers, Food and Drugs Authority, Transport Operators (water delivery), and Market Women.

Customer relationships could be individual, as with the water tanker services that serve many of Accra's semi-urban areas, or with groups of urban farmers having established common storage facilities. The latter, especially, will require trust-building through improvements in the products upon user feedback and ensuring reliability, quality, and timeliness of supplies. The initial target **customers** for the water reuse are farmers; however, down the value chain, other customer segments identified include food vendors, supermarkets, individual and household consumers of vegetables, and market women.

The **cost structure** has CapEx and OpEx. **CapEx** here includes the cost of adding the water polishing system to the wastewater treatment plant at SSGL and construction of a shallow reservoir with water storage tanks. **OpEx** on the other hand includes cost of renting and fuelling truck to deliver water to farmers, cost of sand media for filtering water, and cost of water disinfectant. In addition, other costs for maintenance are operation costs that need to be factored in to the cost structure. To offset these costs, there are two **revenue streams** for the water reuse: charge farmers fees for the water supply and reduce cost of producing (treating) the water. The cost of water is also important as cost must not be beyond what farmers can afford.

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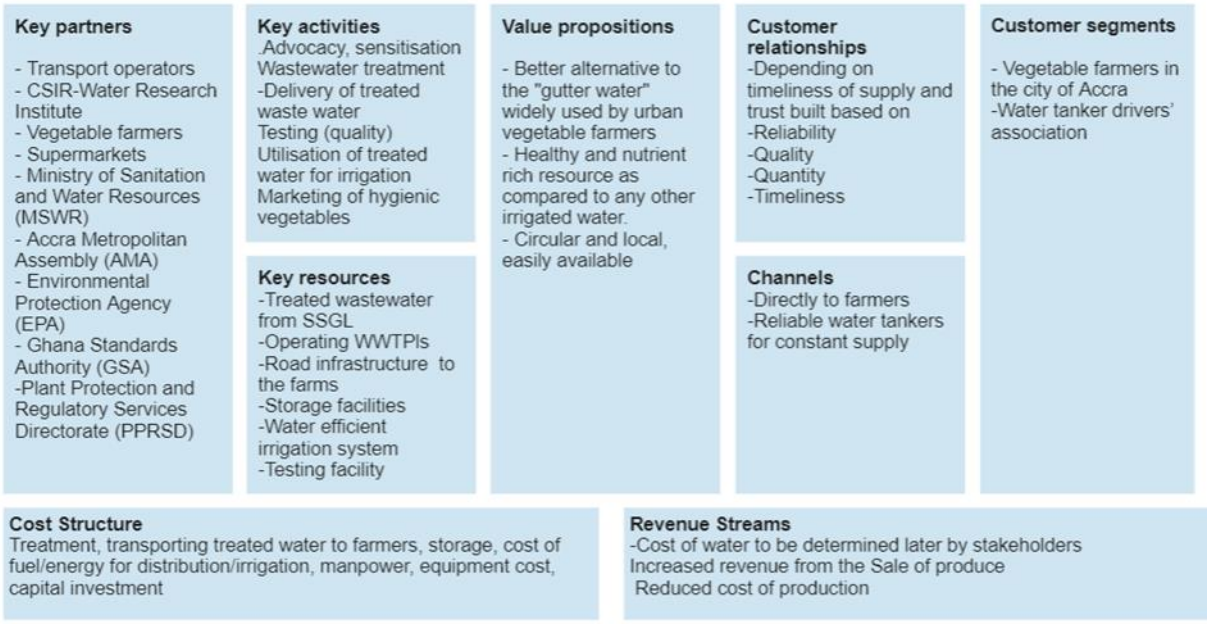


Figure 30: Economic layer, TLBMC for reuse of treated wastewater, Accra.

12.4.2 Environmental layer

As shown in Figure 31, supplies for the water reuse include polytanks for storage of water onsite at the farms, truck that transports the treated wastewater, water pumps for pumping to discharge water from trucks into water storage polytanks, electricity generators, fuel (petrol and diesel). Some of potential environmental consequences of production include emissions from the vehicles, generators and water pumps that use fossil fuel, odour from water if not well treated, and contamination of vegetables by pathogens.

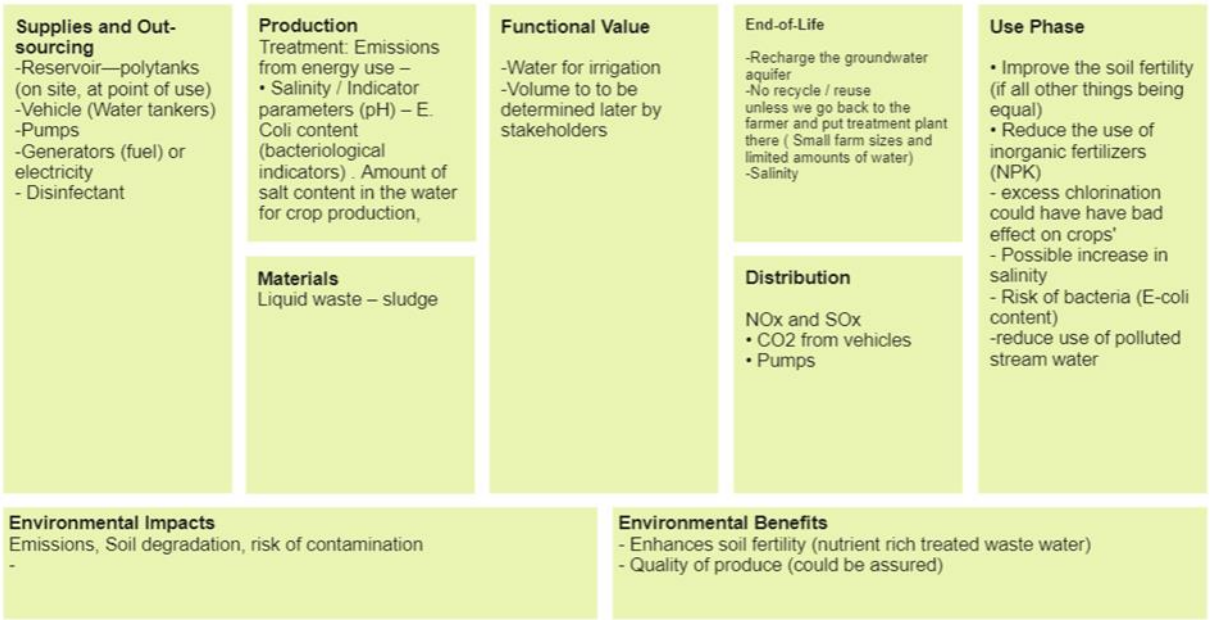


Figure 31: Environmental layer, TLBMC for reuse of treated wastewater, Accra.

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The **functional value** is related to the use of the water for irrigation to produce safe vegetables for the city. The solution owner is yet to determine volume and the cost per cubic litre of water. Meanwhile, **materials** that go into the production of include liquid waste, chlorine for disinfection, sand media, and carbon (biochar used in place of charcoal). In the **use phase** of the treated water, there is improvement in soil fertility due to the nutrients in water. Hence, there is reduction in the use of inorganic fertilizer such as NPK by farmers. The **end-of-life** for the reuse of water includes the recharge of groundwater aquifer. However, there are emissions associated with **distribution**, due to the fact that the treated water is transported by trucks and water pumps are used to discharge the water as well as for irrigation by farmers. Thus, associated emissions include Nitrogen oxides (NO_x), sulphur oxides (SO_x), and Carbon Dioxide (CO_2).

With respect to **environmental impacts**, the water reuse solution has the potential to negatively contribute to emissions, soil degradation, contamination of crops if water is not properly treated, and affect personal health of farmers and consumers. On the other hand, there are **potential positive impacts**. The benefits exist of enhancing soil fertility thereby reducing the use of inorganic fertilizer, and the quality and safety of the vegetables can be assured due to the removal of pathogens and other contaminants during the water treatment stages.

12.4.3 Social layer

Local communities are affected in a positive way by the water reuse solution. As shown in Figure 32, the production of safe and healthy vegetables for the city is one such effect; there a reduction in risks to public health from consumption of vegetables. Furthermore, ensuring that urban farmers, as **end users**, have water to farm throughout the year improves on job security and income levels. **Social values** of the water reuse solution include work and happiness attitude borne out of the confidence that the water being used for farming is safe and increased real estate value.

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Figure 32: Social layer, TLBMC for wastewater reuse, Accra.

Water reuse for urban farming can have a negative **social impact** of using prime residential land for farming. In economic terms, vegetable farming does not yield same value as real estate and may be seen as a drawback. On the other hand, the **social benefits** of water reuse for urban farming are substantial. Some of these include employment opportunities for unemployed youth, reduction in crime rate due to reduced unemployment, improving ecotourism, and enhancement in the aesthetics and air quality of residential areas.

12.4.4 Alignment across the TLBMC layers

The water reuse has potential too affect the income of farmers who can save time used in fetching drain water to irrigate their crops. This saved time can then be used to tend the crops as well as expand cultivation. Hence, the solutions are aligned to the economic layer as well, given that the SMEs' and farmers' expansion of production and cultivation are due to the improved economic levels of the owners. On the environment layer the water reuse will contribute to the reduction in extraction of stream water for irrigating crops. The effect of this also is that healthy vegetables will be produced.

12.5 Maturity, dependencies, and uncertainties

The treatment process and handling of water is mature. However, for the full and wider adoption, there are several uncertainties and dependencies. First, farmers willingness to pay for the water was not positive (when a willingness-to-pay question was asked during the baseline study); and although SSGL is yet to provide a cost for the water production and delivery, is it not certain that farmers will be willing to pay for the water when they can revert to using the untreated wastewater from the drains as before without any punitive repercussions. Hence, there will be a need to enforce city byelaws on crops production with untreated wastewater, to ensure that farmers are compelled to use safe water to irrigate their

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crops, and which could compel them to pay for the safe water from SSGL. There lies the dependence on policy/regulation.

Second, even if farmers were willing to pay for the water, the present means of delivering the water is my means of trucks, which has negative environmental impacts, e.g., due to use of diesel, and which contributes to increasing the cost of the water to the farmer. Hence, the absence of infrastructure for delivering the recovered water to farmers is the main source of infrastructure dependency. For instance, SSGL has to deliver the water by means of truck over a distance of 12km. Third, farmers' access to a premium market is key to increase their incomes, which will in turn enable them to afford to pay for the cost of water delivery (at least). Thus, there is a greater dependency on market access for farmers to ensure they have increased income from reusing water for their farming. The ability to obtain premium pricing from restaurants and supermarkets depends on testing that assures that the produce is safe. The downside to farmers selling to supermarkets and restaurants; however, is that the market women who also depend on the farmers could be denied access to the vegetables.

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13 BM exploration in the Czech case

13.1 Solution in focus

The BM in the Czech case is selling reclaimed water from the wastewater treatment plant for irrigation of Prague public green areas. The water is reclaimed in the Prague wastewater treatment plant, at Cisarsky island in the Vltava River. The plant currently has two treatment facilities. The older waterline consists of a conventional wastewater treatment plant. The new water line (finished in 2019) is an underground mechanical-biological wastewater treatment plant with chemical phosphorus precipitation. The design was limited by space because the whole plant was to be completely covered due to flood protection and esthetical appearance of the area near Prague historical centre. The sludge management processes are a mixture of primary sludge and biological sludge, both from the old and the new waterline.

The plant is near Prague Zoo, the Botanical Garden, the gardens of Troja castle, and Stromovka Park. The roof of the new waterline itself is a green park area. These nearby green areas require artificial irrigation during summer, for which often potable water is used until now. A part of the Stromovka Park is also irrigated with Vltava River water, through a historical water tunnel. The location of the plant relative to these potential users implies relatively low transportation costs. During the demonstration phase, about 20m³ of treated water is planned to be transported for irrigation in green areas per day. The wastewater is disinfected either only with ultraviolet (UV) treatment, which already exists in the new waterline, or also with ultrafiltration (UF) treatment (membranes), which would require new investment. Additional UF treatment is expected to somewhat decrease the levels of undesired substances in reclaimed water, such as micro-pollutants and bacteria.

13.2 BM for water reuse for irrigation in Prague

During the project timeline, most realistic potential early users of reclaimed water were expected to be Stromovka park and tree watering in the nearby 7th District of the city of Prague. Therefore, these early use cases were used as the starting point for the business model exploration.

13.2.1 Economic layer

In terms of economic factors, **key partners** in this BM are the owner of the water infrastructure PVS, the infrastructure operator PVK, and the city of Prague as the owner of the green areas and its gardening department responsible for irrigation (Contribution rganization Lesy hl. Města Prahy). The Ministry of Agriculture is in a key role as a regulator. Moreover, other possible users of reclaimed water (such as various types of urban gardeners and greenery providers) and technology providers (such as monitoring equipment) are important.

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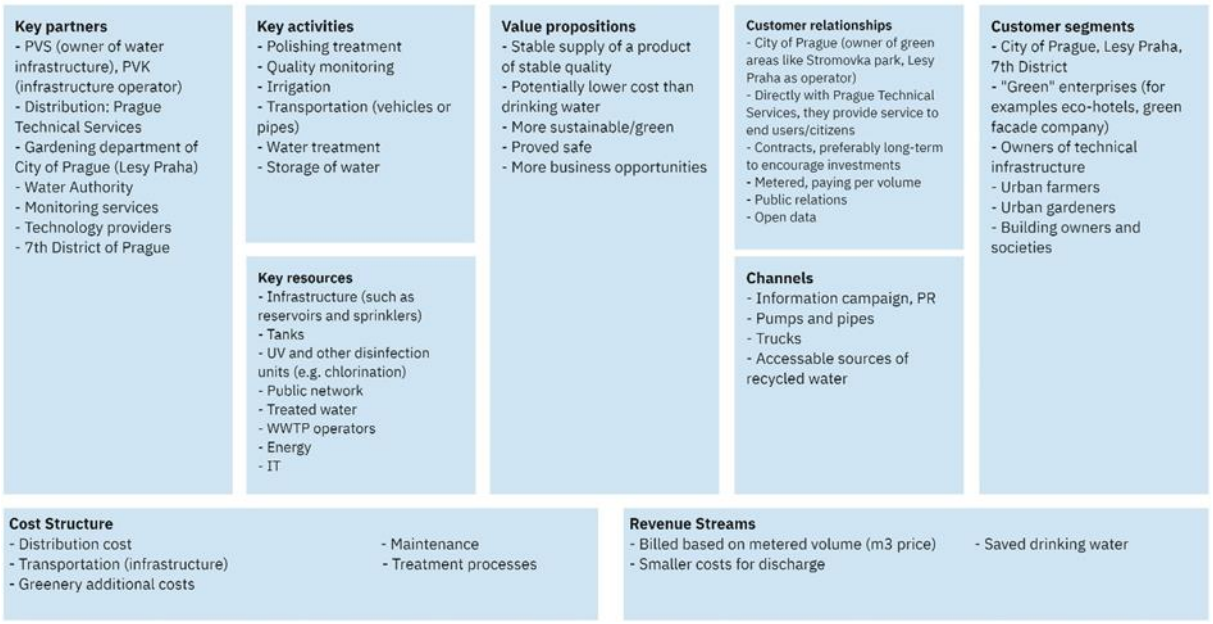


Figure 33: Economic layer, TLBMC for reuse of treated wastewater, Prague.

As shown in Figure 33, **key resources** are the reclaimed wastewater as well as the related physical infrastructure, means of transportation and distribution of reclaimed water (pumps and pipes, possibly trucks), and tanks for storage of water. Reflecting this, the **key activities** in this BM are water treatment, quality monitoring, water transportation and use in irrigation. The key **value proposition** for use of reclaimed water in irrigation is that it increases the availability of a stable water source with stable quality for green areas, which is valuable especially during droughts when there may be water scarcity. According to work done in WP3 of WIDER UPTAKE, the reuse of water for irrigation of green areas after UV treatment is an efficient solution in various indicators (see Table 10 and Table 11). The findings in Table 10 are from the New Water Line. Here, the base alternative stands for continued non-reuse of wastewater for irrigation (discharge to the river), and higher values indicate higher efficiency. In Table 11, concerning irrigation of green areas, the base alternative is continued use of river water in irrigation. Higher values indicate higher efficiency.[17]

Table 10: Preliminary estimates of efficiency of different solutions at the WWTP.

Alternative	Energy efficiency (m^3/kWh)	Economic efficiency (m^3/€)	Water circularity (%)	Nutrient circularity (%)
Base alternative	3980	10122	95	55
Water reuse for irrigation after UV treatment	3980	10122	95	92
Water reuse for irrigation	715	1593	95	92

[17] Source: Adrian Werner, Anurag Bhambhani & Zoran Kapelan (2023). Application of the circularity/efficiency assessment framework – key results. WIDER UPTAKE project, report.

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after UV and UF treatment				

Table 11: Preliminary assessment of the efficiency of different solutions on the use side (irrigation, green areas).

Alternative	Water efficiency (m^2/m^3)	Nutrient efficiency (m^2/kg)	Energy efficiency (m^2/kWh)	Economic efficiency ($m^2/€$)	Water circularity (%)	Nutrient circularity (%)
Base alternative	1.71	68.38	390.41	1.15	8	25
Water reuse for irrigation after UV treatment	4.35	146.48	390.41	2.27	58	65
Water reuse for irrigation after UV and UF treatment	4.35	146.48	390.41	2.27	58	65

Moreover, reclaimed water is expected to be cheaper than potable water for the user, where the latter costs about 2,1€/m³. It also increases circularity in the city and improves resilience of the water system, as well as paves way for further water circularity in future, such as using reclaimed water for street cleaning.

In terms of **costs**, both negative and positive effects are foreseen. Key costs are related to the capital and operational expenditure of the distribution of the reclaimed water. In terms of capital expenditures **(CapEx)**, investments in trucks (or alternatively pipelines) are needed. Additional investments in ultrafiltration treatment would naturally decrease the economic and energy efficiency of reclaimed water. Operational expenditures **(OpEx)** will include personnel, energy, chemicals for disinfection and membrane cleanings, as well as water quality monitoring and maintenance. In terms of **revenues,** the end-user will be billed per cubic meter, creating a new revenue stream for the utility. The wastewater utility will get lower costs from effluent discharge, while for users reclaimed water is cheaper than potable water from tap, but likely more expensive than river water. Key **customer relationships** are between the utility and the city of Prague: its gardening department Lesy Praha. In future, however, also other municipalities, various green enterprises (such as hotels or green façade providers), other urban gardeners, building owners and other communities could be potential **customers** of reclaimed water.

13.2.2 Environmental layer

An overview of the potential environmental impacts of the BM is provided in Figure 34. In terms of environmental factors, the hypothetical maximum of the supply of reused water is the amount of treated water in the new waterline, which typically is around 2m³ per second, and about 100 000m³ per day. Such a large volume of water reuse would of course reduce the demand for potable water. However, in practice it is unlikely that all treated water would be reused. This is also affected by e.g., tank volumes and demand for irrigation water. For

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comparison, a bit less than half a million cubic meters of water is annually used for irrigation of grass areas in Prague (less than 5 days of wastewater treatment in the facility). Hence the partners at the BM workshop assumed a **functional value** of 470 000-665 000 m3/year. However, the city aims to increase the amount of its green areas, which will increase the demand for irrigation water. Nevertheless, there is thus plenty of capacity for other possible uses. In terms of **resources and equipment**, water reuse requires energy, chemicals, and membranes in the treatment process. In the irrigation part, e.g., new sprinkler systems are needed in new greenery areas.

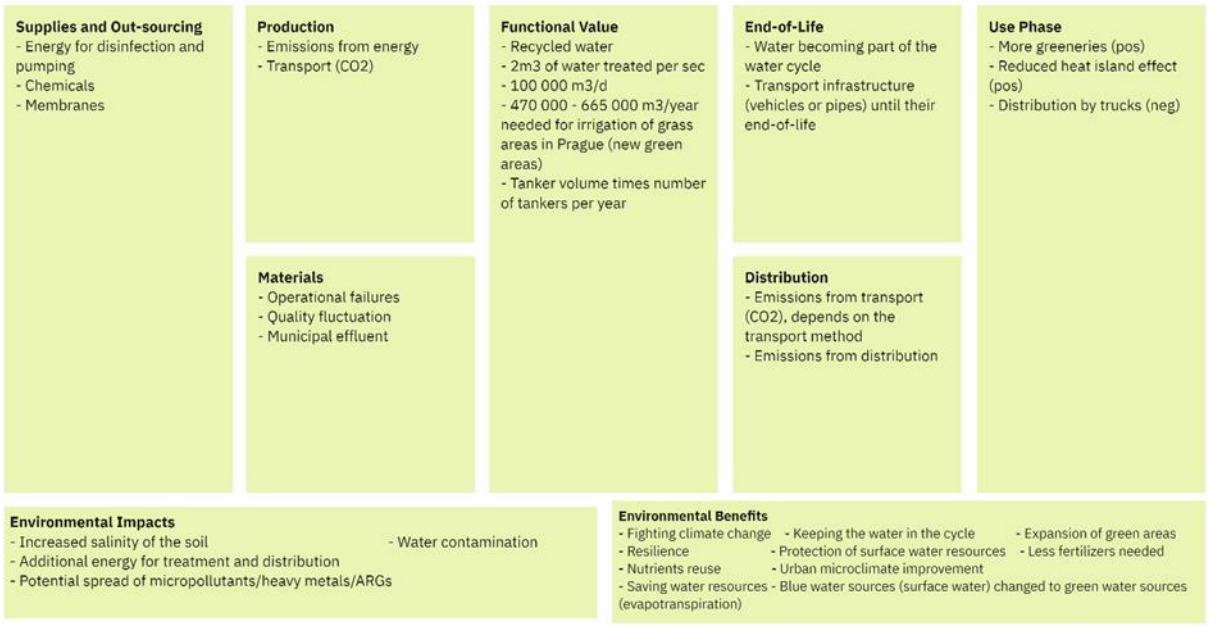


Figure 34: Environmental layer, TLBMC for treated wastewater reuse, Prague.

Environmental benefits are the protection of surface water resources and consequent water resource savings and adaptation to climate change, for instance to extreme weather events such as droughts. Meanwhile, the resilience and the circularity of the water system in increased and the urban microclimate is improved due to more plants. Indeed, increased availability of irrigation water allows the establishment of more greeneries in the city, thus reducing heat island effect in the inner city. As reclaimed water is rich in nutrients, need for artificial fertilizers is reduced in irrigation areas (see Table 11). At the same time, also negative **environmental impacts** can be foreseen. **Distribution** of reclaimed water requires energy and may lead to aerial emissions. Reused water may also increase the salinity of soil, and carry undesirable nutrients, such as micropollutants, heavy metals, and antibiotic resistance genes. Moreover, for instance operational failures and quality fluctuation in wastewater can cause unexpected environmental impacts.

13.2.3 Social layer

In terms of social factors, the key **social value** of the water reuse BM is improving the water security of the irrigation of the city's green areas, increasing liveability and attractiveness of the city especially during hot summer periods. It also creates irrigation opportunities for urban gardeners and other potential users. The availability of irrigation water increases **liveability** also

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by enabling the increase of green areas, reduction of the urban heat island effect, reduction of plant loss during droughts, and improvement of biodiversity in urban areas.



Figure 35: Social layer, TLBMC for reuse of treated wastewater, Prague.

As shown in Figure 35, increased water reuse may also influence **societal culture**, by raising the general environmental awareness and attention of the public e.g., in terms of circularity, water cycles, sustainability, etc. Increased circularity and sustainability may also improve the city's public image and bolster environmentally friendly mindset. Also, novel economic and employment opportunities may follow especially in the case that water reuse is more widely considered as an option also in other uses than city greeneries. At the same time, the general public may also react negatively to water reuse, not accepting the use of wastewater-based resources in public greeneries. For gardeners, some protective gear (such as gloves) may be needed. However, increased circularity and environmental focus may increase the sense of purpose and meaning in daily work. In terms of **governance**, transparency regarding the quality of water may be achieved by publishing water quality data on a publicly available website.

13.2.4 Alignment across the TLBMC layers

The key value propositions of water reuse in Prague are environmental and social in nature. Reused water not only increases circularity by reducing the consumption of potable water and artificial fertilizers, but also increases the resilience of the water system in case of droughts. This means that the city of Prague has better conditions to achieve their goal of increasing green areas in the city as affordable and sustainable irrigation water becomes more available. Increase of green areas then may works as a climate adaptation mechanism by reducing urban heat island effect and making the city more attractive for its residents and visitors. The case is thus a typical CBM in the sense that it can be seen to create economic, environmental, and social value, respectively.

13.3 Maturity, dependencies, and uncertainties

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The technological maturity of the solution is high: wastewater treatment with UV disinfection is already in place, and the preliminary results suggest that the discharged water already today has high enough quality to be used for irrigation of green areas in the city (Puškáčová et al., 2021). Also, the technological solutions for the transport water and irrigation are conventional: transport with trucks, or alternatively by pipelines, with regular sprinkler systems. Challenges related to the BM are thus primarily non-technological in nature.

The BM is **dependent on infrastructural development:** Commitment to a site is needed to allow transport investments to take place. Moreover, the distance of the site from the water treatment plant affects the economic feasibility of water reuse (longer distance, higher costs of transport). In a rough estimate, it is expected that the use site should be within approximately 5km distance from the wastewater facility for truck transport to be feasible. Building fixed (underground) infrastructure such as pipelines would have high sunk costs and commit to long-term water reuse in specific sites to justify such investments. Also, storage issues need to be resolved.

Resource dependency: A challenge of water reuse is that the input factor (sewage water) cannot be controlled by the supplier of reclaimed water (the wastewater utility). A wastewater treatment facility simply must deal with the material that arrives to the facility. Therefore, also the quality of reclaimed water has some variability. However, the preliminary results from WIDER UPTAKE project shows that the water quality after treatment is rather stable and maximum values in different substances would be within the "Class 2" parameters set by the Czech regulations (where "Class 1" has no restrictions for use, and "Class 3" is not allowed for use) for irrigation use, which gives a suitable starting point if reclaimed water is allowed for irrigation use of greenery areas.

Dependency on other value chains: Generally speaking, the city of Prague, which is the owner and the operator of the green areas in the city, has been positive about the idea of reclaimed water for greenery irrigation. The city also has the stated goals of increasing both the level of circular economy and greenery areas within the city – something that water reuse for irrigation could support. It is expected that the city is willing to pay for the reclaimed water if it is cheaper than tap water. The utilities PVS and PVK as organizations are also highly interested in the water reuse ambition. The key partners in the early phase of water reuse are thus generally positive about the BM. In the long term, however, its robustness would be strengthened by having multiple customers for the reclaimed water.

Dependency on policy and regulations: The BM for water reuse for greenery irrigation is currently blocked by national regulations. The Czech Water Act does not currently recognize the terms "reused water" or "reclaimed water" in the context or irrigation. Irrigation with reused water is therefore not allowed. EU Regulation 2020/741 seeks to increase water reuse in EU, and the Ministry of Agriculture is responsible for implementing this regulation in the Czech Republic. However, this has not yet taken place. There is currently a dialogue with the Ministry of Environment, suggesting that treated wastewater for urban reuse can be defined as utility or usage water and thereby allowed if it satisfies the same requirements as in EU Regulation 2020/741. Also, public health authorities will be in key role.

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In the absence of regulations that allow the use of reclaimed water for irrigation, gardeners consider alternative solutions. A key alternative is to use river water. According to preliminary results from WIDER UPTAKE, river water also has less salinity and micropollutants, and is likely cheaper than reclaimed water. However, reclaimed water has a stable quality and more nutrients. However, even if use of reclaimed water is permitted, the BM may still face competition from direct river water use.

When it comes to social acceptance, the general public's understanding of the concept and meaning of water reuse in irrigation, and its safety, will be crucial. The expectation is that the past droughts have created awareness on the importance of water system resilience, and that the public acceptance of using reclaimed water for greenery irrigation is good. Transparency about the quality of the treated will be important – this information is already today available on the website of PVS. Similar acceptance issues naturally apply also to possible customers of reclaimed water. For instance, some kind of certification may be needed to convince users regarding the quality and the safety of the water for irrigation purposes.

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14 Comparative discussion

14.1 Maturity of the identified business models

The identified BMs and their contexts are different and cannot be directly compared. Still, with the ongoing transition in the sector as a whole, it is interesting to take an overarching look at how mature the different models are, e.g., to what extent they are developed and agreed-upon, and how far the partners have reached in terms of implementing them. This depends on both internal factors, related to the specific solution and partnership in focus, and on external factors, e.g., the dependencies and key uncertainties identified for each of the cases. In this section, we focus on the internal factors.

Table 12 highlights four dimensions of maturity. First, the technological maturity of the focal solution is a key factor of influence; if the technology is up and running, it is easier to be specific about other BM elements, than if the solution itself is yet to be finally developed or selected. A second factor is what we have called the extent of the BM. This concept is drawn from governance policy analysis, where extent refers to the completeness of the policy or BM; to which degree it covers all relevant actions and aspects. In the third column, we have included a category termed commitment, referring to whether the actors and stakeholders identified as key partners have bought into or formally committed to the model, e.g., in terms of agreements, specific actions or active engagement with the partnership. Finally, we highlight the time dimension, based on the noted observations on the extent to which the model is operative or when it can be implemented or upscaled.

Table 12: Dimensions of maturity for the studied BMs.

Business model	Technological maturity	Extent	Commitment	Time dimension
Soil improvement products, Sirkula	Basic solution in place Continuous development of new designs	All elements are in place Dialogue on expansion	All actors committed Possibly different future directions	In operation Positive development New factory being planned (1-2 years) (short-term)
Struvite, Hias	Full scale facility 2022 Producing, early-phase	All elements identified Process to prioritise between alternatives	All actors committed	Starting On the market in few months (short-term)
Organic fertiliser, IVAR	Basic solution in place Continuous development of new designs	All elements in place for Vietnam Not clear for Norway	Common interests, but also differences in perspective	Model for Vietnam/international market in place Application in Norway more uncertain (medium term)
Biocomposite for riverbank protection, NPSP	Established for other applications.	At the current stage, all elements are not clear	Key partners committed, but not all are on board	Still demonstration phase Uncertain (medium term)

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	Still experimenting with alternatives			
Water reuse, Sicily	Mature technology, Ultrafiltration unit in place	Elements in place for trial period (not for permanent relations)	Key partners committed	Demonstration Uncertain (medium- or long-term)
Biochar, SSGL	Facility in place	All elements are in place for the demonstration; not for the upscaling	All partners committed, dialogue with policymakers	In operation Positive development (short-term)
Water reuse, Accra	Mature technology Pilot	All elements not in place for the demonstration; not for the upscaling	Positive interest, depending on demonstration results	Demonstration ongoing but uncertain in the long-term
Water reuse, Prague	Mature technology Pilot	All elements not in place	All key partners are not committed yet	Demonstration Uncertain (long-term)

As Table 12 shows, the focal solutions in all the cases are relatively mature. Still, there are certain variations. Reuse of treated wastewater for irrigation in agriculture and urban dwellings is based on known processes and implemented in an increasing number of cases around the world, despite high investment costs and institutional barriers. In Accra and Prague there are pilot facilities, whereas Corleone has the UF facility in place. Biochar from pyrolysis is also a known technology, but the use of sludge as feedstock to provide biochar for the intended use in Ghana is novel. The use of biocomposite for riverbank protection in the Netherlands is also new, and different material solutions are explored in parallel. In the cases of soil improvement from Sirkula and organic fertiliser produced at IVAR, the processes are established, although there is a continuous development of new designs. As noted in Chapter 3, the technological readiness level of struvite extraction from wastewater treatment is 8-9, and at Hias, a full-scale facility has recently been put in place. However, water treatment for reuse is less implemented than the solutions for nutrient and material resources recovery, even though this technology, as such, is as mature and stable.

When it comes to extent, all elements in the BMs are in place for the organic fertilisers at IVAR, as well as for the soil products from Sirkula and struvite from Hias. In the former two, the models are currently up and running, and in the case of Sirkula there are quite specific plans for expansion. In the Netherlands, there is still uncertainty concerning elements such as manufacturing of the riverbank protection panels. For the use of biochar for energy in Ghana, the BM has also been worked out in detail, but the product has yet to be put on the market pending certification. However, for the reuse of treated wastewater for urban agriculture in Accra, key partner roles, customer relationships and channels beyond the pilot stage are not yet defined. In Sicily agreements are being made for an initial trial period, but in terms of more long-term interactions a model has not been fixed, and the same is the case for the BM for reuse in Prague.

In terms of commitment, all key actors are already engaged or involved through intention agreements in the cases linked to Sirkula and Hias. This may be related to the findings on

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industrial networks in these cases. In Hamar, a quite large network of stakeholders was reported. Hias, in particular, as well as Sirkula, have central roles, also in terms of linking stakeholders. Thus, one could say that Hias is not only an anchor firm providing key resources for the IS, but also takes a role as "network orchestrator" which is quite important, according to the literature reviewed in Chapter 6. Furthermore, relations between the core partners are long-established, and most of the identified partners are in geographical proximity of each other. For the organic fertiliser produced at IVAR, all are keen to utilise the digestate in the most sustainable way, but less clear on future directions, e.g., international versus Norwegian market. According to the SNA, this industrial network has a core with many reciprocal ties, but not one single actor standing out as an orchestrator, although Terramarine has an important bridging role.

For biocomposites in the Netherlands, NPSP and Waternet are committed, as are other resource providers that NPSP rely on, whereas a manufacturer is currently lacking. Here, the network is more diverse and flexible. The relationship between NPSP and Waternet is close, but the BM is more loosely linked with the core activities and processes in Waternet. Also, one of the intended partners is located in another country. For water reuse in Sicily all partners for the demonstration stage are committed, and there is an active dialogue to convince relevant policymakers. The latter is reflected in the SNA, which shows a high number of reciprocal ties. However, the user side may be less represented. In the case of biochar from SSGL all key partners are committed. As we have seen, SSGL has a dominating role in the industrial network in this case, whereby they can also "orchestrate" the needed value chain. As regards water reuse for urban agriculture in Accra, closer relationships and more dialogue with the relevant authorities and potential users are needed. For water reuse in Prague there is a positive interest, but the future engagement will depend on the demonstration results (as well as development of regulations).

The time perspective for implementation and upscaling is also variable: As we have seen, the BM for the soil improvement products from Sirkula has entered operation. There is a positive development, and an expansion is foreseen in the near future (1-2 years). The BM for the struvite from Hias has been agreed upon, and the product is expected to enter into the market within few months. For the organic fertilisers from IVAR, the BM in relation to the international market is in place, but with a view to the Norwegian market things are less clearly defined. For biocomposites in the Netherlands, the BM is currently in a testing phase and more uncertain in the medium term, as it depends on results from pilot testing. For water reuse in Sicily, a BM beyond the initial trial period is not worked out in detail but given the recent network developments and investments that already have been made it is expected to come into place, perhaps in a medium-term perspective. For the biochar for energy in Ghana, we have seen, the model is in operation with positive prospects in the short term. For the water reuse in Ghana and Prague, however, institutional changes are required, and the considered BMs are more likely to be implemented in a longer-term perspective.

14.2 BM impacts and alignment levels

Sustainable business model development can be understood and assessed in multiple ways. In the preceding chapters, we first applied the TLBMC, to explore new BMs and discuss the values

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they may generate according to the so-called 'triple bottom line'. Table 13 highlights some of the key aspects identified for each BM, in order to discuss their levels of alignment and provide some overall reflections on their contributions towards sustainability. It does not purport to give a full overview of the impacts of each BM.

Table 13: Selected key aspects and levels of alignment for the studied BMs.

NB. The + and (-) signs are used to give a tentative weight in terms of outcome, from +++ as strongly positive, to (-) (-) (-) as having a quite negative or uncertain outcome. This in order to provide a summary view of the alignment between the TLBMC layers in each case.

Business model	Economic layer	Environmental layer	Social layer	Alignment between layers
Soil improvement, Sirkula	++ Revenue from sales Cost price principle	+++ Replacing peat Eliminating waste	+++ Employment, awareness, inclusion, etc.	High
Struvite, Hias	++ Revenue from sales Avoided costs Cost price principle	+++ Replacing non-substitutable P Limit use of chemicals	+ Impact limited by volume Social value through innovation synergies	Medium
Organic fertilisers, IVAR	+++ Profit, Terramarine IVAR gets rid of digestate Full cost coverage principle	++ Replace non-substitutable P Eliminating waste Transport emissions	+++ Employment Job satisfaction Awareness Food security, employment Vietnam	High
Biocomposite for river bank protection, NPSP	(-) High costs Revenues from some sales	+ Replace tropical hardwood Circular	++ Employment Knowledge development Awareness	Low
Water reuse, Sicily	(-) High costs Lack of funds Revenues uncertain	+++ Increased water security Nutrient-rich Possible micro-pollutants, salinity	+++ Employment Community revival Awareness Job satisfaction	Low
Biochar, SSGL	++ Revenue from sales Price increasingly competitive	+++ Limit deforestation Low-emission energy	+++ Employment Awareness Job satisfaction	High
Water reuse, Accra	(-) (-) Cost of treatment & distribution Revenues uncertain	+++ Secure and stable water supply (for urban agriculture) Nutrient-rich Possible micro-pollutants, salinity	+++ Urban livelihoods Food security Liveability, aesthetics Reduced health risk	Low

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Water reuse, Prague	(-) High costs Smaller costs for discharge Increase fees	+++ Replacing use of potable water Nutrient-rich Reduced heat island effect Possible micro-pollutants, salinity	++ Liveability, aesthetics Awareness	Low

For the majority of the cases, the profitability under current conditions is not high. The soil improvement products sold by Sirkula generate economic value, but their production is integrated with other activities at their recycling centre and the organic raw materials come from sludge and waste treatment processes financed by user fees. The economic value linked to struvite generation at Hias is first and foremost linked to avoided cost. In the case of organic fertilisers from IVAR, Terramarine is able to sustain itself as a commercial company, but the digestate from IVAR has so far been provided free of charge. SSGL may earn money from the biochar production, based on current market trends. In the case of biocomposite from NPSP, the actors willingly admit that the cost of the material is a main challenge. When it comes to water reuse, the preliminary findings from WP3 suggest that some of the solutions are more economically efficient than the current alternatives. However, the distribution of roles, financing, and economic feasibility of the BM for full-scale operations is not yet clear.

When it comes to environmental aspects, all the studied BMs have multiple and clearly identified positive impacts, although the interactions in the Dutch case are quite complex and difficult to assess. The energy used is renewable in most cases, and material circularity is also high, though varying between the different biocomposite designs at NPSP. Both the soil improvement, fertiliser products and treated wastewater reuse provide important nutrients and have positive impacts on microbial life and soil structure. For the water reuse BMs, we have noted the risk of micro-pollutants and increased salinity. In principle, this is a risk also for the soil and fertiliser products, but since this is much less, we did not include it in the table. Moreover, it is important to note that major positive environmental impacts are linked to wider system interactions, e.g., mitigating water scarcity, substituting fossil P, and limiting use of tropical hardwood and deforestation, as well as to direct impacts of the products themselves.

Overall, the identified social impacts are also highly positive. While some, such as employment, can be valuated, others are difficult to measure and assign a monetary value. Most of the noted benefits are local and regional, however some are also national and international. If only current conditions are considered, some of the social impacts linked to the BM for struvite are limited by the fact that the volume of struvite will be limited, hence this is rated a bit lower, but when the synergies with soil improvement from Sirkula and possible expansion to a larger soil factory with Norgro are considered, these may also be substantial.

The ratings from +++ to (-) (-) (-) in Table 13 provide a tentative weighting of the impacts across the different layers for each BM. Based on this and the key aspects highlighted, the last column provides a high-medium-low rating of the level of alignment for each BM. As shown, we argue that the level of alignment is relatively high both for the soil improvement from Sirkula, the organic fertiliser from IVAR and the biochar from SSGL, where the environmental and social

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benefits are combined with business and economic value creation. For the BMs linked to struvite from Hias and biocomposite from NPSP we respectively find a medium and low level of alignment. In the case of Hias, there are potentially high environmental benefits, but the direct social and economic gains will be more limited since the volume of struvite is limited. In the case of NPSP, we also see environmental benefits, but these vary across designs. The social benefits in terms of increased awareness and knowledge-building are potentially, substantial, while possibilities for increased employment is limited, and the economic side is still uncertain.

For the BMs linked to water reuse, we argue that there is a low degree of alignment in all the cases. The potential environmental and social gains identified are high, especially in Sicily and Ghana, but so far these are not matched by any promising economic model, the gains remain only potential. This underscores the argument that valorisation of wastewater resources is needed (Water Europe, 2020; Kundu et al. 2022). In addition, it also implies that unlike the other resources such as biochar and organic fertiliser whose transportation for distribution is relatively easy and/or adaptable, the same cannot be said of the water reuse where additional investments are required for distribution.

14.3 Business Model dependencies

The impacts and alignment levels discussed above are influenced by the interaction between the BMs and their wider ecosystem, amongst others, including the natural resource base, existence or need for development of infrastructure, dependency on existing value chains and their interaction with other existing or emerging BMs. An overview of the most important dependencies identified for each BM is provided below (Table 14).

Table 14: Dependencies identified for the studied BMs.

Business model	Dependency on infrastructure development	Dependency on resources	Dependency on existing value chains	Interaction with other business models
Soil improvement products, Sirkula	No, except production hall	Digestate from Hias is core, Supplemented with other organic waste fractions	Future distribution through Felleskjøpet (Nordic Garden or Norgro)	Synergy with BM for struvite from Hias Sirkula Recycling Park
Struvite, Hias	No	Volume limited by volume of wastewater treated Important to control/limit contaminants in sewage	Soil production of Norgro (and Sirkula), distribution network of Felleskjøpet Cost of precipitation chemicals	Strengthened if other WWTPs also will produce Ecological farming Competing with conventional fertiliser companies (synergy possible in future)
Organic fertiliser, IVAR	No	Digestate from IVAR is core, but may be supplemented with other organic waste fractions	Only Vietnam, for now Utilising cheap return ships from Asia for transport	Struvite harvesting could make product more relevant in Norway More expensive mineral fertilisers may increase demand Currently not good fit with fertiliser plans and

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				equipment of Norwegian farmers
Biocomposite for river bank protection, NPSP	No	Increasing competition for biomass, competing with other uses	Needs partner willing to produce the material and contractors who are willing to use the product	Depends on acceptance from Water Authorities and municipalities, Competes with tropical hardwood and recycled plastics
Water reuse, Sicily	Yes, needs reservoir and distribution network	Yes, depends on volume and quality of wastewater	Cost of chemicals	Pumping groundwater, energy prices
Biochar, SSGL	No, except more storage space	Sludge and sawdust	Wholesaler of charcoal	Presently linked to use of charcoal stoves, competing with charcoal and LPG
Water reuse, Accra	Yes, in medium and longer term; also needs storage and transportation	Yes, depends on volume and quality of wastewater	Cost of chemicals	Water tanker services
Water reuse, Prague	Yes, in medium and longer term	Yes, depends on volume and quality of wastewater	Cost of chemicals	Possible synergy with blue-green infrastructure providers

As Table 14 shows, the different BMs have a varying dependency on infrastructure development. This can be linked to the mode of distribution of the products in the different BMs. The soil products from Sirkula, the struvite from Hias, the organic fertilizer from IVAR/Terramarine, biochar from SSGL, and the biocomposite products from NPSP can all be transported by truck, ship, or other means of transportation. In the BMs where water is the end-product – in Accra, Prague, and Sicily – there is a need for infrastructure development in the medium to long term in order to make the BM viable. Transporting water by trucks or tractors is feasible in the demonstration phase but seems less feasible when the BMs are to provide water for reuse in a larger scale. The infrastructure needed will include distribution systems (pipes and pumps) as well as storage reservoirs (basins or ponds).

The BMs are to a varying degree influenced by resource dependencies. The cases concerning water reuse in Prague, Sicily and Accra are all dependent on certain volumes and qualities of wastewater to be successful. Correspondingly, the IS networks for these solutions have been characterised as resource dependent in chapter 5, where the core actors SSGL, AMAP and PVS are described as anchor firms in the IS networks as they control the water resource needed for the BM to be successful. The BM for struvite from Hias, likewise, is dependent on the volume and quality of wastewater inflow. The BM for the soil improvement from Sirkula is not dependent on the biomass from Hias to the same extent, but by providing biogas, digestate and struvite Hias still holds a position as main anchor for the industrial symbiosis network in the Hamar case. It is an interesting observation that the key actors in the networks, who have a high degree of power within the network through their control over resources, themselves are dependent on or restricted by the volume and quality of the resource/s. While Terramarine and Sirkula may utilise biomass from different sources, their activity depends on the availability of biomass which in a longer-term perspective may be demanded for multiple uses and

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eventually limited. One could say the same about the biochar in Ghana, but here SSGL is in control of both the resource and the product, and due to their "quasi-monopoly" position they have a big and increasing pool of resources to draw on. The BM for biocomposite material in the Netherlands needs access to input materials such as reeds/grass, calcite, recovered toilet paper. In contrast to the other IS networks. Thus, NPSP's network is wide and open, with interaction and potential for sourcing different raw materials from different actors and promoting their materials for different applications. This gives flexibility, but the conceived BM also gives them less control over their supply chain. In their case, as the Netherlands is a CE frontrunner, increasing competition for certain kinds of biomass is already noted.

Furthermore, Table 14 provides an overview of the different BMs' dependency on existing value chains, that is, on e.g., existing distribution channels, supplies, and access to relevant production partners. The water reuse BMs in Sicily and Ghana, for example, are strongly dependent on access to and prices of chemicals needed for treating the wastewater up to the standard required for reuse. NPSP's BM for biocomposites depends on getting manufacturers that are willing and have the capacity to produce the biocomposite riverbank protection panels, and contractors who are willing to install the new solution rather than the traditional wooden riverbank protections. Although dependency can have negative connotations, there are also cases where the studied BMs can utilize their connections in existing value chains as an advantage. One example is how the organic fertilizer from IVAR/Terramarine takes advantage of cheap transport utilising return ships from Asia, thus improving the economic sustainability of the BM. Another example, related to market access, is SSGL's BM for biochar where they use established wholesalers in the market for regular charcoal to promote and sell their new product. Similarly, both Sirkula and Hias aim to use the existing network of Felleskjøpet, through Norgro and/or Nordic Garden, to reach potential customers.

In terms of interaction with other BMs, IVAR's BM for organic fertiliser production could be influenced if IVAR starts producing struvite in a way similar to Hias. Phosphorous (P) recovery could make the fertilizer product less attractive in areas with high need of P such as Vietnam since it would contain less P (or more P would need to be added). On the other hand, less P could make the fertiliser more relevant for areas less in need of P, such as many parts of Norway. At the same time, the struvite may replace fossil P. Currently, the struvite from Hias will be used as an alternative and in this sense compete with BMs for production of mineral fertilizers in applications where the slow-release properties of struvite are an advantage. Depending on volumes and a range of other factors, a symbiotic relationship may also develop in a longer-term perspective, where the struvite is used as input to conventional fertiliser production. The biochar produced by SSGL is currently being used in charcoal stoves. As a substitute for wood-based charcoal, the biochar is evidently in competition with traditional charcoal. Additionally, charcoal as fuel for stoves used by MSMEs and in households is currently being challenged by LPG, which does not affect air quality in the same way as charcoal. As such, the BM on biochar is in competition with both charcoal and LPG. In the Netherlands, the BM for biocomposites is currently competing with the dominant solution for riverbank protection, which is hardwood panels, as well as alternatives based on recycled plastics. As such they rely on reaching the potential end-user of their solution, water boards and municipalities, and convince contractors to include biocomposite solutions in the tenders that are put by the water boards and municipalities.

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14.4 Wider trends and uncertainties

The respective BMs and their impacts will also be influenced by wider societal trends. How the anticipated climate change impacts on water resources play out will influence the demand for treated wastewater reuse. In semi-arid Sicily, severe water stress is and will remain an increasing challenge. While the Czech Republic already is classified as water-stressed, Ghana is expected to be water-stressed by 2025, due to climate change (Kankam-Yeboah et al., 2013), and the country continues to rely on wood to produce charcoal. According to the Energy Commission of Ghana, 52.1% of the total wood supply was used for charcoal production in 2021 compared to 28% in 2000 (Energy Commission, 2022, p. 41). The Czech Republic is already water stressed, and a multi-year drought in the country in 2014 –2019 has significantly increased public awareness.

Demographic development will also influence the demand for alternative water sources. Projections for Sicily show a population reduction of 9.9% towards 2050. However, AMAP does not foresee a reduction of demand, due to the impending climate changes which will limit the supply from natural sources. While the Czech Republic has a fairly stable population, the population of Prague is expected to increase by 20% by 2050 (OECD, 2018). The national population growth rate for Ghana towards 2050 is projected to be 1.38%, with a strong urbanisation trend: While the population of Accra Greater Accra Metropolitan Area currently is around 4 million, it is expected to grow to 9.6 million by 2050.[18] This is likely to give a strong increase in the demand for water as well as energy.

In the Czech Republic, water reuse for industrial purposes has high public acceptance.[19] However, less is known about the public acceptance of reuse for urban blue-green infrastructure. Among key stakeholders, there is some scepticism concerning use of treated wastewater for agriculture, which also seems to influence attitudes towards urban reuse. Increased documentation of the risks and benefits will hopefully increase the level of acceptance. In the Sicilian case, the picture is mixed: Local farmers are willing to participate in an initial trial phase. However, previous studies suggest that the willingness to pay for treated wastewater among Sicilian farmers up to now has been limited (Ventura et al, 2019). Despite prevailing regulatory barriers (discussed below) other key stakeholders show increasing interest. However, some note that meeting current wastewater treatment and discharge requirements should be priority number one, as Sicily is struggling with backlogs in this area.

The reuse of treated wastewater for agriculture is fairly new in Ghana, although the use wastewater for urban agriculture has been practiced in the major cities of the country for several years. Hence, the key stakeholders note a need for awareness-creation and market studies. The urban farmers targeted in the demonstration phase are very positive. Other key stakeholders also see a good potential, but documentation of the risks and benefits of water

[18] https://www.who.int/initiatives/urban-health-initiative/pilot-projects/accra
[19] Veolia (2020), press release (in Czech), https://www.veolia.cz/cs/media/tiskove-zpravy/cesi-vnimaji-recyklaci-vody-jako-jeden-z-nejefektivnejsich-nastroju-v-boji.

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reuse in this complex urban setting is crucial. When it comes to biochar, the level of public and end-user acceptance seems high, related to the fact that the local, largely informal charcoal production is an issue in public debate. As many, especially lower-income, households and MSMEs rely on charcoal energy for certain uses and the price of charcoal is increasing, competitive alternatives are welcomed. In this case, too, the relevant authorities request more documentation, as will be provided by WIDER UPTAKE.

The BMs linked to resource recovery are influenced by climate change impacts on water resources and demographic factors in more indirect ways. As noted in Chapter 4, global food demand is projected to increase drastically towards 2050. This, together with anticipated climate change impacts, will influence the geographical distribution, balance between livestock and plant-based production, and related resource use in agriculture. Although depletion of phosphate rock reserves may not be the most important limitation for global food production, a decline of the production in China is projected and it is anticipated that the global supply will rely increasingly on individual countries, most notably Morocco (Walan et al., 2014). At the same time, UNEP (2021) estimates that 40% of drained peatlands needs to be rewetted by 2050, in order to transform global peatlands from a net source of greenhouse gas emissions to a net sink. Thus, increased conservation and valorisation of peat is foreseen.

How these trends will play out in different regions is uncertain, but there is a strong likelihood that they will increase the demand for fertilisers and soil improvement based on renewable resources. In the Hamar case, as we have seen, there is increasing acceptance of the soil improvement products from Sirkula. In Norway, spreading of sludge on farmland (under certain, regulated conditions) is widespread and there is less scepticism towards sludge-based products than what has been seen in the EU. There is an increasing concern to conserve peatlands, as well as increased public awareness about CE and the need to limit the use of virgin resources. Struvite is less known but promoted by the Centre for Ecological Farming in Norway, as well as the professional gardeners that have tested it so far. The social acceptance of organic fertilisers of the type produced by IVAR in Norway has been influenced by the prevailing practice of spreading sludge, which farmers are used to getting for free. Agricultural advisors introduced to the product in in an early phase were also sceptical about its usability. Still, as we have seen, there is reason to believe that cost is the most decisive factor, so acceptability may increase as the costs of mineral fertilisers are rising, in Norway as well as in the international market.

In the Netherlands, the level of awareness and public acceptance for solutions made from recovered resources is considered as high. The Dutch water boards, who are the most likely end-users of the biocomposite riverbank solutions, are leading the way through tenders that promote environmentally friendly solutions and products. However, the solution relies on increased market acceptance for it to be feasible. Thus, despite some uncertainties, the are several wider development trends that seem to be conducive for the studied BM. In line with the literature reviewed in chapter 6 (Donner et al., 2020; Vermun et al., 2019, Guldmann and Huulgaard, 2020) we find that limited acceptability may be a challenge, for wastewater reuse and also to some extent for organic fertilisers. However, the local stakeholders also see a positive development in this area.

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14.5 Impact of policy and regulations

As noted in Chapter 7, policy and regulations can be both drivers and barriers to CE based on water resources. The BMs for soil improvement, struvite and organic fertilisers in Norway find support in the national CE strategy, as well as in strategies for bioeconomy development both at the national and regional level. Stricter enforcement of the national discharge requirements linked to the Urban Wastewater Treatment Directive since 2019 is also forcing the sector to focus more on different secondary treatment options and associated opportunities for resource recovery, especially for larger utilities and in sensitive areas. With the new version of the Directive, proposed in late 2022, the requirements are tightened and applied also for smaller utilities, and tertiary treatment is required for more of the larger ones in Norway. This may create a window of opportunity, where more wastewater treatment plants can pool resources and for example to achieve larger volumes of struvite. However, the huge investments that are required could also lead to sub-optimal technology choices, and there is a lobby that is more focus on exemption provisions than sustainable innovation.

The EU Fertilising Products Regulation (EU FPR) (Regulation (EU) No 2019/1009), in force from 2022, harmonises the requirements for fertilisers, setting harmonised limits for a range of contaminants. It is also associated with a delegated act, which adds 'STRUBIAS' materials (biochars / pyrolysis materials, struvite / recovered phosphate salts, and ash-based products) to the list of materials allowed in fertilising products, defines criteria for their production and use, and concludes that they are safe and efficient for use in farming. This, together with the recent approval of 'STRUBIAS' also for ecological farming in the EU, is boosting the interest in soil and fertiliser products that contain struvite. On the other hand, the decision to not include sludge-based fertilisers under EU FPR constitutes a barrier for the organic fertiliser from IVAR, in that it cannot be CE marked and traded in the EU, even if it is allowed in Norway and a number of other countries. Another barrier, for the time being, is that the national Regulation of Fertilisers of Organic Origin (2003/951) has been under revision since 2009. This, together with an underlying struggle about the rules and requirements for sludge and sludge-based products, is causing uncertainty regarding the future framework conditions and market for such products.

In the case of the BM for biocomposites by NPSP no major regulatory barriers were identified. Circular solutions linked to water resources as well as for building and construction are actively encouraged in the national CE strategy, and resource recovery and CE are also identified as important in the 2030 Vision document of the Dutch Water Boards. While conducive policies are in place and use of instruments such as green public procurement and partnering are encouraged, operational barriers remain.

As for the biochar BM in Ghana, there is not much direct policy support, e.g., the Renewable Energy Master Plan for Ghana includes charcoal, briquets, woodlot cultivation and some biogas and 'waste-to-energy', as well as improved biomass cookstoves. However, although biochar is not specifically mentioned, the opportunity is there (given assurance of quality and safety) to develop it under “charcoal.” Thus, while there is relevant legislation, policymakers have paid little attention to how charcoal is produced and sold locally (Brobbey et al., 2015), except for export. Thus, there are no major hindrances in terms of legislation; rather, a regulatory gap and

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need for recognition. While there are different kinds of green economy projects/programmes being pursued in Ghana (Ali et al., 2021), treated wastewater reuse is not an area of focus for any of them. This poses a barrier for the BM linked to water reuse. Future implementation will also depend on the direction and operationalisation of AMA's Integrated Master Plan for Accra and its suburbs, e.g., in terms of spatial planning and scope for urban agriculture, as well as development of infrastructure. In addition, SSGL continues to construct wastewater treatment plants (WWTPs) nationwide and it is therefore anticipated that the BM linked to water reuse in Ghana would be taken into consideration as the company constructs more WWTPs.

For the BM for reuse of treated wastewater in Prague, there are also barriers and uncertainties linked to governance and regulations. The water sector is regulated in accordance with the relevant EU directives, including Directive 2000/60/EC, which establishes a framework for the EU action in the field of water policy. However, the local stakeholders note that there are coordination challenges, as the responsibility for the national regulations is spread across several institutions. The strategic framework for CE under development for the Czech Republic addresses solid organic waste (OECD, 2021), but does not provide a prominent place for wastewater. As in Ghana, national water policy does not yet recognize “reclaimed water” and “water reuse” as legal categories. Until this gap is addressed, it is difficult to upscale reuse solutions.

Both in the Czech and Sicilian cases, the quality requirements for reuse water are a main concern since quality monitoring is a major cost component. Here, the implementation of EU's Regulation on minimum requirements for water reuse (EU 2020/741) is a crucial factor. In the case of Sicily, the existing regulation (Ministerial Decree 185/2003) currently requires testing of fifty-four parameters for quality, health, and safety of the treated wastewater that is to be used for irrigation in agriculture. According to AMAP this testing regime is unconducive for the BM on water reuse in Sicily, as the testing regime will be too costly considering the price they can ask for the treated wastewater. However, the EU Regulation (EU 2020/741) requires testing of thirty-two parameters. A change of the national regulation is therefore hoped for. In this regard, the recent publication of a regional law (L.R. SICILIA 22/03/2022, N. 4 “Norme in materia di riutilizzo delle acque reflue urbane”) on the reuse of urban wastewater, stating that the Region will guarantee the reuse of treated wastewater in accordance with the EU regulation (EU 2020/741), is an important achievement.

Thus, remaining regulatory barriers are of influence, especially for the BMs linked to reuse of treated wastewater. Important processes are ongoing, both at the EU, national and regional levels, and the degree of coordination and policy push onwards will affect the feasibility and sustainability of the studied BMs strongly.

15 Conclusion

This report has provided an assessment of the industrial networks and BMs being developed in relation to the demonstration cases in WIDER UPTAKE. Key concepts from the literature on Industrial Symbiosis and Social Network Analysis were applied to characterise the industrial

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networks in each case and discuss some of their strengths and weaknesses, with a view to further development.

To build common understanding and stimulate further development of Circular Business Models, a BM exploration process centred on two workshops was conducted for each of the cases. The Triple Layered Business Model Canvas was used as basis, supplemented by a discussion of future scenarios, linked to the types of dependencies and uncertainties highlighted in the Ecologies of Business Model Experimentation (EBME) framework.

The finding underscore that Circular Economy based on water resources has certain, special characteristics, since it is closely integrated with the natural water cycle. Water must be abstracted within ecological boundaries, and reuse and recovery of resources must be done with a strong focus on minimizing risks, optimising positive impacts in relation to ecosystems and human health, and with returns on investments (social or economic).

The studied BMs have a strong potential, especially in terms of the environmental benefits and social value that can be created. Besides energy efficiency and circularity, positive impacts on soil and plant growth and reduced use of chemicals are significant environmental impacts for the models linked to water reuse and nutrient recovery. On the social side, employment effects are observed for all the BMs, as well as impacts in terms of environmental awareness and knowledge development. Direct impacts on food security and energy security are also found in some of the cases.

The economic feasibility and value generated vary including: the cases of organic fertilisers from IVAR/Terramarine; soil improvement products from Sirkula and biochar from SSGL where there is a strong potential in the short term; to the struvite case where economic gains are mainly from avoided costs and biocomposites, where cost competitiveness remains a major challenge; and the BMs for water reuse, where preliminary results indicate economic efficiency for some of the cases, but the revenue streams are difficult to specify until remaining regulatory and organisational barriers are addressed.

Importantly, a large part of the economic, environmental, and social value creation that is foreseen is related to wider system interactions, e.g., saving groundwater for potable use, substituting fossil phosphorus as a critical raw material, limiting deforestation and use of tropical hardwood, as well as providing livelihoods for poor urban dwellers and reversing land abandonment. We also note that most of these benefits arise from the coupling of resource flows, across enterprises and sectors. This underscores that Industrial Symbiosis holds important potential and may be associated with higher sustainability gains than individual circularities, especially when the volume of resources is limited.

While some are based on green-twinning (e.g., NPSP and Waternet) and pre-existing organisational relationships (e.g., Hias and Sirkula), the industrial networks we have seen may to some extent be characterised as anchor-tenant systems, e.g., AMAP, PVS, SSGL, Hias and IVAR provide wastewater, sludge or digestate from their main processes that constitute a main input for the new BMs. This may result in a form of resource dependency, which should be well managed to avoid potential lock-in effects.

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The industrial networks in the studied cases also have different characteristics. The ones in Hamar, Sicily and Accra have dominating core actors who also tend to take a role as network orchestrators. In Stavanger we find a core with several stakeholders and Terramarine as an important bridging actor, while the Prague case, for now, did not exhibit any clear core, and the Dutch network appears more open and flexible. Those relating to reuse have a higher share of public stakeholders, reflecting the institutional challenges that need to be solved, whereas those related to resource recovery involve more private sector ties. The degree of network collaboration underscores the need for broad stakeholder engagement and cross-sector collaboration in transitioning towards CE and a water-smart society.

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